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Master Thesis

2D Electron Spectroscopy in Hot Environments: A Redfield Master Equation Approach

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Declaration of Authorship

I, Leopold Bodamer, declare that this thesis titled, "2D Electron Spectroscopy in Hot Environments: A Redfield Master Equation Approach" and the work presented in it are my own. I confirm that:

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- Where any part of this thesis has previously been submitted for a degree or any other qualification at this University or any other institution, this has been clearly stated.
- Where I have consulted the published work of others, this is always clearly attributed.
- Where I have quoted from the work of others, the source is always given. With the exception of such quotations, this thesis is entirely my own work.
- I have acknowledged all main sources of help.
- Where the thesis is based on work done by myself jointly with others, I have made clear exactly what was done by others and what I have contributed myself.

I confirm that this printed version is identical to the digital version of my thesis.
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Date:

Contents

Notation and conventions

Throughout this thesis, natural units are used such that $\hbar = 1$. Frequencies, energies, and inverse times are therefore expressed in the same units.

The following notation conventions apply:

- **Vectors.** Vectors in real space are denoted by bold symbols, e.g. $\boldsymbol{E}$ (electric field), $\boldsymbol{P}$ (polarization), $\boldsymbol{k}$ (wave vector), $\boldsymbol{d}$ (transition dipole moment), and $\boldsymbol{e}$ (polarization unit vector). The magnitude of a vector $\boldsymbol{d}$ is written as $d = |\boldsymbol{d}|$ when required.

- **Operators and states.** Operators acting on Hilbert spaces carry a hat, e.g. $\hat{H}$ for the Hamiltonian and $\hat{\boldsymbol{\mu}}$ for the dipole operator. Density operators are written as $\hat{\rho}$. Matrix elements in a chosen basis are denoted by plain symbols, e.g. $\rho_{ij} = \langle i|\hat{\rho}|j\rangle$.

- **Quantum mechanical pictures.** Different dynamical representations (Schrödinger, interaction, Heisenberg) are indicated by subscripts. For example, $\hat{O}_I(t)$ denotes an operator in the interaction picture. To avoid ambiguity with this convention, the interaction Hamiltonian is written as $\hat{H}_{\text{int}}$.

- **Superoperators.** Linear maps acting on operators (e.g. propagators or Liouvillians) are written in calligraphic font, such as $\mathcal{U}$, $\mathcal{L}$, or $\mathcal{M}$. Their action on density operators is written explicitly as $\mathcal{L}[\hat{\rho}]$.

- **Field projections.** For a fixed laboratory polarization direction $\boldsymbol{e}$, vector fields are often projected onto this axis. The scalar field envelope is then defined as

  $$\varepsilon(t) = \boldsymbol{e} \cdot \boldsymbol{E}(t),$$

  and correspondingly $\hat{\mu} = \hat{\boldsymbol{\mu}} \cdot \boldsymbol{e}$. Whenever such projections are used, the underlying vector character is understood but suppressed in the notation.

- **Complex quantities.** The symbols Re and Im denote the real and imaginary parts of a complex quantity, respectively.

EBERHARD KARLS UNIVERSITÄT TÜBINGEN

Abstract

Mathematisch-Naturwissenschaftliche Fakultät
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Master of Science

2D Electron Spectroscopy in Hot Environments: A Redfield Master Equation Approach

by Leopold BODAMER

This thesis investigates the application of two-dimensional photon-echo spectroscopy to model quantum systems, with a focus on open quantum system dynamics. A comprehensive theoretical framework is developed that combines the response-function formalism of nonlinear spectroscopy in a non-perturbative setting with master-equation approaches for open quantum systems (QuTiP). Starting from minimal reference models, including a single qubit and a four-level system formed by two coupled qubits, the methods are validated and the approach is applied to a system of N qubits arranged on a cylindrical geometry inspired by the structural motifs of microtubules. The results demonstrate the capability of two-dimensional spectroscopy to probe coherence and energy-transfer dynamics in these systems, providing insights into their quantum behavior and interactions with the environment. This work lays the groundwork for future experimental and theoretical studies in the field of quantum biology and quantum information science.

Chapter 1

Introduction

REFERENCE: sounds good but are refs good?

Quantum dynamics in molecular and condensed-matter systems unfold on femtosecond timescales and are shaped by the interplay of coherent inter-site couplings and environmental fluctuations. In many systems of current interest—light-harvesting complexes, molecular aggregates, conjugated polymers—the relevant electronic degrees of freedom are embedded in warm, structurally complex surroundings that drive rapid decoherence and dissipation [1, 2]. Understanding how these environment-induced processes compete with coherent energy transfer, and identifying their signatures in experimentally accessible observables, is a central challenge in the physics of open quantum systems.

Conventional linear spectroscopy—absorption and emission—reveals excitation energies and oscillator strengths but compresses all dynamical information into one-dimensional lineshapes, where homogeneous dephasing, population relaxation, and static disorder produce overlapping contributions that are difficult to disentangle [3]. Nonlinear spectroscopy overcomes this limitation. In particular, two-dimensional electronic spectroscopy (2DES) correlates excitation and detection frequencies by controlling the time delays between a sequence of ultrashort laser pulses [4–6]. The resulting frequency–frequency maps separate homogeneous from inhomogeneous broadening, resolve inter-state couplings through cross peaks, and track energy-transfer pathways as a function of the waiting time. On the theory side, the measured signal can be expressed through multi-time response functions of the transition-dipole operator, providing a direct link between spectroscopic observables and open-quantum-system dynamics [3].

This thesis develops a computational framework for simulating 2DES signals of open quantum systems in parameter regimes relevant to warm, structured environments, and applies it to model systems of increasing complexity.

1.1 Scope and approach

The open-system dynamics are modelled with the Redfield master equation [1, 2, 7], which provides a transparent microscopic link from a system–bath Hamiltonian—characterised by bath correlation functions and spectral densities—to relaxation and dephasing rates. Because Redfield theory is not guaranteed to preserve complete positivity, its application is restricted to regimes where the weak-coupling and Markov assumptions are well motivated. Results are benchmarked against Lindblad-form generators to assess the impact of the secular approximation and to ensure physical consistency.

On the spectroscopy side, the third-order response-function formalism [3, 5] provides the conceptual framework. Numerically, however, a non-perturbative time-domain strategy is adopted: the driven open-system equation of motion is propagated with the laser pulses included explicitly, and lower-order contributions are eliminated through phase cycling over auxiliary propagations [8, 9]. This approach naturally accounts for finite pulse durations and pulse-overlap effects without resorting to the impulsive limit.

1.2 Model systems and physical questions

The formalism is applied to a hierarchy of model systems designed to isolate specific spectroscopic signatures:

- **Two-level system.** A minimal benchmark for validating the numerical framework and for disentangling the 2D lineshape contributions of homogeneous dephasing, population relaxation, and static disorder.
- **Dimer.** Two coupled sites that produce cross peaks, enabling a direct comparison of coherent excitonic mixing and incoherent population transfer during the waiting time.
- ***N*-site cylindrical ensemble.** A multi-chromophore system whose geometry is motivated by the protofilament arrangement in microtubules [10], serving as a structured test case for geometry-dependent coupling networks and their manifestation in 2D spectra.

The central physical question is how environment-induced relaxation and dephasing, as parametrised by the spectral density and temperature, compete with coherent couplings to determine peak positions, lineshapes, and waiting-time dynamics in two-dimensional spectra.

1.3 Thesis outline

Chapter 2 introduces the open-quantum-system framework, deriving the Redfield master equation and discussing bath correlation functions and spectral densities. Chapter 3 develops the nonlinear-spectroscopy formalism, from linear response and the Kubo formula through third-order response functions to the construction of rephasing and nonrephasing 2D spectra. Chapter 4 specifies the Hamiltonians, dipole operators, and bath models for the systems listed above.

Supporting material is collected in the appendices: Appendix A (numerical implementation and software architecture), Appendix B (Maxwell-to-signal derivation for coherent 2D Fourier-transform spectroscopy), Appendix C (Dyson-series derivation of the n-th order polarization), and Appendix D (detection schemes and complex-signal reconstruction).

Chapter 2

Open Quantum Systems

Any real-world quantum system is not perfectly isolated. Instead, it is coupled to surrounding degrees of freedom such as electromagnetic modes, lattice vibrations, or solvent fluctuations, though the coupling is often weak but never exactly zero. In contrast to an ideal closed system with unitary Schrödinger evolution,

$$i\hbar\frac{\partial}{\partial t}\left|\psi(t)\right\rangle = \hat{H}(t)\left|\psi(t)\right\rangle, \tag{2.1}$$

an open system generally exhibits effective non-unitary dynamics once the environment is traced out. This coupling drives relaxation toward an environment-defined equilibrium and leads to two key mechanisms: **decoherence**, the loss of phase relations and dephasing, and **dissipation**, energy exchange and thermalization [1, 2].

Concrete examples include atoms subject to vacuum electromagnetic field fluctuations [1], quantum dots in solid-state settings coupled to phonons [11], molecular systems interacting with surrounding solvent degrees of freedom [3], and quantum computers whose coherence can be rapidly degraded by environmental noise [12].

For molecular chemistry and biological complexes, the relevant quantum degrees of freedom are embedded in warm, wet, and structurally complex environments, so strong decoherence and fast relaxation are expected. Yet these systems must still obey quantum laws, so the task is to explain classical-looking behavior within a fully quantum description. [13].

This chapter provides the theoretical foundation for including environmental effects in the quantum-dynamical models used throughout the thesis. A practical weak-coupling description is introduced that connects microscopic Hamiltonians to relaxation/dephasing rates and to bath models used later in applications and simulations. In Sec. 2.1 the Redfield master equation is introduced and derived from microscopic system–bath dynamics. In Sec. 2.2 bath correlation functions and their Fourier/spectral representations are discussed, including how emission and absorption arise via detailed balance. In Sec. 2.3 harmonic-oscillator (bosonic) environments are considered and explicit thermal correlators are computed, establishing the link to spectral densities. Finally, Sec. 2.4 summarizes common phenomenological spectral densities such as Ohmic-type and Drude–Lorentz and their physical interpretation.

2.1 The Redfield Master Equation

In this thesis the Redfield master equation is used as the weak-coupling description of open-system dynamics. Once the environment is specified by its correlation function or equivalently a spectral density, the reduced dynamics follows directly from the microscopic system–bath model and can be simulated efficiently even for multi-level molecular and excitonic systems [1–3]. More accurate methods such as numerically exact path-integral [14] or hierarchy techniques [15] can be too costly for the parameter scans and system sizes considered here. It is important to note that the Redfield equation, while preserving trace and Hermiticity, does not guarantee complete positivity of the density matrix, which is a fundamental requirement for physical quantum states [7]. So care must be taken when determining when the Redfield equation is useful [16–18].

Originally developed by A.G. Redfield in 1957 and 1965 for nuclear magnetic resonance relaxation phenomena [16], the Redfield equation describes the time evolution of the reduced density matrix, formally defined in Sec. 2.1.1, of a quantum system weakly coupled to a thermal environment. Unlike phenomenological approaches that introduce dissipation and dephasing *ad hoc*, the Redfield equation relates them to environmental correlation functions or spectral densities [1, 2, 19].

The following requirements must be fulfilled by the final derived Redfield equation:

1. The equation should be linear in the system density matrix $\frac{d}{dt}\hat{\rho}_S(t) = F(\hat{\rho}_S(t))$. This is the reduced equation of motion.

2. The equation should be Markovian, meaning that the evolution of the system density matrix at time t only depends on the state of the system at time t and not on its past history.

3. The equation should be trace-preserving. $\mathrm{Tr}[\hat{\rho}_S(t)] = \mathrm{Tr}[\hat{\rho}_S(0)]$ for all times t.

The Redfield equation is now derived from the microscopic dynamics of the system–environment composite, following the approach presented in [20] and standard textbooks such as Breuer and Petruccione [1]. The steps are presented in a more explicit way. In particular, the elimination of the first-order contribution is treated rigorously. The environmental correlation functions and spectral densities that characterize the bath properties and determine the system's relaxation behavior are then examined.

2.1.1 Mathematical Background

Before the derivation of the Redfield equation, the basic mathematical tools and notation used throughout are briefly collected: the density-operator formalism for mixed states, the trace and partial trace for extracting reduced system dynamics, and the corresponding expectation values.

Density Matrix Formalism

The density matrix formalism provides a powerful framework to describe the dynamics of quantum systems, especially when dealing with mixed states and decoherence effects. The density matrix $\hat{\rho}$ is defined as [21]

$$\hat{\rho} = \sum_i p_i \left|\psi_i\right\rangle \left\langle\psi_i\right|, \quad p_i \geq 0, \quad \sum_i p_i = 1. \tag{2.2}$$

It represents a statistical mixture, the average of an ensemble with different realizations of pure quantum states ψ_i with classical probabilities p_i. In other words, one cannot tell with certainty the quantum state of the system. This is particularly useful in the context of open quantum systems, where the system is entangled with an environment and one loses information about the system of interest to the total system. The conditions in Eq. (2.2) ensure the following properties that a physical density matrix must satisfy

$$\begin{aligned} \hat{\rho}^\dagger &= \hat{\rho}, \\ \left\langle\psi\right| \hat{\rho} \left|\psi\right\rangle &\geq 0 \quad \forall \left|\psi\right\rangle, \\ \mathrm{Tr}(\hat{\rho}) &= 1. \end{aligned}$$

In this formalism, for a canonical basis $\{|i\rangle\}$, the diagonal elements ρ_{ii} represent populations and the off-diagonal elements ρ_{ij} represent coherences between states. Coherences are phase relations, which are crucial for interference. For example for a two level system a clear phase relation and a pure quantum state would have off diagonal elements of $\rho_{ij} = 1/2$. This state is often referred to as a fully coherent superposition of the two states.

When coupling a system to an environment, the environment is responsible for decoherence. The state evolves over time to a purely statistical mixture of states.

The mean value or expectation value of an operator $\hat{O}$ can be calculated by additionally averaging over the classical probabilities p_i to obtain

$$\begin{aligned}
\langle \hat{A} \rangle &= \sum_i p_i \langle \psi_i | \hat{A} | \psi_i \rangle \\
&= \sum_n \sum_i p_i \langle \psi_i | \hat{A} | \phi_n \rangle \langle \phi_n | \psi_i \rangle \\
&= \sum_n \langle \phi_n | \sum_i | \psi_i \rangle p_i \langle \psi_i | \hat{A} | \phi_n \rangle \\
&= \mathrm{Tr}(\hat{\rho}\, \hat{A}).
\end{aligned}$$

Here, an arbitrary basis $\{|\phi_n\rangle\}$ and a corresponding completeness relation $\sum_n |\phi_n\rangle \langle \phi_n| = \mathbb{1}$ was used.

Partial Trace

The partial trace simplifies a quantum system description by removing certain parts through averaging. It can be viewed as the inverse operation to forming a tensor-product space. This is useful when only one part of a composite system is of interest. For a bipartite Hilbert space $\mathcal{H}_{\mathcal{A}} \otimes \mathcal{H}_{\mathcal{B}}$, the partial trace over $\mathcal{H}_{\mathcal{B}}$ is the unique linear map $\mathrm{Tr}_B : \mathcal{B}_1(\mathcal{H}_{\mathcal{A}} \otimes \mathcal{H}_{\mathcal{B}}) \to \mathcal{B}_1(\mathcal{H}_{\mathcal{A}})$ satisfying $\mathrm{Tr}[(\mathrm{Tr}_B \hat{X})\, \hat{A}] = \mathrm{Tr}[\hat{X}(\hat{A} \otimes \mathbb{1}_B)]$ for all $\hat{A} \in \mathcal{B}(\mathcal{H}_{\mathcal{A}})$, where $\mathcal{B}(\mathcal{H})$ is the algebra of bounded operators on $\mathcal{H}$, and $\mathcal{B}_1$ denotes trace-class operators, namely those X with finite trace norm[1, 22]

$$\|\hat{X}\|_1 = \mathrm{Tr}\sqrt{\hat{X}^\dagger \hat{X}}$$

This ensures expectation values of local observables $\hat{A} \otimes \mathbb{1}_B$ are preserved. In an orthonormal basis $\{|b_i\rangle\}$ of $\mathcal{H}_{\mathcal{B}}$, it acts as

$$\mathrm{Tr}_B \hat{X} = \sum_i (\mathbb{1}_A \otimes \langle b_i|)\, \hat{X}\, (\mathbb{1}_A \otimes |b_i\rangle). \tag{2.3}$$

With this definition thermal expectation values take the form

$$\langle \hat{A} \rangle = \mathrm{Tr}[\hat{\rho}\, \hat{A}] = \frac{1}{Z} \sum_n e^{-\beta E_n} A_{nn}, \tag{2.4}$$

with the inverse temperature

$$\beta = \frac{1}{k_\mathrm{B} T}.$$

2.1.2 System–Environment Setup

A quantum system of interest interacting with an environment with effectively infinitely many degrees of freedom is considered. The total Hilbert space $\mathcal{H}_T$ is the tensor product of the Hilbert space of the system $\mathcal{H}_S$ and that of the environment $\mathcal{H}_E$:

$$\mathcal{H}_T = \mathcal{H}_S \otimes \mathcal{H}_E.$$

The evolution of the total system is governed by the Liouville–von Neumann equation

$$\frac{d}{dt} \hat{\rho}_T(t) = -i[\hat{H}_T, \hat{\rho}_T(t)], \tag{2.5}$$

where $\hat{\rho}_T(t)$ is the density matrix of the total system and $\hat{H}_T$ is the total Hamiltonian. Units with $\hbar = 1$ are used throughout this thesis. Equation (2.5) is the mixed-state analogue of the Schrödinger equation Eq. (2.1) (see Sec. 2.1.1). Without loss of generality, the total Hamiltonian $\hat{H}_T \in \mathcal{B}(\mathcal{H}_T)$ can be decomposed as

$$\hat{H}_T = \hat{H}_S \otimes \mathbb{1}_E + \mathbb{1}_S \otimes \hat{H}_E + \alpha \hat{H}_{\text{int}},$$

where $\hat{H}_S$ and $\hat{H}_E$ act on the individual Hilbert spaces $\mathcal{H}_S$ and $\mathcal{H}_E$, respectively, while $\hat{H}_{\text{int}}$ acts on the composite space $\mathcal{H}_T$ and describes the system–environment interaction. The dimensionless parameter α is the coupling strength parameter.

The open-system setup is sketched in Fig. 2.1.

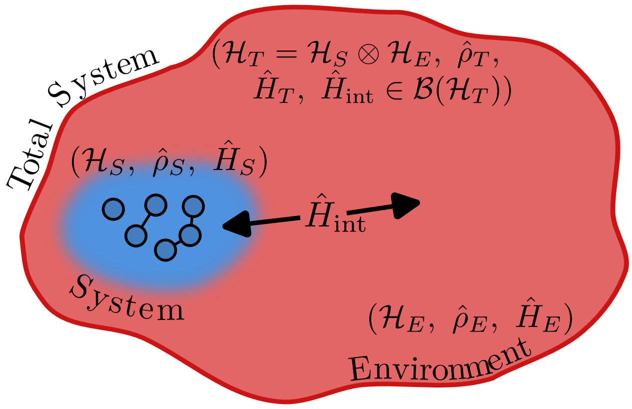

Figure 2.1: Schematic open-system setup. A system S is coupled to an environment E via $\hat{H}_{\text{int}}$. Schematically the system is indicated to contain six interacting particles such as atoms or molecules, while the environment has infinitely many degrees of freedom.

The interaction Hamiltonian is typically written in the form

$$\hat{H}_{\text{int}} = \sum_i \hat{S}_i \otimes \hat{E}_i, \tag{2.6}$$

where $\hat{S}_i$ are system operators and $\hat{E}_i$ are environment operators. These will be concretized later in Sec. 2.3.

2.1.3 Interaction Picture

To describe the system dynamics, the interaction picture is used, where operators evolve with respect to the free Hamiltonian $\hat{H}_0 = \hat{H}_S + \hat{H}_E$. Any operator $\hat{O}$ in the Schrödinger picture takes the form

$$\hat{O}_I(t) = e^{i(\hat{H}_0)t}\,\hat{O}\,e^{-i(\hat{H}_0)t},$$

in the interaction picture. Throughout this thesis, hats indicate operators, while the subscript I indicates an interaction-picture quantity.

This transformation isolates the effect of the system-bath interaction $\hat{H}_{\text{int}}$, allowing systematic perturbative treatment of weak coupling. The same interaction-picture machinery will later be applied to the light–matter interaction in spectroscopy (see Chapter 3 and Appendix C), but at the level of the already-reduced system.

States now evolve only according to the interaction Hamiltonian $\hat{H}_{\text{int}}$, and Eq. (2.5) becomes

$$\frac{d}{dt}\hat{\rho}_{T,I}(t) = -i\alpha[\hat{H}_{\text{int},I}(t), \hat{\rho}_{T,I}(t)], \tag{2.7}$$

which can be formally integrated as

$$\hat{\rho}_{T,I}(t) = \hat{\rho}_{T,I}(0) - i\alpha \int_0^t ds[\hat{H}_{\text{int},I}(s), \hat{\rho}_{T,I}(s)].$$

Inserting this back into Eq. (2.7) yields

$$\frac{d}{dt}\hat{\rho}_{T,I}(t) = -i\alpha \left[\hat{H}_{\text{int},I}(t), \hat{\rho}_{T,I}(0)\right] - \alpha^2 \int_0^t \left[\hat{H}_{\text{int},I}(t), \left[\hat{H}_{\text{int},I}(s), \hat{\rho}_{T,I}(s)\right]\right] ds,$$

The iteration can be repeated, leading to a series expansion in powers of α

$$\frac{d}{dt}\hat{\rho}_{T,I}(t) = -i\alpha \left[\hat{H}_{\text{int},I}(t), \hat{\rho}_{T,I}(0)\right] - \alpha^2 \int_0^t \left[\hat{H}_{\text{int},I}(t), \left[\hat{H}_{\text{int},I}(s), \hat{\rho}_{T,I}(s)\right]\right] ds + \mathcal{O}(\alpha^3). \tag{2.8}$$

The expansion is truncated at second order, justified by the weak coupling assumption $\alpha \ll 1$, which represents the **Born approximation**.

Equation (2.8) still contains the full history through $\hat{\rho}_{T,I}(s)$ inside the integral and is therefore *non-Markovian*. As a first step toward a time-local Born–Markov description, one assumes that the total state varies slowly on the bath correlation time and replaces $\hat{\rho}_{T,I}(s) \to \hat{\rho}_{T,I}(t)$ inside the integrand. This is a time-local substitution at second order and yields

$$\frac{d}{dt}\hat{\rho}_{T,I}(t) = -i\alpha\big[\hat{H}_{\text{int},I}(t), \hat{\rho}_{T,I}(0)\big] - \alpha^2 \int_0^t ds\, \big[\hat{H}_{\text{int},I}(t), [\hat{H}_{\text{int},I}(s), \hat{\rho}_{T,I}(t)]\big], \tag{2.9}$$

which is now local in $\hat{\rho}_{T,I}(t)$ (but still retains an explicit upper integration limit t so is not yet Markovian). The remaining Markov step is the extension of the upper limit to ∞ using short bath memory.

The exact equation of motion for the reduced state $\hat{\rho}(t)$ involves the full many-body dynamics of the environment, which is generally intractable. Therefore, the next step is to trace out the environmental degrees of freedom.

2.1.4 Reduced Dynamics and Born Approximation

Since the dynamics of the system alone is of interest, the reduced density matrix is defined using the partial trace operation in Eq. (2.3):

$$\hat{\rho}_{S,I}(t) = \text{Tr}_E[\hat{\rho}_{T,I}(t)].$$

Thus Eq. (2.9) becomes

$$\frac{d}{dt}\hat{\rho}_{S,I}(t) = -i\,\alpha\,\text{Tr}_E\big[\,\hat{H}_{\text{int},I}(t), \hat{\rho}_{T,I}(0)\,\big] - \alpha^2 \int_0^t ds\,\text{Tr}_E\big[\,\hat{H}_{\text{int},I}(t), [\hat{H}_{\text{int},I}(s), \hat{\rho}_{T,I}(t)]\,\big]$$

Vanishing First-Order Term

The first order term in α in Eq. (2.10) will vanish after tracing over the environment if the interaction operators satisfy

$$\langle \hat{E}_i \rangle_0 \equiv \text{Tr}_E[\hat{E}_i\, \hat{\rho}_E(0)] = 0, \tag{2.10}$$

This is achieved by assuming that the total system starts in a **separable** product state of the form

$$\hat{\rho}_{T,I}(0) = \hat{\rho}_{S,I}(0) \otimes \hat{\rho}_E(0), \tag{2.11}$$

Some textbooks impose the condition Eq. (2.10) as a starting assumption. Here a step further is taken. When it is not initially satisfied, it can *always* be enforced by redefining the Hamiltonian through a shift

$$\hat{H}_T = \hat{H}'_S + \hat{H}_E + \alpha \hat{H}'_{\text{int}},$$

with

$$\hat{H}'_{\text{int}} = \sum_i \hat{S}_i \otimes \hat{E}'_i, \qquad \hat{E}'_i = \hat{E}_i - \langle \hat{E}_i \rangle_0, \qquad \langle \hat{E}_i \rangle_0 \equiv \mathrm{Tr}_E[\hat{E}_i\, \hat{\rho}_E(0)],$$

and a correspondingly shifted system Hamiltonian

$$\hat{H}'_S = \hat{H}_S + \alpha \sum_i \hat{S}_i \langle \hat{E}_i \rangle_0.$$

This "renormalization" merely redefines system energy levels and does not affect dissipative structure. Hence, without loss of generality the shift is assumed to be performed so that $\langle \hat{E}_i \rangle_0 = 0$. Explicitly, the first-order term in Eq. (2.10) vanishes:

$$\sum_i \mathrm{Tr}_E\big[\hat{S}_i \otimes \hat{E}_i, \hat{\rho}_{S,I}(0) \otimes \hat{\rho}_E(0)\big] = \sum_i \big(\hat{S}_i \hat{\rho}_{S,I}(0) - \hat{\rho}_{S,I}(0)\hat{S}_i\big)\, \mathrm{Tr}_E[\hat{E}_i\, \hat{\rho}_E(0)] = 0,$$

which implements that a static bath mean field can be absorbed into $\hat{H}_S$. The reduced equation of motion becomes

$$\begin{aligned} \frac{d}{dt}\hat{\rho}_{S,I}(t) &= -i\,\alpha\, \mathrm{Tr}_E\big[\, \hat{H}_{\text{int},I}(t), \hat{\rho}_{T,I}(0)\,\big] - \alpha^2 \int_0^t ds\, \mathrm{Tr}_E\big[\, \hat{H}_{\text{int},I}(t), [\hat{H}_{\text{int},I}(s), \hat{\rho}_{T,I}(t)]\,\big] \\ &= -\alpha^2 \int_0^t ds\, \mathrm{Tr}_E\big[\, \hat{H}_{\text{int},I}(t), [\hat{H}_{\text{int},I}(s), \hat{\rho}_{T,I}(t)]\,\big]. \end{aligned} \tag{2.12}$$

To make the time-translation structure explicit, set $s' = t - s$ (equivalently, $ds' = -ds$). When s runs from 0 to t, s' runs from t to 0. Reversing the limits removes the minus sign, so the integration range remains $[0, t]$. Using this change and expanding the double commutator,

$$\big[A, [B, X]\big] = ABX - AXB - BXA + XBA,$$

Eq. (2.12) can be rewritten in the form

$$\begin{aligned} \frac{d}{dt}\hat{\rho}_{S,I}(t) = \alpha^2 \int_0^t ds\, \mathrm{Tr}_E \Big\{ & \hat{H}_{\text{int},I}(t) \big[\hat{H}_{\text{int},I}(t-s)\hat{\rho}_{T,I}(t) - \hat{\rho}_{T,I}(t)\, \hat{H}_{\text{int},I}(t-s)\big] \\ & - \big[\hat{H}_{\text{int},I}(t-s)\hat{\rho}_{T,I}(t) - \hat{\rho}_{T,I}(t)\, \hat{H}_{\text{int},I}(t-s)\big] \hat{H}_{\text{int},I}(t) \Big\}. \end{aligned}$$

Next, the assumption of Eq. (2.11) is strengthened by requiring that throughout the evolution the total state remains close to a factorized form $\hat{\rho}_{T,I}(t) \approx \hat{\rho}_{S,I}(t) \otimes \hat{\rho}_{E,I}(t)$. This can again be stated as the **Born approximation** of weak coupling. Collecting all system operators and all environment operators, and inserting the interaction Hamiltonian Eq. (2.6) explicitly, the following expression is obtained. Operators at time $t - s$ are tracked with index i' and at time t with i

$$\begin{aligned} \frac{d}{dt}\hat{\rho}_{S,I}(t) = \alpha^2 \sum_{i,i'} \int_0^t ds \Big\{ & \mathrm{Tr}_E\big[\hat{S}_{i,I}(t)\hat{S}_{i',I}(t-s)\hat{\rho}_{S,I}(t) \otimes \hat{E}_{i,I}(t)\hat{E}_{i',I}(t-s)\hat{\rho}_{E,I}(t)\big] - \\ & \mathrm{Tr}_E\big[\hat{S}_{i,I}(t)\hat{\rho}_{S,I}(t)\hat{S}_{i',I}(t-s) \otimes \hat{E}_{i,I}(t)\hat{\rho}_{E,I}(t)\hat{E}_{i',I}(t-s)\big] - \\ & \mathrm{Tr}_E\big[\hat{S}_{i',I}(t-s)\hat{\rho}_{S,I}(t)\hat{S}_{i,I}(t) \otimes \hat{E}_{i',I}(t-s)\hat{\rho}_{E,I}(t)\hat{E}_{i,I}(t)\big] + \\ & \mathrm{Tr}_E\big[\hat{\rho}_{S,I}(t)\hat{S}_{i',I}(t-s)\hat{S}_{i,I}(t) \otimes \hat{\rho}_{E,I}(t)\hat{E}_{i',I}(t-s)\hat{E}_{i,I}(t)\big] \Big\}. \end{aligned}$$

Since the trace only acts on the environment, the system operators can be taken out of the trace, and the two-point correlation functions are defined using Eq. (2.25)

$$C_{ii'}(t,s) \equiv \langle \hat{E}_{i,I}(t)\hat{E}_{i',I}(t-s)\rangle = \mathrm{Tr}_E\big[\hat{E}_{i,I}(t)\hat{E}_{i',I}(t-s)\,\hat{\rho}_{E,I}(t)\big]. \tag{2.13}$$

The term 'two-point' indicates that the bath operators $\hat{E}_i$ and $\hat{E}_{i'}$ are evaluated at two different times. Up to this point, the derivation has been general and no assumption of stationarity was required. The correlator can depend on both times through $\hat{\rho}_{E,I}(t)$. Now the bath is taken to be stationary, i.e. $[\hat{H}_E, \hat{\rho}_E(0)] = 0$. This implies $\hat{\rho}_{E,I}(t) = \hat{\rho}_E(0)$ and therefore

$$C_{ii'}(t,s) = C_{ii'}(t-s).$$

To make this dependence explicit we introduce the time difference

$$\tau \equiv t - s,$$

and subsequently write $C_{ii'}(\tau)$ in place of $C_{ii'}(t,s)$.

Units. With $\hat{H}_{\mathrm{int}} = \sum_i \hat{S}_i \otimes \hat{E}_i$ and $\hbar = 1$, $\hat{H}_{\mathrm{int}}$ has units of time^{-1}, hence $[\hat{S}_i][\hat{E}_i] = \text{time}^{-1}$. If $\hat{S}_i$ is chosen dimensionless, then $[\hat{E}_i] = \text{time}^{-1}$ and therefore $[C_{ii'}] = \text{time}^{-2}$.

If the bath operators are Hermitian, i.e. $\hat{E}_i^\dagger = \hat{E}_i$, and the bath state is stationary (i.e., $[\hat{H}_E, \hat{\rho}_E(0)] = 0$), the cyclic property of the trace gives the conjugation relation

$$C^*_{ii'}(\tau) = \tilde{C}_{ii'}(\tau) \equiv \langle \hat{E}_{i',I}(t-\tau)\hat{E}_{i,I}(t)\rangle,$$

In the interaction picture, stationarity implies $\hat{\rho}_{E,I}(t) = \hat{\rho}_{E,I}(0) = \hat{\rho}_E(0)$ and the correlators depend only on the time difference $\tau = t - s$. With $\hat{E}_{i,I}(t) = e^{i\hat{H}_E t}\hat{E}_i e^{-i\hat{H}_E t}$ this implies the further symmetry condition

$$C^*_{ii'}(-\tau) = C_{ii'}(\tau).$$

and thus finally obtain the desired form of the Redfield equation

$$\frac{d}{dt}\hat{\rho}_{S,I}(t) = \alpha^2 \sum_{i,i'} \int_0^t ds \Big\{ C_{ii'}(t-s)\big[\hat{S}_{i,I}(t), \hat{S}_{i',I}(t-s)\hat{\rho}_{S,I}(t)\big] + \text{H.c.} \Big\}. \tag{2.14}$$

This form shows explicitly that the system evolution depends only on bath correlation functions evaluated along the free bath dynamics.

2.1.5 Markov Approximation

However Eq. (2.14) is still not Markovian, since it carries an explicit integration over time t. An additional approximation is therefore introduced. Besides weak coupling, assume that the bath correlation functions decay on a characteristic time scale τ_E that is much shorter than the relevant system evolution time τ_S, i.e. $\tau_E \ll \tau_S$. In particular, $C_{ii'}(\tau)$ becomes negligible for $\tau \gtrsim \tau_E$, so the integrand has support only on a short memory window. If the reduced state varies slowly on this window, one may approximate it as constant over the integration range and extend the upper limit to infinity, which yields a Markovian generator for $t \gg \tau_E$.

$$\int_0^t ds\, C_{ii'}(t-s) f(s) \;\longrightarrow\; \int_0^\infty d\tau\, C_{ii'}(\tau) f(t), \qquad (t \gg \tau_E),$$

where the slow variation of $\hat{\rho}_{S,I}(t)$ and of the system operators on the time scale τ_E has been used. Applying this rule to Eq. (2.12) yields a *time-local* Markovian Redfield generator. In the interaction picture the operators still carry explicit t-dependence, but the kernel is time-translation invariant and depends only on τ, so the generator has no explicit dependence on absolute time beyond the system operators themselves

$$\frac{d}{dt}\hat{\rho}_{S,I}(t) = -\alpha^2 \sum_{i,i'} \int_0^\infty d\tau \left(C_{ii'}(\tau)[\hat{S}_{i,I}(t), \hat{S}_{i',I}(t-\tau)\hat{\rho}_{S,I}(t)] + \text{H.c.} \right). \tag{2.15}$$

Here $\hat{S}_{i,I}(t-\tau)$ denotes the free interaction-picture evolution under $\hat{H}_S$, so the explicit $(t-\tau)$ dependence reflects free system dynamics rather than any memory in the kernel. This step encapsulates the loss of memory: the generator now includes a very short time window τ_E where the system operators can be approximated as constant (frozen at time $\hat{\rho}_{S,I}(t)$). Equation Eq. (2.15) is the Redfield master equation in the interaction picture under the Born (weak-coupling) and Markov (short-memory) approximations. In this form, all environmental effects are encoded in the bath correlation functions $C_{ii'}(\tau)$, which are analyzed in the next section.

2.2 Environmental Correlators and Spectral Properties

In what follows, the origin and key properties of bath correlation functions (defined by Eq. (2.13)) are outlined, their spectral Fourier representation is introduced, and the simultaneous encoding of emission and absorption processes is discussed. In particular, the decay of $C_{ii'}(\tau)$ on the bath correlation time τ_E quantifies how rapidly the environment "forgets", and the Markov approximation used above is justified when τ_E is much shorter than the relevant system evolution time scales. The connection between $C_{ii'}(\tau)$ and the spectral density is direct and experimentally accessible. Modern noise spectroscopy techniques enable direct measurement of bath spectral properties from coherence-decay observations without requiring pulse sequences or prior knowledge of the system-bath coupling constants—providing parameter-free experimental access to the bath structure that drives decoherence in the e.g. molecular systems [23].

2.2.1 Fourier Representation and Secular Approximation

For a stationary thermal bath in equilibrium, where its state is given by the Gibbs state (see Sec. 2.3.2), the Kubo–Martin–Schwinger (KMS) condition relates correlations at positive and negative times via a shift into the complex time plane,

$$C_{ii'}(\tau) \equiv \langle \hat{E}_{i,I}(\tau)\hat{E}_{i',I}(0)\rangle = \langle \hat{E}_{i,I}(0)\hat{E}_{i',I}(\tau + i\beta)\rangle \equiv C_{i'i}(+\tau + i\beta). \tag{2.16}$$

A short proof of this relation in the eigenbasis of the bath Hamiltonian is as follows

$$\begin{aligned}
C_{ii'}(\tau + i\beta) &= \mathrm{Tr}(\hat{\rho}_\beta\, \hat{E}_i\, \hat{E}_{i',I}(\tau + i\beta)) \\
&= \frac{1}{Z}\sum_{m,n} e^{-\beta E_n} \langle n|\, \hat{E}_i\, |m\rangle\, \langle m|\, \hat{E}_{i',I}(\tau + i\beta)\, |n\rangle \\
&= \frac{1}{Z}\sum_{m,n} e^{-\beta E_n} e^{i(E_m - E_n)(\tau + i\beta)} (\hat{E}_i)_{nm} (\hat{E}_{i'})_{mn} \\
&= \frac{1}{Z}\sum_{m,n} e^{-\beta E_m} e^{i(E_n - E_m)\tau} (\hat{E}_{i'})_{nm} (\hat{E}_i)_{mn} \\
&= C_{i'i}(\tau).
\end{aligned}$$

Eigenoperator Decomposition

To proceed analytically and to connect with relaxation pathways, the system coupling operators are decomposed into eigenoperators of the system Hamiltonian. Let $\hat{H}_S = \sum_\epsilon \epsilon\, \hat{\Pi}_\epsilon$ be the spectral resolution with projectors $\hat{\Pi}_\epsilon$. The Bohr frequencies are defined as $\omega = \epsilon' - \epsilon$ and the corresponding interaction operators in the eigenbasis are

$$\hat{S}_i(\omega) = \sum_{\epsilon'-\epsilon=\omega} \hat{\Pi}_\epsilon \, \hat{S}_i \, \hat{\Pi}_{\epsilon'}.$$

In the interaction picture these acquire simple oscillatory phases

$$\hat{S}_{i,I}(t) = \sum_\omega e^{-i\omega t}\hat{S}_i(\omega), \qquad \hat{S}_{i',I}(t-\tau) = \sum_{\omega'} e^{-i\omega'(t-\tau)}\hat{S}_{i'}(\omega'). \tag{2.17}$$

Note that a time-independent system Hamiltonian $\hat{H}_S$ is assumed here. The derivation can in principle be extended to time-dependent $\hat{H}_S(t)$, for example in spectroscopy where external driving fields lead to an explicitly time-dependent system Hamiltonian [3] (see Appendix C). Generalizations of Markovian master-equation derivations to driven systems are discussed in [24].

Inserting Eq. (2.17) into Eq. (2.15) produces sums over oscillatory factors $e^{-i(\omega-\omega')t}$ multiplying integrals of $C_{ii'}(\tau)e^{i\omega'\tau}$. The inner commutator in Eq. (2.15) becomes

$$[\hat{S}_{i,I}(t),\, \hat{S}_{i',I}(t-\tau)\, \hat{\rho}_{S,I}(t)] = \sum_{\omega,\omega'} e^{-i(\omega-\omega')t} e^{+i\omega'\tau} \big[\hat{S}_i(\omega),\, \hat{S}_{i'}(\omega')\hat{\rho}_{S,I}(t)\big],$$

and the total Redfield equation reads

$$\frac{d}{dt}\hat{\rho}_{S,I}(t) = -\alpha^2 \sum_{i,i'}\sum_{\omega,\omega'} e^{-i(\omega-\omega')t}\left(\int_0^\infty d\tau\, C_{ii'}(\tau)\, e^{+i\omega'\tau}\right)\big[\hat{S}_i(\omega),\, \hat{S}_{i'}(\omega')\hat{\rho}_{S,I}(t)\big] + \text{H.c.}.$$

The correlator is now expressed in the frequency domain and the noise-power spectrum of the environment is defined [25]. To avoid confusion with the system coupling operators $\hat{S}_i$, we denote the bath spectrum by $\mathcal{S}(\omega)$ (common in the literature as $S(\omega)$).

$$\mathcal{S}(\omega) \equiv \int_{-\infty}^{\infty} d\tau\, C(\tau)\, e^{+i\omega\tau}, \tag{2.18}$$

Units. Since $[C] = \text{time}^{-2}$ for dimensionless $\hat{S}_i$ and $\hbar = 1$, the Fourier transform implies $[\mathcal{S}(\omega)] = [C]\,\text{time} = \text{time}^{-1}$.

This quantity tells how strongly bath fluctuations at frequency ω contribute to noise. For a stationary bath correlation function $C(\tau)$, the noise spectrum $\mathcal{S}(\omega)$ is by the Wiener–Khinchin relation the Fourier transform of $C(\tau)$ and real and non-negative $\mathcal{S}(\omega) \geq 0$ [26].

The KMS condition Eq. (2.16) can now be translated to frequency domain, where it imposes the detailed-balance symmetry

$$\mathcal{S}(-\omega) = e^{-\omega\beta}\mathcal{S}(\omega), \tag{2.19}$$

which ensures that upward and downward transition rates obey Boltzmann ratios.

The exact one-sided Fourier (Laplace-) transform of the bath correlation function can be split into the power spectrum and an energy shift $\lambda(\omega)$ as

$$\begin{aligned}\chi_{ii'}(\omega) &\equiv \int_0^\infty d\tau\, C_{ii'}(\tau)\, e^{+i\omega\tau} \\ &= \tfrac{1}{2}\mathcal{S}_{ii'}(\omega) + i\,\lambda_{ii'}(\omega),\end{aligned}$$

with

$$\mathcal{S}_{ii'}(\omega) = \chi_{ii'}(\omega) + \chi^*_{i'i}(\omega).$$

which reduce Sec. 2.2.1 to the compact form

$$\frac{d}{dt}\hat{\rho}_{S,I}(t) = -\alpha^2 \sum_{i,i'} \sum_{\omega,\omega'} e^{-i(\omega-\omega')t} \chi_{ii'}(\omega') \left[\hat{S}_i(\omega),\, \hat{S}_{i'}(\omega')\hat{\rho}_{S,I}(t)\right] + \text{H.c.}.$$

Going back to the Schrödinger picture, this yields

$$\frac{d}{dt}\hat{\rho}_S(t) = -i[\hat{H}_S + \hat{H}_{\text{LS}}, \hat{\rho}_S(t)] + \mathcal{D}_{\text{R}}[\hat{\rho}_S(t)],$$

with the Lamb-shift Hamiltonian

$$\hat{H}_{\text{LS}} = \sum_{\omega} \sum_{i,i'} \lambda_{ii'}(\omega)\, \hat{S}_i^\dagger(\omega)\hat{S}_{i'}(\omega),$$

which will be absorbed into a renormalized system Hamiltonian and disregarded in the subsequent derivation. The Redfield dissipator is

$$\mathcal{D}_{\text{R}}[\hat{\rho}] = \frac{1}{2} \sum_{\omega,\omega'} \sum_{i,i'} e^{-i(\omega-\omega')t} \mathcal{S}_{ii'}(\omega') \left(\hat{S}_{i'}(\omega')\, \hat{\rho}\, \hat{S}_i^\dagger(\omega) - \hat{S}_i^\dagger(\omega)\hat{S}_{i'}(\omega')\, \hat{\rho}\right) + \text{H.c.}. \tag{2.20}$$

Matrix Elements and Redfield Tensor

In the eigenbasis of the Hamiltonian $\{|a\rangle\}$, taking $\langle a| \cdot |b\rangle$ of the master equation and grouping terms gives

$$\frac{d}{dt}\rho_{ab}(t) = -i\,\omega_{ab}\rho_{ab}(t) + \sum_{c,d} R_{abcd}\,\rho_{cd}(t),$$

where the Bloch–Redfield tensor, also implemented in QuTiP, is defined as

$$\begin{aligned} R_{abcd} = -\frac{1}{2} \sum_{i,i'} \Big\{ &\delta_{bd} \sum_n S^i_{an} S^{i'}_{nc}\, \mathcal{S}_{ii'}(\omega_{cn}) - S^{i'}_{ac} S^i_{db}\, \mathcal{S}_{ii'}(\omega_{ca}) \\ &+ \delta_{ac} \sum_n S^i_{dn} S^{i'}_{nb}\, \mathcal{S}_{ii'}(\omega_{dn}) - S^{i'}_{ac} S^i_{db}\, \mathcal{S}_{ii'}(\omega_{db}) \Big\}. \end{aligned} \tag{2.21}$$

with S^i the system coupling operators and $\mathcal{S}_{ii'}(\omega)$ the bath spectra[25].

Secular Approximation

The oscillatory prefactors $e^{-i(\omega-\omega')t}$ in Eq. (2.20) average to zero on coarse-grained times Δt satisfying $\tau_E \ll \Delta t \ll 1/|\omega - \omega'|$ whenever $\omega \neq \omega'$. Keeping only terms with $\omega_{ab} = \omega_{cd}$ in Eq. (2.21) removes such fast-rotating couplings and ensures a block-diagonal structure in the Redfield tensor R_{abcd}.

This yields the *secular* Redfield equation in Lindblad form

$$\begin{aligned} \frac{d}{dt}\hat{\rho}_S(t) &= -i[\hat{H}_S + \hat{H}_{\text{LS}}, \hat{\rho}_S(t)] \\ &\quad + \sum_{i,i'} \sum_{\omega} \gamma_{ii'}(\omega) \Big(\hat{S}_{i'}(\omega)\hat{\rho}_S(t)\hat{S}_i^\dagger(\omega) - \tfrac{1}{2}\{\hat{S}_i^\dagger(\omega)\hat{S}_{i'}(\omega), \hat{\rho}_S(t)\}\Big), \end{aligned}$$

where $\gamma_{ii'}(\omega) = 2\text{Re}\,\chi_{ii'}(\omega)$. This form is guaranteed to preserve complete positivity provided the matrix $[\gamma_{ii'}(\omega)]_{i,i'}$ is positive semidefinite for each frequency ω. Without the secular approximation, Eq. (2.20) need not generate a completely positive dynamical map, explaining the caution required when applying the full Redfield equation.

For a thermal bath in equilibrium, the bath correlation functions satisfy the KMS condition. In the weak-coupling limit with a full secular approximation, the resulting Lindblad generator has the Gibbs state as a stationary state (cf. Equation (2.24)) [27, 28]

$$\mathcal{L}_{\text{sec}}[\hat{\rho}_{\text{Gibbs}}] = 0.$$

By contrast, the non-secular (Bloch-)Redfield generator retains terms with $\omega \neq \omega'$, which couple populations and coherences in the energy basis. Without full secularization, quantum detailed balance and Gibbs stationarity are not guaranteed, even for a single thermal bath satisfying KMS [1, 29]. In weak coupling, the stationary state typically differs from $\hat{\rho}_{\text{Gibbs}}$ by corrections of second order in the system–bath coupling (and may exhibit bath-induced population shifts and coherences).

An important exception is pure dephasing, where the coupling operator commutes with the system Hamiltonian, $[\hat{A}, \hat{H}_S] = 0$. In that case there is no energy exchange and $\hat{\rho}_\beta$ remains stationary also without secularization [1].

Uncorrelated Hermitian Couplings

If the baths are uncorrelated and $S_i = S_i^\dagger$:

$$\begin{aligned}
&\mathcal{S}_{ii'}(\omega) = \delta_{ii'}\mathcal{S}_i(\omega) \implies \\
&R_{abcd} = -\frac{1}{2}\sum_i \Big\{ \delta_{bd}\sum_n S^i_{an}S^i_{nc}\mathcal{S}_i(\omega_{cn}) - S^i_{ac}S^i_{db}\mathcal{S}_i(\omega_{ca}) \\
&\qquad + \delta_{ac}\sum_n S^i_{dn}S^i_{nb}\mathcal{S}_i(\omega_{dn}) - S^i_{ac}S^i_{db}\mathcal{S}_i(\omega_{db}) \Big\}.
\end{aligned}$$

2.2.2 Physical Interpretation: Emission and Absorption

Emission and absorption are both automatically included because $\mathcal{S}_{ii'}(\omega)$ contains positive- and negative-frequency components related by Eq. (2.19). Positive frequencies ($\omega > 0$) describe system energy loss (emission), while negative frequencies ($\omega < 0$) describe system energy gain (absorption). Detailed balance follows immediately: the ratio of absorption to emission contributions at frequency $\omega > 0$ is $e^{-\omega\beta}$. In the high-temperature limit ($\beta \to 0$), this simplifies to $\mathcal{S}(-\omega) \approx \mathcal{S}(\omega)$, making the power spectrum symmetric around $\omega = 0$ and rendering emission and absorption rates nearly equal.

The explicit relation between $J(\omega)$, the correlator, and the full spectrum is derived in the next section (see Eq. (2.33)).

2.3 Harmonic Oscillator Baths and Explicit Correlation Functions

The previous section established that the reduced dynamics is governed by the bath correlation functions $C_{ij}(\tau)$. To make the Redfield equation practical, these correlators must be evaluated for a concrete microscopic model. Here the focus is on the harmonic-oscillator bath, a widely used model that captures electromagnetic field fluctuations, phonons, and other bosonic environments.

The task now is to calculate the bath correlation function $C(\tau)$ for a concrete model. Recall that $C(\tau) = \langle E(t+\tau)E(t)\rangle$ measures how bath fluctuations at two different times are correlated in thermal equilibrium. For the harmonic oscillator bath, these correlators can be computed exactly, revealing how $C(\tau)$ connects to experimentally relevant spectral densities and governs the system's approach to thermal equilibrium.

For these explicit calculations, the following mathematical result is useful.

2.3.1 Geometric Series

The infinite geometric series

$$\mathcal{G} = a + ar + ar^2 + ar^3 + \cdots = \sum_{n=0}^{\infty} ar^n,$$

converges to

$$\mathcal{G} = \frac{a}{1-r} \quad \text{for } |r| < 1. \tag{2.22}$$

Differentiating Eq. (2.22) with respect to r gives

$$\sum_{n=0}^{\infty} nr^n = \frac{r}{(1-r)^2}, \quad |r| < 1. \tag{2.23}$$

The formal correlation functions introduced above are now specialized to the paradigmatic bosonic bath. This leads naturally to spectral-density representations and the standard phenomenological models such as Ohmic, sub-/super-Ohmic, and Drude–Lorentz.

2.3.2 Bosonic Environment and Gibbs State

A bosonic bath is modeled as a (large) collection of independent harmonic oscillators in thermal equilibrium:

The thermal state of the bosonic bath is given by the Gibbs density matrix

$$\hat{\rho}_{\text{Gibbs}} = \frac{e^{-\beta \hat{H}_E}}{\mathrm{Tr}[e^{-\beta \hat{H}_E}]}, \tag{2.24}$$

so that for any bath operator $\hat{A}$,

$$\langle \hat{A} \rangle = \mathrm{Tr}(\hat{\rho}_{\text{Gibbs}} \hat{A}) = \frac{1}{Z} \sum_n e^{-\beta E_n} A_{nn}, \tag{2.25}$$

in the energy eigenbasis of $\hat{H}_E$, where $Z = \mathrm{Tr}[e^{-\beta \hat{H}_E}]$ and $\beta = 1/(k_B T)$ is the inverse temperature of the environment. The bath Hamiltonian consists of independent harmonic oscillators

$$\hat{H}_E = \sum_k \omega_k \left(\hat{b}_k^\dagger \hat{b}_k + \tfrac{1}{2} \right), \tag{2.26}$$

and the average energy of mode k is

$$E_k = \omega_k (n_k + \tfrac{1}{2}),$$

with $n_k = \langle \hat{b}_k^\dagger \hat{b}_k \rangle$ the Bose–Einstein occupation number.

The zero-point contribution $E_{\text{vac}} = \frac{1}{2} \sum_k \omega_k$ adds only a mode-independent constant to the bath energy. Constant energy offsets commute with all observables, so they neither modify thermal populations nor enter the correlation functions that drive the system dynamics. This term is therefore dropped in the dynamical equations (standard for normal ordering, where raising operators are placed to the left of lowering operators). For completeness, the $\frac{1}{2}$ is retained in the partition function Z_k for normalization, but it cancels out of all correlation functions and rates. In other words, only energy *differences* influence the evolution.

Single Mode Partition Function and Occupation

For a single mode k, the thermal state reads

$$\hat{\rho} = \frac{e^{-\beta\omega_k \hat{b}_k^\dagger \hat{b}_k}}{Z_k},$$

with partition function

$$\begin{aligned} Z_k &\equiv \mathrm{Tr}\left[e^{-\beta \hat{H}_E}\right] = \sum_{m=0}^{\infty} \langle m| \, e^{-\beta\omega_k(\hat{b}_k^\dagger \hat{b}_k + \frac{1}{2})} \, |m\rangle \\ &= \frac{e^{-\beta\omega_k/2}}{1 - e^{-\beta\omega_k}}. \end{aligned}$$

The Bose–Einstein average occupation number follows using Eq. (2.23) and is given by

$$\begin{aligned} n_k = \langle \hat{b}_k^\dagger \hat{b}_k \rangle_{\mathrm{th}} &= \mathrm{Tr}\left[\hat{b}_k^\dagger \hat{b}_k \frac{e^{-\beta \hat{H}_E}}{Z_k}\right] \\ &= \frac{\mathrm{Tr}\left[\hat{b}_k^\dagger \hat{b}_k e^{-\beta\omega_k \hat{b}_k^\dagger \hat{b}_k}\right]}{Z_k} \\ &= \frac{\sum_{m=0}^{\infty} \langle m| \, \hat{b}_k^\dagger \hat{b}_k e^{-\beta\omega_k \hat{b}_k^\dagger \hat{b}_k} \, |m\rangle}{\frac{e^{-\beta\omega_k/2}}{1-e^{-\beta\omega_k}}} \\ &= \frac{e^{-\beta\omega_k}}{1 - e^{-\beta\omega_k}} \\ &= \frac{1}{e^{\beta\omega_k} - 1}. \end{aligned} \tag{2.27}$$

Since the bath is composed of many independent modes, this calculation can be done for every mode, and the total partition function factorizes to

$$Z_{\mathrm{bath}} = \prod_k Z_k = \prod_k \frac{e^{-\beta\omega_k/2}}{1 - e^{-\beta\omega_k}}.$$

With the thermal properties of the bath established, attention can be turned to its coupling to a small system of interest. The bath correlation functions from Eq. (2.13) are therefore calculated for the simple case of a single coupling operator $\hat{E}$. The derivation follows and slightly expands [30].

2.3.3 Microscopic Form of the Bath Correlator

With linear system–bath coupling to oscillator displacements, the Caldeira–Leggett model [1, 31] gives the single bath coupling operator as

$$\hat{E} = \sum_{n=1}^{\infty} c_n \hat{x}_n, \qquad \hat{x}_n = \sqrt{\frac{1}{2m_n\omega_n}}(\hat{b}_n + \hat{b}_n^\dagger).$$

In the interaction picture the bath operators evolve as $\hat{E}_I(t) = e^{i\hat{H}_E t}\hat{E}e^{-i\hat{H}_E t}$ with $\hat{H}_E$ from Eq. (2.26), yielding

$$\hat{E}_I(0) = \sum_n c_n \sqrt{\frac{1}{2m_n\omega_n}}(\hat{b}_n + \hat{b}_n^\dagger), \tag{2.28}$$

$$\hat{E}_I(\tau) = \sum_n c_n \sqrt{\frac{1}{2m_n\omega_n}}\left(\hat{b}_n e^{-i\omega_n\tau} + \hat{b}_n^\dagger e^{i\omega_n\tau}\right). \tag{2.29}$$

The thermal correlation function for this single relevant bath operator is

$$C(\tau) = \langle \hat{E}_I(\tau)\hat{E}_I(0)\rangle.$$

Inserting Eqs. (2.28) and (2.29) yields a double sum over bath modes,

$$\begin{aligned} C(\tau) &= \sum_{n,m} \frac{c_n c_m}{\sqrt{4m_n m_m \omega_n \omega_m}} \Big\langle \big(\hat{b}_n e^{-i\omega_n\tau} + \hat{b}_n^\dagger e^{i\omega_n\tau}\big)\big(\hat{b}_m + \hat{b}_m^\dagger\big)\Big\rangle, \\ &= \sum_{n,m} \frac{c_n c_m}{\sqrt{4m_n m_m \omega_n \omega_m}} \Big(e^{-i\omega_n\tau}\langle \hat{b}_n \hat{b}_m^\dagger\rangle + e^{i\omega_n\tau}\langle \hat{b}_n^\dagger \hat{b}_m\rangle \Big), \end{aligned}$$

where mixed terms such as $\langle \hat{b}_n \hat{b}_m\rangle$ vanish for a thermal state of uncoupled oscillators. Doing similar calculations to Eq. (2.27) gives

$$\langle \hat{b}_n \hat{b}_m^\dagger\rangle = \delta_{nm}(n_n + 1) \quad \text{and} \quad \langle \hat{b}_n^\dagger \hat{b}_m\rangle = \delta_{nm} n_n,$$

with $n_n = \langle \hat{b}_n^\dagger \hat{b}_n\rangle = 1/(e^{\beta\omega_n} - 1)$ given by Eq. (2.27). Using these results, the discrete form of the bath correlation function is obtained as

$$C(\tau) = \sum_n \frac{c_n^2}{2m_n\omega_n}\Big[(n_n+1)e^{-i\omega_n\tau} + n_n e^{i\omega_n\tau}\Big].$$

2.3.4 Spectral Density Representation

To connect this discrete expression with the continuum form, note that the correlator can be expanded by writing $e^{\pm i\omega_n\tau} = \cos(\omega_n\tau) \pm i\sin(\omega_n\tau)$. This yields

$$\begin{aligned} C(\tau) &= \sum_n \frac{c_n^2}{2m_n\omega_n}\Big[(n_n+1)(\cos(\omega_n\tau) - i\sin(\omega_n\tau)) + n_n(\cos(\omega_n\tau) + i\sin(\omega_n\tau))\Big] \\ &= \sum_n \frac{c_n^2}{2m_n\omega_n}\Big[(2n_n+1)\cos(\omega_n\tau) - i\sin(\omega_n\tau)\Big]. \end{aligned}$$

Since $2n_n + 1 = 2/(e^{\beta\omega_n} - 1) + 1 = \coth(\beta\omega_n/2)$ (recall $\coth(x/2) = (e^x+1)/(e^x-1)$), this can be rewritten as

$$C(\tau) = \sum_n \frac{c_n^2}{2m_n\omega_n}\Big[\coth\Big(\frac{\beta\omega_n}{2}\Big)\cos(\omega_n\tau) - i\sin(\omega_n\tau)\Big]. \tag{2.30}$$

To move from discrete modes to a continuum description, introduce the spectral density, which encodes both the mode density and coupling strength,

$$J(\omega) = \pi\sum_n \frac{c_n^2}{2m_n\omega_n}\delta(\omega - \omega_n),$$

which summarizes how strongly each bath frequency couples to the system. Here it is assumed that the bath has discrete modes ω_n and each couples to the system with strength $\propto c_n$. By construction, the delta-function representation implies the distributional identity that for any smooth test function $f(\omega)$,

$$\sum_n \frac{c_n^2}{2m_n\omega_n} f(\omega_n) = \frac{1}{\pi}\int_0^\infty d\omega\, J(\omega) f(\omega). \tag{2.31}$$

This simply means that the discrete mode sum can be replaced by a continuum integral once $J(\omega)$ is defined. We then choose the specific function $f(\omega) = \coth(\beta\omega/2)\cos(\omega\tau) - i\sin(\omega\tau)$ because it is exactly the factor appearing in the discrete correlator. Applying the identity with this f yields a representation of the bath correlator in a continuum limit

$$C(\tau) = \int_0^\infty d\omega \, \frac{J(\omega)}{\pi} \Big[\coth\Big(\frac{\beta\omega}{2}\Big) \cos(\omega\tau) - i \sin(\omega\tau) \Big]. \tag{2.32}$$

The cosine term is even in τ and captures symmetrized noise, while the sine term is odd and encodes dissipation. In the macroscopic limit, the mode set becomes dense, and the spectral density $J(\omega)$ can be expressed as $J(\omega) = \rho(\omega) g(\omega)^2$, where $\rho(\omega)$ is the density of states and $g(\omega)$ the form factor. The spectral density thus fully characterizes the environmental coupling: it determines which bath frequencies exist and how strongly each frequency couples to the system. The real part of $C(\tau)$ describes symmetrized noise, while the imaginary part describes dissipation [1, 2, 32].

The parametric choice of $J(\omega)$, i.e., the bath spectral density, fixes whether noise is dominated by low-frequency components and whether the bath is smooth or has sharp features [1, 2]. This will be illustrated visually in Sec. 2.4, see in particular Fig. 2.2 and Fig. 2.4.

The bath correlation function $C(\tau)$ is the key quantity determining the system dynamics in the Redfield equation. It is fully specified by the spectral density $J(\omega)$ and the temperature-dependent factor $\coth(\beta\omega/2)$, which weights each frequency according to thermal occupation. The spectral density is also directly related to the noise power spectrum (cf. Equation (2.18)):

$$\mathcal{S}(\omega) = \int_{-\infty}^{\infty} d\tau \, C(\tau) e^{+i\omega\tau} = 2\pi J(|\omega|) \begin{cases} n_{\text{th}}(\omega) + 1, & \omega > 0, \\ n_{\text{th}}(|\omega|), & \omega < 0, \end{cases} \tag{2.33}$$

Here $J(\omega)$ is defined for $\omega > 0$, and the thermal occupation is $n_{\text{th}}(\omega) = 1/(e^{\beta\omega} - 1)$. The piecewise form makes explicit that $\omega > 0$ corresponds to emission and $\omega < 0$ to absorption, with the detailed-balance ratio fixed by $e^{-\beta\omega}$. Thus, specifying either the correlator $C(\tau)$, the noise power spectrum $\mathcal{S}(\omega)$, or the spectral density $J(\omega)$ together with the temperature β suffices to fully characterize the environmental coupling. The spectral density is often the most convenient choice, as it is a real-valued function of a single variable and can be directly related to microscopic models.

In the following, some of the most common choices for $J(\omega)$ are introduced.

2.4 Common choices of Spectral Densities

The main qualitative differences between bath models are set by the low-frequency scaling of $J(\omega)$ and by how high-frequency modes are regularized (cutoff). Both directly impact correlation times and the Markov regime.

2.4.1 Ohmic and Related Spectral Densities

The Redfield equation can also be connected to classical phenomenological models of dissipation. For an Ohmic spectral density, the low-frequency behavior is linear, and the spectral density is given by [2]

$$J(\omega) \propto \gamma\omega,$$

where γ is the damping constant. This corresponds to a classical friction-like dissipation. To ensure physical behavior, a cutoff is introduced

$$J(\omega) = \alpha \frac{\omega^s}{\omega_c^{s-1}} e^{-\omega/\omega_c}, \tag{2.34}$$

which enforces convergence of integrals such as Eq. (2.33). Here α is a dimensionless coupling constant, ω_c is the cutoff frequency, and s controls the low-frequency behavior of the bath, thereby effecting the timescale of the environment. The parameter s classifies the spectral density into three regimes: Ohmic ($s = 1$), *sub-Ohmic* ($s < 1$), and *super-Ohmic* ($s > 1$) [2, 25]. Sub-Ohmic baths exhibit slower fluctuations and longer memory, while super-Ohmic baths suppress low-frequency noise and emphasize fast, high-frequency modes. To make this separation explicit in the numerical illustrations below, it is convenient to measure all frequencies in units of a characteristic system scale ω_0 and to express bath parameters as dimensionless ratios (ω_c/ω_0, T/ω_0, α/ω_0). As shown in Fig. 2.2, the sub-Ohmic correlation function remains nonzero for longer times than the Ohmic and super-Ohmic cases.

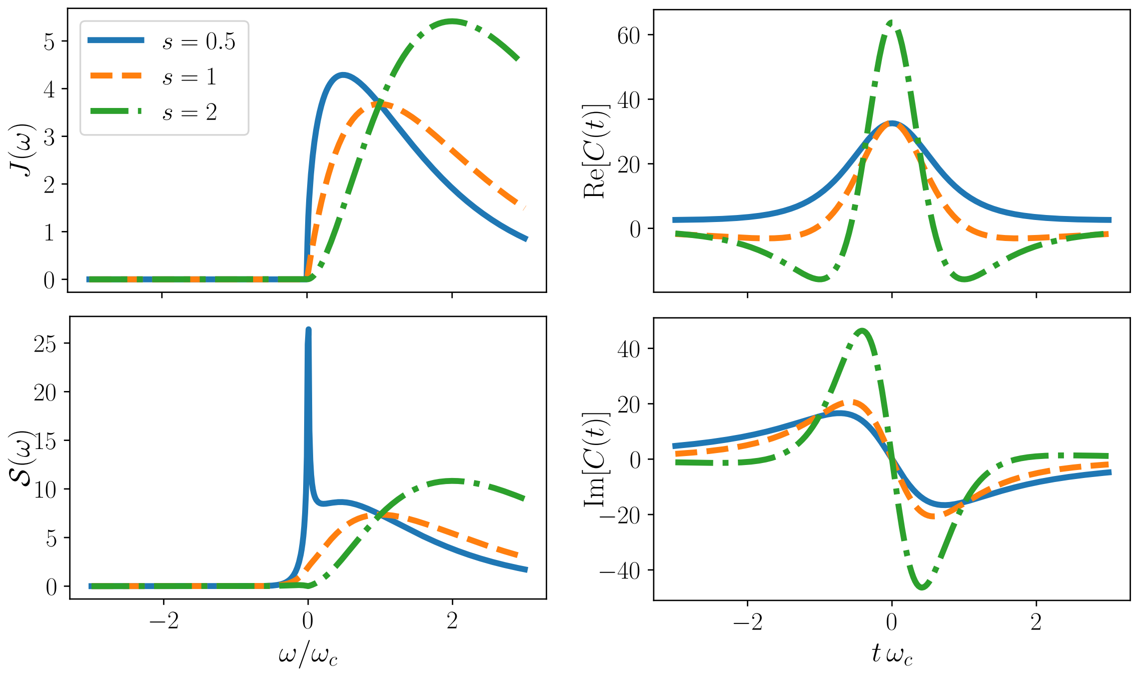

Figure 2.2: Comparison of sub-Ohmic ($s = 0.5$), Ohmic ($s = 1$), and super-Ohmic ($s = 2$) spectral densities, power spectra, and correlation functions. Parameters: $\omega_c/\omega_0 = 10$, $T/\omega_0 = 1$, $\alpha/\omega_0 = 1$.

For an Ohmic bath, increasing ω_c pushes bath weight to shorter times and reduces the bath memory time. This is precisely the physical content behind the Markov approximation and is illustrated in Fig. 2.3.

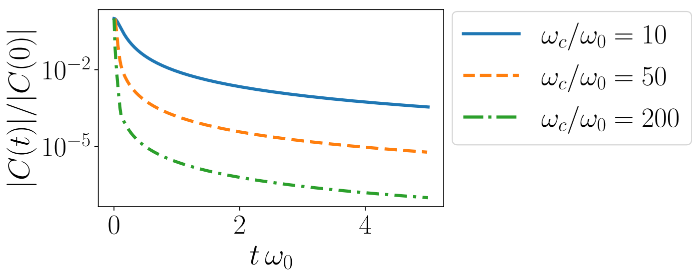

Figure 2.3: Normalized bath correlation magnitude $|C(t)|/|C(0)|$ for an Ohmic bath ($s = 1$), showing faster decay for larger cutoff ω_c/ω_0. Parameters: $T/\omega_0 = 10$, $\alpha/\omega_0 = 10^{-3}$.

Another widely used model with linear low-frequency behavior is the Drude–Lorentz spectral density, which introduces a Lorentzian cutoff

$$J(\omega) = 2\lambda \frac{\omega\gamma}{\omega^2 + \gamma^2},$$

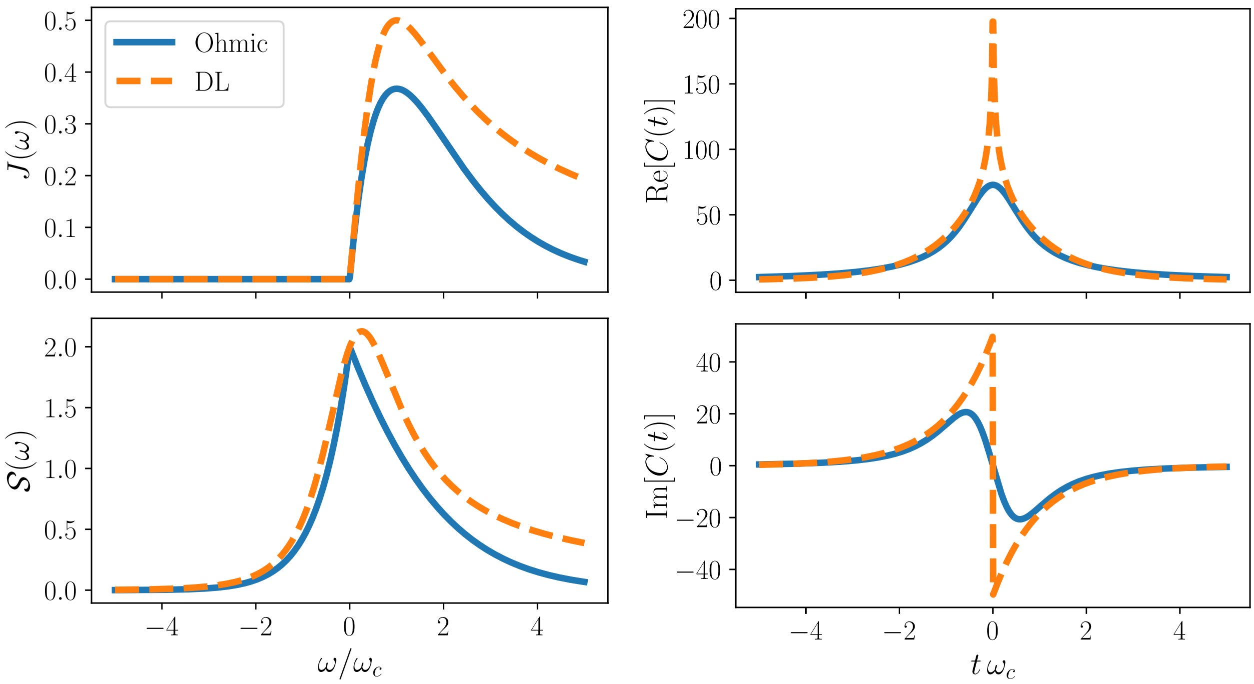

Figure 2.4: Comparison of representative bath models (Ohmic and Drude–Lorentz) showing spectral densities, associated power spectra, and time-domain correlation functions for coupling strength $\alpha = 0.1$, cutoff $\omega_c = 100$, and temperature $T = 100$. To make both models as comparable as possible, the Drude–Lorentz parameters are chosen consistently with the Ohmic-type convention via $\gamma = \omega_c$ and $\lambda = \alpha\,\omega_c/2$.

where λ is the reorganization energy, and γ is interpreted as a cutoff. This model represents an overdamped Brownian oscillator. A visual comparison between the Ohmic and Drude–Lorentz spectral densities, together with their corresponding correlation functions and power spectra, is shown in Fig. 2.4. The Drude–Lorentz model exhibits a longer high-frequency tail than the exponentially cut off Ohmic case.

Interestingly, when the electronic transition frequency substantially exceeds the characteristic bath frequencies, the *high-frequency tails* of $J(\omega)$ become dominant for determining population relaxation kinetics and must be carefully specified, even if spectral densities with different functional forms appear nearly identical in their peak structures [33].

2.4.2 Structured Spectral Densities

Beyond these overdamped models, underdamped Brownian-oscillator spectral densities are often used to represent discrete vibrational structure, or microscopic chain models such as the Rubin model, whose continuum limits recover Ohmic/Drude-type behavior [30, 34]. In practice, structured environments are frequently approximated by a sum of such components, such as Drude plus Brownian terms, which is also a standard starting point for hierarchical-equations-of-motion (HEOM) approaches [35, 36].

A bath is *structured* when $J(\omega)$ exhibits pronounced features near such transitions, such as sharp resonances, band edges, or steep slopes [1]. By contrast to the Ohmic cases, a resonant Lorentzian peak centered at a finite frequency ω_0 may be written as

$$J(\omega) = \lambda\,\frac{\gamma}{(\omega-\omega_0)^2+\gamma^2}, \tag{2.35}$$

where γ controls the linewidth and λ the coupling strength. Such a spectrum concentrates weight near ω_0 and corresponds physically to a damped mode (e.g. a molecular vibration or cavity-like resonance) coupled to a broader background bath. The associated bath correlation function contains oscillatory contributions of the form $e^{-\gamma t}\cos(\omega_0 t)$, reflecting the finite-frequency structure. The distinction between structured and unstructured baths is independent of Markovianity.

In summary, environmental effects enter the Redfield framework through $J(\omega)$, which determines the bath correlation function $C(t) = \langle E(t)E(0)\rangle$ and thus the memory time τ_E (Eqs. (2.32) to (2.33)). Broad, slowly varying spectra typically support a Markovian description, whereas sharply structured features near system transitions can generate oscillatory and long-lived correlations, requiring more careful modeling beyond simple Markovian Redfield theory.

Chapter 3

Nonlinear Spectroscopy and Two-Dimensional Electronic Spectra

3.1 Motivation and Historical Development

Spectroscopy, in the broadest sense, studies how matter interacts with energy—most commonly through absorption, emission, or scattering of radiation [37]. Atoms and molecules occupy discrete energy levels. Therefore, the frequency of light absorbed or emitted during a transition satisfies the Planck relation $\Delta E = h\nu = hc/\lambda$. Early optical spectroscopy exploited this fact. The stationary line spectra of atoms provided the first "fingerprints" of quantum energy levels [38–40]. Quantum mechanics later related these to transitions between eigenstates of the microscopic Hamiltonian [41, 42]. In practice, transitions are labeled by the wavenumber $\tilde{\nu} = 1/\lambda$ [cm^{-1}], which is directly proportional to energy. The electronic transitions studied in this thesis lie in the visible and near-UV range ($\tilde{\nu} \sim 13\,000$–$25\,000\,\mathrm{cm}^{-1}$). In this regime, the relevant coherence and relaxation dynamics unfold on femtosecond timescales.

For decades, spectroscopy was essentially *linear*: a weak field probes the sample and one records intensity as a function of frequency. Linear absorption and emission reveal excitation energies, oscillator strengths, and broadening mechanisms. However, they compress dynamical information—such as coherence loss, population transfer, and spectral diffusion—into one-dimensional lineshapes. As a result, different microscopic mechanisms are difficult to disentangle.

The development of lasers in the 1960s made it possible to generate intense, coherent light fields. This opened the door to *nonlinear* optics. The induced polarization $\boldsymbol{P}(t)$ contains not only components linear in the applied field $\boldsymbol{E}(t)$,

$$\boldsymbol{P}^{(1)}(t) = \varepsilon_0 \int_0^\infty \mathrm{d}\tau\, \chi^{(1)}(\tau)\cdot\boldsymbol{E}(t-\tau),$$

where $\chi^{(1)}$ is, in general, a (second-rank) susceptibility tensor. There are also quadratic, cubic, and higher-order contributions,

$$\boldsymbol{P}^{(n)}(t) \propto \boldsymbol{E}(t)^n, \qquad n \geq 2.$$

These additional terms are responsible for phenomena such as second-harmonic generation, four-wave mixing, and stimulated Raman scattering. Conceptually, they probe *multi-time* correlation functions of the system. Therefore, they encode dynamical information that is inaccessible in the linear-response regime [3, 43].

Two-dimensional Fourier-transform spectroscopy represents a powerful experimental tool in nonlinear spectroscopy. By correlating excitation and detection frequencies, it separates homogeneous and inhomogeneous contributions to spectral line shapes. Inhomogeneous broadening arises from static disorder in the molecular environment (e.g. different binding sites or solvent configurations), which distributes the transition frequencies across the ensemble. Homogeneous broadening, in contrast, reflects the intrinsic linewidth of a single chromophore, determined by coherence loss and population relaxation. Two-dimensional spectra reveal the homogeneous linewidth even in strongly inhomogeneously broadened systems [44, 45]. Additionally, cross peaks directly visualize couplings between transitions and energy-transfer pathways.

To connect these experimental capabilities to the theoretical framework developed in Chapter 2, the response-function formalism is employed. A systematic link is provided from the applied pulse sequence, through the induced polarization in the sample, to the measured signal. The response functions themselves are multi-time correlation functions of the dipole operator, evaluated with respect to the equilibrium density matrix.

This formalism separates the problem into matter and field contributions: response functions encode the system's dynamics, while the fields enter through causal convolutions. This separation makes the perturbative order explicit, naturally incorporates phase-matching conditions, and interfaces with open-quantum-system master equations—as the most important example for this thesis, the Redfield equation (2.15), for which the concrete derivation can be found in Sec. 2.1.

To keep the chapter self-contained, derivations are split into a *field side* (how $\boldsymbol{P}^{(3)}$ radiates the measured signal) and a *matter side* (how the incident fields generate $\boldsymbol{P}^{(n)}$). Appendix B covers the field side and Appendix C the matter side.

The remainder of this chapter is organized as follows: The remainder of this chapter is organized as follows: Section 3.2 introduces the semiclassical light–matter Hamiltonian in the dipole approximation; Sec. 3.3 defines the macroscopic polarization and its role in the signal (field-side details in Appendix B); Sec. 3.4 states the response-function results used later (matter-side derivations in Appendix C); and Sec. 3.5–3.7 applies the formalism to three-pulse experiments, phase matching, phase cycling, and 2D spectrum construction.

3.2 Light–Matter Interaction in the Dipole Approximation

Throughout this chapter, a semiclassical description of light is adopted. The electromagnetic field is treated as a classical, time-dependent driving field, while the matter degrees of freedom are described quantum mechanically. The central object is the density operator of the matter, whose evolution determines the optical response.

The microscopic dynamics of the full (molecule + environment) system are governed by the Liouville–von Neumann equation (cf. Equation (2.5))

$$\frac{d}{dt}\hat{\rho}_T(t) = -i\,[\hat{H}(t), \hat{\rho}_T(t)], \tag{3.1}$$

with total Hamiltonian

$$\hat{H}(t) = \hat{H}_S(t) + \hat{H}_E + \hat{H}_{\mathrm{int}}.$$

Here $\hat{H}_E$ is the environment Hamiltonian and $\hat{H}_{\mathrm{int}}$ the system–environment coupling. The semiclassical light field enters exclusively through the time-dependent system Hamiltonian $\hat{H}_S(t)$.

3.2.1 Minimal coupling and dipole approximation

Starting from the minimal-coupling Hamiltonian for a particle of charge q and mass m in the Coulomb gauge,

$$\hat{H} = \frac{1}{2m}\Big[\hat{\boldsymbol{p}} + q\,\boldsymbol{A}(\hat{\boldsymbol{r}}, t)\Big]^2 + V(\hat{\boldsymbol{r}}) - q\,\phi(\hat{\boldsymbol{r}}, t),$$

where $\boldsymbol{A}(\hat{\boldsymbol{r}}, t)$ is the classical vector potential, $\phi(\hat{\boldsymbol{r}}, t)$ the scalar potential, and $V(\hat{\boldsymbol{r}})$ the static binding potential of the particle. In the long-wavelength (electric-dipole) approximation for neutral optical transitions, the familiar length-gauge interaction is obtained [41, 46], $\hat{H}_S(t) = \hat{H}_0 - \hat{\boldsymbol{\mu}} \cdot \boldsymbol{E}(t)$, where $\hat{H}_0$ is the field-free system Hamiltonian and $\hat{\boldsymbol{\mu}}$ the electric dipole operator. The electric field is assumed to vary negligibly over the molecular dimensions. For notational simplicity, a fixed linear polarization direction $\boldsymbol{e}$ is chosen. Defining $E(t) = \boldsymbol{e} \cdot \boldsymbol{E}(t)$ and $\hat{\mu} = \hat{\boldsymbol{\mu}} \cdot \boldsymbol{e}$, the interaction reduces to

$$\hat{H}_S(t) = \hat{H}_0 - \hat{\mu}\, E(t).$$

This corresponds to choosing the field polarization parallel to the relevant dipole component.

3.3 Polarization and macroscopic response

Maxwell's equations show that the emitted signal field is sourced by the macroscopic polarization $\mathbf{P}(\mathbf{r}, t)$ of the medium. This is derived in Appendix B.

A brief summary is as follows: In Maxwell's equations, $\mathbf{P}(\mathbf{r}, t)$ acts as a source term,

$$\nabla \times \nabla \times \boldsymbol{E}(\boldsymbol{r}, t) + \frac{1}{c^2}\frac{\partial^2}{\partial t^2}\boldsymbol{E}(\boldsymbol{r}, t) = -\mu_0 \frac{\partial^2}{\partial t^2}\boldsymbol{P}(\boldsymbol{r}, t).$$

In the perturbative regime,

$$\boldsymbol{P} = \boldsymbol{P}^{(1)} + \boldsymbol{P}^{(3)} + \cdots,$$

and in four-wave mixing the detected signal can be obtained via phase matching. The heterodyne cross term makes the detected signal linear in the emitted field

$$\boldsymbol{E}_S^{(3)}(\boldsymbol{r}, t) \propto i\, \boldsymbol{P}^{(3)}(\boldsymbol{r}, t).$$

The response-function formalism then provides $\boldsymbol{P}^{(3)}$ as a causal convolution of the intrinsic matter response $S^{(3)}$ with the incident fields (Appendix C). The microscopic task is therefore to compute the expectation value of the dipole operator.

For a single molecule coupled to an environment,

$$\langle \hat{\boldsymbol{\mu}}(t) \rangle = \mathrm{Tr}_{SE}\{\hat{\boldsymbol{\mu}}\, \hat{\rho}_T(t)\}.$$

If the dipole operator acts only on system degrees of freedom, tracing out the environment yields

$$\langle \hat{\boldsymbol{\mu}}(t) \rangle = \mathrm{Tr}_S\{\hat{\boldsymbol{\mu}}\, \hat{\rho}_S(t)\}, \qquad \hat{\rho}_S(t) = \mathrm{Tr}_E\, \hat{\rho}_T(t).$$

The total density operator $\hat{\rho}_T(t)$ evolves unitarily according to (3.1). By contrast, the reduced density operator $\hat{\rho}_S(t)$ does not, in general, satisfy a unitary equation of motion. Eliminating the bath degrees of freedom leads to an effective master equation

$$\frac{d}{dt}\hat{\rho}_S(t) = \mathcal{L}(t)\hat{\rho}_S(t),$$

whose derivation and approximations (Born–Markov, secular vs. non-secular) are discussed in Chapter 2. In the following, dissipative effects such as population relaxation and dephasing enter exclusively through this reduced dynamics.

For a representative molecule, the microscopic polarization is therefore defined as

$$\boldsymbol{P}(t) = \mathrm{Tr}_S\big[\hat{\boldsymbol{\mu}}\, \hat{\rho}_S(t)\big].$$

In a chosen basis $\{|n\rangle\}$ of the system Hilbert space this becomes

$$\boldsymbol{P}(t) = \sum_{n,m} \boldsymbol{\mu}_{nm}\, \rho_{S,mn}(t), \qquad \boldsymbol{\mu}_{nm} = \langle n|\, \hat{\boldsymbol{\mu}}\, |m\rangle.$$

Oscillating polarization is generated by optical coherences, i.e. off-diagonal elements of the density matrix. If the relevant electronic states carry no permanent dipoles in the chosen eigenbasis, i.e. $\boldsymbol{\mu}_{nn} = \mathbf{0}$, then populations do not contribute.

For an ensemble of identical, noninteracting molecules with number density $N(\mathbf{r})$,

$$\mathbf{P}(\mathbf{r}, t) \simeq N(\mathbf{r})\big\langle \boldsymbol{P}(t) \big\rangle_{\mathrm{or}},$$

where $\langle \cdot \rangle_{\mathrm{or}}$ denotes orientational averaging.

3.4 Response Function Formalism

The response-function formalism connects the applied field $\boldsymbol{E}(t)$ to the induced polarization $\boldsymbol{P}^{(n)}(t)$. The derivations are in Appendix C; here we only state the results used later.

3.4.1 Linear Response and the Kubo Formula

Linear response introduces the machinery (interaction picture, Dyson expansion, left/right Liouville actions) that generalizes to higher orders.

First-order polarization and Kubo formula

Specializing the general convolution and response-function definitions in Appendix C (Eq. (C.10), Eq. (C.11)) to $n = 1$ yields the Kubo formula used throughout the chapter:

$$\chi^{(1)}(\tau) = i\,\Theta(\tau)\,\big\langle\big[\hat{\boldsymbol{\mu}}_I(\tau), \hat{\boldsymbol{\mu}}_I(0)\big]\big\rangle_{\text{eq}}. \tag{3.2}$$

This is the Kubo formula for the linear susceptibility [3]. Importantly, $\chi^{(1)}$ is a *second-rank tensor* (in general anisotropic). In components,

$$\chi^{(1)}_{ij}(\tau) = i\,\Theta(\tau)\,\big\langle\big[\hat{\mu}_{i,I}(\tau), \hat{\mu}_{j,I}(0)\big]\big\rangle_{\text{eq}},$$

and $\chi^{(1)}\cdot\boldsymbol{E}$ denotes the standard tensor–vector contraction.

Diagrammatic interpretation

Even at first order, Eq. (3.2) decomposes into two Liouville-space pathways. For $\tau > 0$:

$$J_{1,ij}(\tau) \equiv \mathrm{Tr}\big\{\hat{\mu}_{i,I}(\tau)\,\hat{\mu}_{j,I}(0)\,\hat{\rho}_{\text{eq}}\big\}, \qquad J^*_{1,ij}(\tau) \equiv \mathrm{Tr}\big\{\hat{\mu}_{i,I}(\tau)\,\hat{\rho}_{\text{eq}}\,\hat{\mu}_{j,I}(0)\big\},$$

so that

$$S^{(1)}_{ij}(\tau) = i\big[J_{1,ij}(\tau) - J^*_{1,ij}(\tau)\big], \qquad \tau > 0.$$

The two contributions correspond to whether the interaction acts on the ket (left) or bra (right) branch of the density operator and are visualized as double-sided Feynman diagrams (Fig. 3.1).

Two-level system

For optical transitions ($\tilde{\nu}_0 \approx 16\,000\,\mathrm{cm}^{-1}$), $\omega_0/(k_B T) \approx 77$ at room temperature gives $p_1/p_0 \approx 10^{-34}$. Initial condition: electronic ground state.

For a two-level system with $|g\rangle$ and $|e\rangle$ the equilibrium state is written as

$$\hat{\rho}_{\text{eq}} = |g\rangle\langle g|\,.$$

and the dipole operator is written in terms of the transition dipole moment d and the dimensionless operator $\hat{m}$ as

$$\hat{\mu} = d\,\hat{m} \quad \text{with} \quad \hat{m} = |e\rangle\langle g| + |g\rangle\langle e|\,.$$

Suppressing polarization indices, the response becomes

$$S^{(1)}(\tau) = i\,d^2\big[J_1(\tau) - J_1^*(\tau)\big], \qquad \tau > 0, \tag{3.3}$$

with $J_1(\tau) = \mathrm{Tr}\{\hat{m}_I(\tau)\,\hat{m}_I(0)\,\hat{\rho}_{\text{eq}}\}$ and $J_1^*(\tau) = \mathrm{Tr}\{\hat{m}_I(\tau)\,\hat{\rho}_{\text{eq}}\,\hat{m}_I(0)\}$.

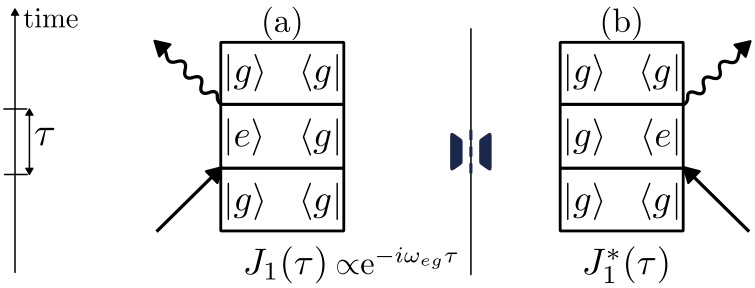

Figure 3.1: First-order double-sided Feynman diagrams to linear response. (a) Interaction on the ket (left action). (b) Interaction on the bra (right action). The commutator in Eq. (3.2) is their difference [43].

Coherence propagator

The coherence created by the first interaction evolves as

$$G_{eg}(\tau) = \exp\bigl[-(i\omega_{eg} + \Gamma_{eg})\tau\bigr],$$

where $\omega_{eg} = E_e - E_g$ and Γ_{eg} is the optical dephasing rate.

Rotating-wave approximation

It is convenient to decompose the real field into positive- and negative-frequency components,

$$\boldsymbol{E}(t) = \boldsymbol{E}^{(+)}(t) + \boldsymbol{E}^{(-)}(t), \qquad \boldsymbol{E}^{(-)}(t) = \bigl[\boldsymbol{E}^{(+)}(t)\bigr]^*, \tag{3.4}$$

and similarly

$$\hat{\boldsymbol{\mu}} = \hat{\boldsymbol{\mu}}^{(+)} + \hat{\boldsymbol{\mu}}^{(-)}.$$

Here $\hat{\boldsymbol{\mu}}^{(+)}$ lowers the excitation number and $\hat{\boldsymbol{\mu}}^{(-)} = (\hat{\boldsymbol{\mu}}^{(+)})^\dagger$ raises it. Within the rotating-wave approximation (RWA), only the near-resonant terms $\hat{\boldsymbol{\mu}}^{(+)} \cdot \boldsymbol{E}^{(-)}$ and $\hat{\boldsymbol{\mu}}^{(-)} \cdot \boldsymbol{E}^{(+)}$ are retained [3, 47]. Each light–matter interaction can couple to the positive-frequency part $\boldsymbol{E}^{(+)}(t) \propto e^{-i\omega t}$ or the negative-frequency part $\boldsymbol{E}^{(-)}(t) \propto e^{+i\omega t}$, indicated by the arrow direction in the Feynman diagrams [43]. Writing a resonant pulse as $E(t) = E_0(t)\, e^{-i\omega t} + E_0^*(t)\, e^{+i\omega t}$ and convolving with $S^{(1)}(t_1) \sim e^{-(\Gamma_{eg}+i\omega_{eg})t_1}$, one obtains a slowly varying term and a rapidly oscillating term $\propto e^{-i2\omega t_1}$. Near resonance ($\omega \approx \omega_{eg}$) the latter averages to zero—this is the rotating-wave approximation. Diagrammatically, only the arrow choice consistent with absorption-like excitation of the ground state survives [43]. The opposite choice would correspond to a "de-excitation" of ρ_{gg} which is not possible.

Frequency-domain susceptibility and absorption

The linear susceptibility is the Fourier transform of the causal response function,

$$\chi^{(1)}(\omega) = \int_{-\infty}^{\infty} \mathrm{d}\tau\, S^{(1)}(\tau) e^{i\omega\tau} = \int_0^{\infty} \mathrm{d}\tau\, S^{(1)}(\tau) e^{i\omega\tau},$$

where $S^{(1)}(\tau < 0) = 0$ has been used. The linear polarization in frequency space then becomes

$$\boldsymbol{P}^{(1)}(\omega) = \varepsilon_0\, \chi^{(1)}(\omega) \cdot \boldsymbol{E}(\omega).$$

In terms of the level pathways $J_1(\tau)$ and $J_1^*(\tau)$ defined in Eq. (3.3), this may be written explicitly as

$$\chi^{(1)}(\omega) = i\, d^2 \int_0^{\infty} \mathrm{d}\tau\, \bigl[J_1(\tau) - J_1^*(\tau)\bigr]\, e^{i\omega\tau}.$$

To connect to conventional lineshapes, the pathway is approximated by the coherence propagator, $J_1(\tau) \propto G_{eg}(\tau)$ with $G_{eg}(\tau) = \exp[-(i\omega_{eg} + \Gamma_{eg})\tau]$. This yields

$$\int_0^\infty \mathrm{d}\tau\, G_{eg}(\tau)\, \mathrm{e}^{i\omega\tau} = \int_0^\infty \mathrm{d}\tau\, \exp\big[-(\Gamma_{eg} + i(\omega_{eg} - \omega))\tau\big] = \frac{1}{\Gamma_{eg} + i(\omega_{eg} - \omega)},$$

and therefore the familiar complex Lorentzian susceptibility

$$\chi^{(1)}(\omega) \propto d^2 \frac{1}{\Gamma_{eg} - i(\omega - \omega_{eg})},$$

whose imaginary part gives the absorptive Lorentzian line shape centered at ω_{eg} with half-width Γ_{eg}. The absorption coefficient is $\alpha(\omega) \propto \operatorname{Im}\chi^{(1)}(\omega)$. For multilevel systems the total absorption is a sum over all allowed $g \to e$ transitions.

More generally, the cumulant-expansion lineshape theory expresses the absorption as [3]

$$I(\omega) \propto \operatorname{Re} \int_0^\infty \mathrm{d}t\, \exp\Big\{ i(\omega - \omega_{eg})t - g(t) - \Gamma t \Big\},$$

where $g(t)$ is the lineshape function determined by the environment correlation function (Chapter 2). The simple Lorentzian limit corresponds to $g(t) = 0$.

3.4.2 Nonlinear Response and Third-Order Polarization

At high field intensities, the response of the medium becomes nonlinear. In general, the macroscopic polarization can be expanded as a power series in the incident electric field. Schematically (suppressing explicit Cartesian indices),

$$\boldsymbol{P} = \varepsilon_0 \left(\chi^{(1)} \boldsymbol{E} + \chi^{(2)} : \boldsymbol{E}\boldsymbol{E} + \chi^{(3)} \vdots \boldsymbol{E}\boldsymbol{E}\boldsymbol{E} + \dots \right),$$

where ε_0 is the vacuum permittivity and $\chi^{(n)}$ denotes the n-th-order susceptibility tensor. The symbols $:$ and $\vdots$ indicate the implied tensor contractions. In the linear regime, the induced polarization is directly proportional to the applied field ($\chi^{(1)}$), describing refraction and absorption. Higher-order terms give rise to a variety of nonlinear optical phenomena [47].

Since even-order susceptibilities vanish in centrosymmetric media, the lowest-order nonlinear response relevant here is third order ($\chi^{(3)}$) [43]. The most important $\chi^{(3)}$ process for this thesis is **four-wave mixing (FWM)** [47], in which three incident fields generate a signal field. The resulting photon-echo and free-induction-decay contributions are the basis of 2D spectroscopy.

The response-function framework separates intrinsic matter response from field ordering and connects microscopic dynamics to experimentally selected signals via phase matching or phase cycling. The matter-side derivation is in Appendix C; here we only quote the third-order results used later. The tensor form is in Appendix C.8, and we work with a fixed polarization (scalar component).

Third-order response function $S^{(3)}$

The third-order polarization is of central interest for four-wave-mixing and 2D spectroscopy. From Appendix C (Eq. (C.12), Eq. (C.13)) the standard causal convolution and response function are

$$\boldsymbol{P}^{(3)}(t) = \int_0^\infty \mathrm{d}\tau_3 \int_0^\infty \mathrm{d}\tau_2 \int_0^\infty \mathrm{d}\tau_1\, \boldsymbol{S}^{(3)}(\tau_3, \tau_2, \tau_1)\, E(t - \tau_3)\, E(t - \tau_3 - \tau_2)\, E(t - \tau_3 - \tau_2 - \tau_1), \quad (3.5)$$

with

$$\boldsymbol{S}^{(3)}(\tau_3, \tau_2, \tau_1) = (\mathrm{i})^3\, \mathrm{Tr}\Big\{ \hat{\boldsymbol{\mu}}\, \mathcal{U}_0(\tau_3)\, \mathcal{V}\, \mathcal{U}_0(\tau_2)\, \mathcal{V}\, \mathcal{U}_0(\tau_1)\, \mathcal{V}[\hat{\rho}_{\mathrm{eq}}] \Big\}. \quad (3.6)$$

The equivalent nested-commutator form and pathway expansion follow from the commutator structure in Appendix C (Eq. (C.3)) and are not repeated here.

Time-domain picture and impulsive limit

The convolution form Eq. (3.5) makes the *causal* time ordering explicit. For fixed delays (τ_1, τ_2, τ_3), the three interactions with the field occur at the times

$$t - \tau_3, \qquad t - \tau_3 - \tau_2, \qquad t - \tau_3 - \tau_2 - \tau_1,$$

and the system evolves freely for durations τ_1, τ_2, and τ_3 between these events. Reading the operator product in Eq. (3.6) from right to left corresponds precisely to this physical sequence: start from ρ_{eq}, apply a dipole interaction, evolve freely, apply another interaction, and so on, followed by the dipole readout at time t.

The physical interpretation of this causal ordering, and the impulsive-limit simplification for ultrashort pulses, was already discussed when relating the detected signal field to $\boldsymbol{P}^{(3)}$ in Sec. 3.2. This section is therefore kept brief and the above expressions are used mainly as the starting point for pathway decompositions below.

In experiments with finite pulse durations or partial pulse overlap, strict time ordering is blurred and the signal becomes a weighted convolution with the pulse envelopes. This can redistribute pathway amplitudes. In the nonperturbative simulations used here, these effects are naturally included.

Two-level pathway construction

To see concretely how the compact operator expression expands into distinct contributions, consider a two-level system with ground state $|g\rangle$ and excited state $|e\rangle$, and assume an initial equilibrium state $\rho_{\text{eq}} = |g\rangle\langle g|$. Let the dipole (projected onto the field polarization) be

$$\hat{\mu} = d\,\hat{m}, \qquad \hat{m} \equiv |e\rangle\langle g| + |g\rangle\langle e|\,,$$

and define the commutator superoperator $\mathcal{M}[\hat{A}] \equiv [\hat{m}, \hat{A}]$. Then $\mathcal{V} = d\,\mathcal{M}$ and (up to an overall prefactor) the third-order response can be written in the same structural form as Eq. (3.6):

$$S^{(3)}(\tau_3, \tau_2, \tau_1) \propto \operatorname{Tr}\Big\{\hat{m}\,\mathcal{U}_0(\tau_3)\,\mathcal{M}\,\mathcal{U}_0(\tau_2)\,\mathcal{M}\,\mathcal{U}_0(\tau_1)\,\mathcal{M}[|g\rangle\langle g|]\Big\}.$$

First interaction. Using $\mathcal{M}[\hat{A}] = \hat{m}\hat{A} - \hat{A}\hat{m}$,

$$\mathcal{M}[|g\rangle\langle g|] = |e\rangle\langle g| - |g\rangle\langle e|\,.$$

Thus the first interaction creates a superposition of optical coherences. The two terms already illustrate the pathway logic: each commutator produces a *left* and a *right* action (ket vs. bra), with a relative minus sign.

Free evolution during τ_1. In the secular approximation (coherences do not mix), the coherences evolve independently with the coherence Green function $G_{eg}(\tau) = \exp[-(i\omega_{eg} + \Gamma_{eg})\tau]$:

$$\mathcal{U}_0(\tau_1)[|e\rangle\langle g|] \approx G_{eg}(\tau_1)\ |e\rangle\langle g|\,, \qquad \mathcal{U}_0(\tau_1)[|g\rangle\langle e|] \approx G^*_{eg}(\tau_1)\ |g\rangle\langle e|\,.$$

Second interaction. Applying $\mathcal{M}$ again converts these coherences into population operators,

$$\mathcal{M}[|e\rangle\langle g|] = |g\rangle\langle g| - |e\rangle\langle e|\,, \qquad \mathcal{M}[|g\rangle\langle e|] = |e\rangle\langle e| - |g\rangle\langle g|\,,$$

with optical phase factors carried by $G_{eg}(\tau_1)$ and $G^*_{eg}(\tau_1)$.

Waiting time and readout. During τ_2 the density operator is (in this simple model) a linear combination of populations. The third commutator converts populations back into coherences, which

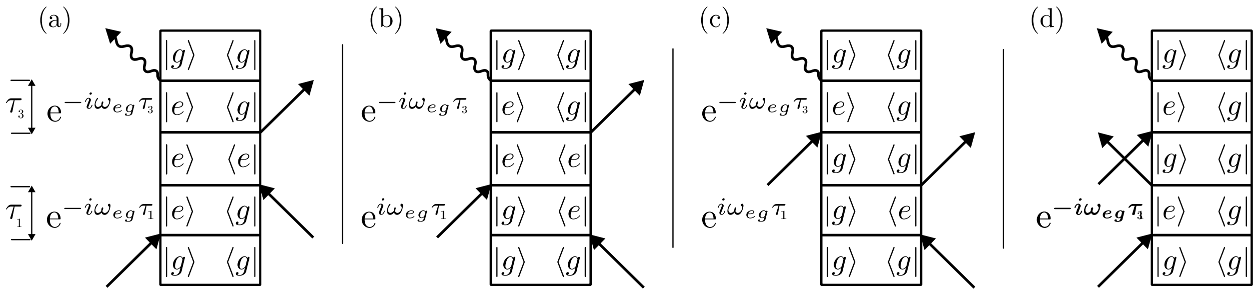

Figure 3.2: Double-sided Feynman diagrams for the four third-order pathway classes S_1–S_4 in Eqs. (3.7) to (3.14). Each diagonal arrow indicates a field interaction acting on the ket (left branch) or bra (right branch) of the density matrix. The complex-conjugate pathways S_α^* correspond to exchanging ket and bra actions as was already done in Fig. 3.1.

evolve during τ_3 and are finally converted into a measurable polarization by the left multiplication with $\hat{m}$ and the trace.

Why there are 2^3 terms. Each of the three commutators produces two contributions (left and right action). Starting from a single operator $|g\rangle\langle g|$, this generates $2^3 = 8$ distinct terms in the third-order response.

Writing the nested commutator explicitly,

$$S^{(3)}(\tau_3,\tau_2,\tau_1) = (-i)\mathrm{Tr}\Big\{\hat{\mu}_I(\tau_1+\tau_2+\tau_3)\,[\hat{\mu}_I(\tau_1+\tau_2),[\hat{\mu}_I(\tau_1),[\hat{\mu}_I(0),\hat{\rho}_{\mathrm{eq}}]]]\Big\},$$

one obtains the eight operator products

$$\begin{aligned}
S^{(3)}(\tau_3,\tau_2,\tau_1) = {} & \mathrm{Tr}\big\{\hat{\mu}_I(\tau_1+\tau_2+\tau_3)\,\hat{\mu}_I(\tau_1+\tau_2)\,\hat{\mu}_I(\tau_1)\,\hat{\mu}_I(0)\,\hat{\rho}_{\mathrm{eq}}\big\} && \Rightarrow\quad S_4 && (3.7)\\
& - \mathrm{Tr}\big\{\hat{\mu}_I(\tau_1+\tau_2+\tau_3)\,\hat{\mu}_I(\tau_1+\tau_2)\,\hat{\mu}_I(\tau_1)\,\hat{\rho}_{\mathrm{eq}}\,\hat{\mu}_I(0)\big\} && \Rightarrow\quad S_1^* && (3.8)\\
& - \mathrm{Tr}\big\{\hat{\mu}_I(\tau_1+\tau_2+\tau_3)\,\hat{\mu}_I(\tau_1+\tau_2)\,\hat{\mu}_I(0)\,\hat{\rho}_{\mathrm{eq}}\,\hat{\mu}_I(\tau_1)\big\} && \Rightarrow\quad S_2^* && (3.9)\\
& + \mathrm{Tr}\big\{\hat{\mu}_I(\tau_1+\tau_2+\tau_3)\,\hat{\mu}_I(\tau_1+\tau_2)\,\hat{\rho}_{\mathrm{eq}}\,\hat{\mu}_I(0)\,\hat{\mu}_I(\tau_1)\big\} && \Rightarrow\quad S_3 && (3.10)\\
& - \mathrm{Tr}\big\{\hat{\mu}_I(\tau_1+\tau_2+\tau_3)\,\hat{\mu}_I(\tau_1)\,\hat{\mu}_I(0)\,\hat{\rho}_{\mathrm{eq}}\,\hat{\mu}_I(\tau_1+\tau_2)\big\} && \Rightarrow\quad S_3^* && (3.11)\\
& + \mathrm{Tr}\big\{\hat{\mu}_I(\tau_1+\tau_2+\tau_3)\,\hat{\mu}_I(\tau_1)\,\hat{\rho}_{\mathrm{eq}}\,\hat{\mu}_I(0)\,\hat{\mu}_I(\tau_1+\tau_2)\big\} && \Rightarrow\quad S_2 && (3.12)\\
& + \mathrm{Tr}\big\{\hat{\mu}_I(\tau_1+\tau_2+\tau_3)\,\hat{\mu}_I(0)\,\hat{\rho}_{\mathrm{eq}}\,\hat{\mu}_I(\tau_1)\,\hat{\mu}_I(\tau_1+\tau_2)\big\} && \Rightarrow\quad S_1 && (3.13)\\
& - \mathrm{Tr}\big\{\hat{\mu}_I(\tau_1+\tau_2+\tau_3)\,\hat{\rho}_{\mathrm{eq}}\,\hat{\mu}_I(0)\,\hat{\mu}_I(\tau_1)\,\hat{\mu}_I(\tau_1+\tau_2)\big\} && \Rightarrow\quad S_4^*. && (3.14)
\end{aligned}$$

Along the same lines as in linear response, the terms labeled S_α^* are the complex conjugates of S_α for Hermitian $\hat{\mu}$ and a real equilibrium state. In typical 2DES calculations (RWA) one therefore groups the response as

$$S^{(3)}(\tau_3,\tau_2,\tau_1) = (-i)\sum_{\alpha=1}^{4}\Big[S_\alpha(\tau_3,\tau_2,\tau_1) - S_\alpha^*(\tau_3,\tau_2,\tau_1)\Big].$$

These four pathway classes are conveniently represented as double-sided Feynman diagrams, shown in Fig. 3.2.

Liouville-space pathways and double-sided Feynman diagrams

The superoperator representation makes the origin of *pathways* transparent. Because $\mathcal{V} = (\mathcal{V}_L - \mathcal{V}_R)$, the product $\mathcal{V}_{j_3}\mathcal{V}_{j_2}\mathcal{V}_{j_1}$ expands into a sum of terms with different choices of left/right actions at the three interaction times. Each such term corresponds to a distinct time-ordered evolution of the density matrix elements (populations and coherences) and is conveniently visualized by a *double-sided Feynman diagram* (also called a Liouville-space pathway diagram).

The basic rules are:

- Two vertical lines represent the ket (left) and bra (right) branches of $\hat{\rho}_S$.
- Time flows upward. The three field interactions occur at τ_1, $\tau_1+\tau_2$, and $\tau_1+\tau_2+\tau_3$.
- An interaction acting on the ket contributes $+1$ to the global sign. An interaction on the bra contributes -1. This sign structure follows from the commutator decomposition $\mathcal{V} = (\mathcal{V}_L - \mathcal{V}_R)$ inherited from the nested commutators in the response function.
- Under the rotating-wave approximation one further assigns whether the interaction involves $\boldsymbol{E}^{(+)}$ or $\boldsymbol{E}^{(-)}$ (absorption vs. emission). This fixes which optical coherence is created and thus determines the overall phase factor and the phase-matching direction.

For interpretation, the eight third-order pathways are regrouped into three physical classes: ground-state bleach (GSB), stimulated emission (SE), and excited-state absorption (ESA). This classification is standard in 2DES and will be used when discussing peak assignments [3, 5].

Relaxation during the waiting time

During the waiting interval T, population relaxation modifies the third-order pathways. For a two-level system, the relevant dissipative channel is spontaneous decay $|e\rangle \to |g\rangle$ at rate k_{ge}, described by the Lindblad jump operator $\hat{L} = \sqrt{k_{ge}}\,|g\rangle\langle e|$ (cf. Chapter 2). The resulting equations of motion for the density-matrix elements are

$$\dot{\rho}_{ee} = -k_{ge}\,\rho_{ee}, \qquad \dot{\rho}_{gg} = +k_{ge}\,\rho_{ee}, \qquad \dot{\rho}_{eg} = -(k_{ge}/2 + \gamma_\varphi)\,\rho_{eg}, \tag{3.15}$$

where γ_φ accounts for additional pure dephasing. Solving these gives the population propagators

$$U_{ee,ee}(T) = \mathrm{e}^{-k_{ge}T}, \qquad U_{gg,ee}(T) = 1 - \mathrm{e}^{-k_{ge}T}, \qquad U_{gg,gg}(T) = 1, \tag{3.16}$$

and the total dephasing rate of the optical coherence is $\Gamma_{eg} = k_{ge}/2 + \gamma_\varphi$, with $T_1 = 1/k_{ge}$ and $T_2 = 1/\Gamma_{eg}$.

Consequently, every pathway that passes through ρ_{ee} during T splits into a contribution that remains in ee (multiplied by $U_{ee,ee}$, yielding standard SE/ESA signals that decay with the lifetime) and a contribution that has relaxed to gg (multiplied by $U_{gg,ee}$, a relaxation-induced pathway acting on the refilled ground state). In the numerical simulations used later, this splitting is generated automatically by the Liouvillian propagator.

Multilevel generalization

Realistic systems typically have many optically accessible excited states, and a central goal of this thesis is to extend the simulation framework from a minimal two-level model to multilevel Hamiltonians. A convenient (and widely used) idealisation that already captures the essential physics of ground-state bleach (GSB), stimulated emission (SE), and excited-state absorption (ESA) is a three-manifold "band" picture:

- a single ground state $|g\rangle$.
- a manifold of singly excited states $\{|n\rangle\}$.
- a manifold of higher excited (doubly excited) states $\{|f\rangle\}$.

Optically allowed transitions connect $|g\rangle \leftrightarrow |n\rangle$ and $|n\rangle \leftrightarrow |f\rangle$. In third-order spectroscopy the Liouville pathways do not contain populations of the f manifold (they end in coherences such as ρ_{fn}), so relaxation out of f typically enters effectively through additional dephasing/broadening of the $n \leftrightarrow f$ coherences.

Propagators in the secular approximation. In the secular approximation, the waiting-time evolution is expressed in terms of Liouville-space propagator elements of the generic form

$$U_{nn,mm}(T), \qquad U_{nm,nm}(T), \qquad U_{gg,nn}(T), \tag{3.17}$$

describing population transfer within the excited manifold ($nn \to mm$), coherence dephasing within the excited manifold ($nm \to nm$), and refilling of the ground state from excited populations ($nn \to gg$). ESA pathways additionally involve propagators for f-manifold coherences (e.g. $U_{fn,fm}(T)$).

Recipe for constructing multilevel signals. Given these propagators, multilevel spectra are obtained by the same "stitching" procedure as in the two-level case:

1. choose the relevant pulse sequence (and thus the allowed pathway classes and phase matching / phase cycling).
2. for each pathway, write the product of four dipole matrix elements (e.g. d_{ng}, d_{mg}, and, for ESA, d_{fn}) and three propagators (coherence–population–coherence) appropriate to that pathway.
3. Fourier transform over the detection time to obtain contributions as functions of ω_{det} (and, for 2D spectra, also over the coherence time to obtain ω_{coh}).
4. sum over all intermediate state indices $n, m, f, \ldots$.

This multilevel bookkeeping is exactly what becomes necessary once the system Hamiltonian contains multiple excited eigenstates (or excitonic bands) and the open-system Liouvillian generates non-trivial transfer and dephasing within these manifolds. The explicit multilevel formulation will therefore serve as the bridge from the pedagogical two-level picture to the full numerical simulations targeted in the later thesis chapters.

General dissipative channels (preview). For later use it is helpful to note that a multilevel relaxation model can be written compactly in Lindblad form by introducing one jump operator per allowed transition $|n\rangle \to |m\rangle$,

$$\hat{L}_{mn} = \sqrt{k_{mn}}\, |m\rangle \langle n| \,,$$

and, if needed, pure-dephasing operators for each level,

$$\hat{L}_n^{(\varphi)} = \sqrt{\gamma_{\varphi,n}}\, |n\rangle \langle n| \,.$$

These ingredients define the Liouvillian whose propagator elements $U_{ab,cd}(t)$ enter directly in the pathway products described above.

In practice, one evaluates $S^{(3)}$ by summing the relevant pathways (often grouped into ground-state bleach, stimulated emission, and excited-state absorption). The experimentally separated rephasing and nonrephasing signals (Sec. 3.5) correspond to different subsets of these pathways, selected either by phase matching or by phase cycling.

IMPORTANT: Revisit this transition and streamline the third-order pathway walkthrough; clarify how the formal response connects to the experimental three-pulse geometry.

3.5 Three-Pulse Experiments and Phase Matching

Having established the third-order response function $S^{(3)}(\tau_3, \tau_2, \tau_1)$ as the material kernel encoding all microscopic dynamics, the experimental realization is now specified: a sequence of three ultrashort pulses with controlled delays and wavevectors. Phase matching isolates specific pathway subsets (rephasing vs nonrephasing), and phase cycling further selects desired signal contributions.

IMPORTANT: Add explicit connection: third-order response (Section 3.4.2) provides the material kernel; this section specifies the incident field geometry that probes it experimentally.

IMPORTANT: hier mini text

3.5.1 Wavevector phase-matching condition

In nonlinear optical processes, multiple light waves interact within a material to generate new frequency components. In a plane-wave picture, momentum conservation of the incident and generated fields implies that the signal wavevector $\boldsymbol{k}_S$ is given by a signed sum of the input wavevectors,

$$\boldsymbol{k}_S = \pm\boldsymbol{k}_1 \pm \boldsymbol{k}_2 \pm \boldsymbol{k}_3. \tag{3.18}$$

This is the (wavevector) *phase-matching condition.* The signs track whether the corresponding interaction involves the analytic (positive-frequency) part of the field or its complex conjugate (negative-frequency part) as introduced in Eq. (3.4). Equivalently, in Liouville space they reflect whether the field interaction acts on the ket (left action) or bra (right action) branch of the density matrix: absorption-like interactions contribute $+\boldsymbol{k}_i$, while emission-like/complex-conjugate interactions contribute $-\boldsymbol{k}_i$.

Different sign combinations correspond to different phase-matched signal directions and therefore to different subsets of nonlinear pathways. In noncollinear geometries these directions can be spatially separated, while in collinear geometries the same contributions can be isolated via their phase dependence (phase cycling, Sec. 3.6). The two most important third-order phase-matching conditions for this thesis (rephasing and nonrephasing) are discussed below [45].

3.5.2 Representation of three ultrashort pulses

The experimentally relevant case of three ultrashort pulses with carrier wavevectors $\boldsymbol{k}_1, \boldsymbol{k}_2, \boldsymbol{k}_3$ and carrier frequency ω_L is now considered. In the slowly varying envelope approximation,

$$\boldsymbol{E}(\boldsymbol{r}, t) = \sum_{j=1}^{3} \Big[\boldsymbol{E}_j(t - t_j)\, e^{i(\boldsymbol{k}_j\cdot\boldsymbol{r} - \omega_L t + \phi_j)} + \text{c.c.} \Big], \tag{3.19}$$

where t_j is the arrival time of pulse j, ϕ_j is its carrier phase-kick, and $\boldsymbol{E}_j(t)$ is a slowly varying envelope (assumed identical spectral center ω_L for simplicity). The complex analytic signal is

$$\boldsymbol{E}^{(+)}(\boldsymbol{r}, t) = \sum_{j=1}^{3} \boldsymbol{E}_j(t - t_j)\, e^{i(\boldsymbol{k}_j\cdot\boldsymbol{r} - \omega_L t + \phi_j)}.$$

The setup can be visualized in a noncollinear boxcar geometry as shown in Fig. 3.3. Here three pulses with wavevectors $\boldsymbol{k}_1$, $\boldsymbol{k}_2$, and $\boldsymbol{k}_3$ interact with the sample and generate a signal emitted into the phase-matched direction $\boldsymbol{k}_S$.

3.5.3 Intuition: carrier phase-kicks and pulse area (delta-pulse picture)

IDEA: TODO create a suitable figure -> move this to an appendix or additional meterial as it is not central to the thesis

The phase parameters ϕ_j in Eq. (3.19) are often introduced as bookkeeping devices for phase cycling, but they also have a simple physical meaning: for a resonantly driven transition they set the

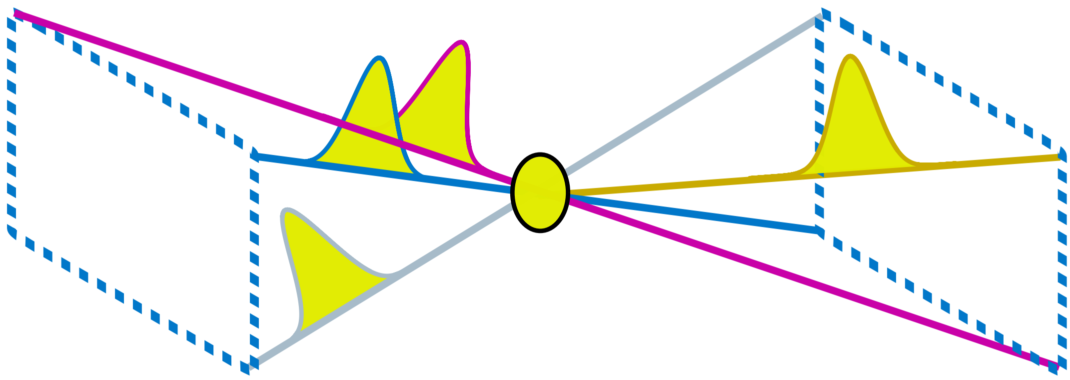

Figure 3.3: Schematic four-wave-mixing setup in a noncollinear boxcar geometry. Three input pulses with wavevectors $\boldsymbol{k}_1, \boldsymbol{k}_2, \boldsymbol{k}_3$ generate a third-order signal field radiated in a phase-matched direction $\boldsymbol{k}_S$.

optical phase of the induced coherence and, equivalently, the *rotation axis* of the pulse on the Bloch sphere. This is most transparent for an effective two-level system $\{|g\rangle, |e\rangle\}$ in the rotating-wave approximation. Writing the complex Rabi frequency of pulse j as

$$\Omega_j(t) \equiv \boldsymbol{\mu}_{eg} \cdot \boldsymbol{E}_j(t), \tag{3.20}$$

the interaction Hamiltonian in the laser-rotating frame can be expressed (up to standard conventions) as

$$\hat{H}_{\mathrm{int},j}(t) \;\propto\; \Omega_j(t-t_j)\Big(e^{-i\phi_j}\;|e\rangle\langle g| + e^{+i\phi_j}\;|g\rangle\langle e|\Big). \tag{3.21}$$

Thus, changing ϕ_j does not change the pulse *intensity envelope* but rotates the complex drive between its in-phase and quadrature components. On the Bloch sphere this corresponds to a rotation around an axis in the equatorial (x–y) plane with azimuth angle ϕ_j (cf., e.g., the tutorial discussion in Ref. [48]).

The *pulse area*

$$\theta_j \;\equiv\; \int_{-\infty}^{\infty} \mathrm{d}t\, \Omega_j(t-t_j) \tag{3.22}$$

sets the corresponding rotation angle. In the idealized delta-pulse limit, one replaces $\Omega_j(t-t_j) \to \theta_j\, \delta(t-t_j)$, and the pulse acts as an instantaneous unitary

$$\hat{U}_j \;=\; \exp\!\left[-\frac{i\,\theta_j}{2}\left(\cos\phi_j\,\hat{\tau}_x + \sin\phi_j\,\hat{\tau}_y\right)\right], \tag{3.23}$$

where $\hat{\tau}_{x,y,z}$ denote the Pauli matrices, which rotates the Bloch vector by angle θ_j about the equatorial axis at azimuth ϕ_j. For example, a $\pi/2$ pulse ($\theta_j = \pi/2$) prepares a maximal coherence whose phase is set by ϕ_j, while a π pulse ($\theta_j = \pi$) inverts population. In multi-pulse spectroscopy, the phases imprinted on successive coherences interfere. This is exactly why the macroscopic third-order polarization decomposes into phase-tagged components $\propto e^{i(l\phi_1+m\phi_2+n\phi_3)}$ in Eq. (3.24), enabling pathway selection by phase matching or phase cycling.

3.5.4 Third-order polarization in space and time

Inserting Eq. (3.19) into the third-order convolution Eq. (3.5), and keeping only terms resonant in RWA, one finds that the third-order polarization contains contributions oscillating with phase factors

$$\exp\Big[i(\pm\boldsymbol{k}_1 \pm \boldsymbol{k}_2 \pm \boldsymbol{k}_3)\cdot\boldsymbol{r} - i\omega_L t + i(\pm\phi_1 \pm \phi_2 \pm \phi_3)\Big],$$

where each sign corresponds to an interaction with $\boldsymbol{E}^{(+)}$ (ket) or $\boldsymbol{E}^{(-)}$ (bra). Thus the third-order polarization can be written as

$$\boldsymbol{P}^{(3)}(\boldsymbol{r},t) = \sum_{l,m,n\in\{-1,0,+1\}} e^{i(\boldsymbol{k}_S\cdot\boldsymbol{r}-\omega_L t+\Phi_{lmn})}\,\boldsymbol{P}^{(3)}_{l,m,n}(t), \tag{3.24}$$

with

$$\boldsymbol{k}_S = l\,\boldsymbol{k}_1 + m\,\boldsymbol{k}_2 + n\,\boldsymbol{k}_3, \qquad \Phi_{lmn} = l\phi_1 + m\phi_2 + n\phi_3.$$

Only those combinations that satisfy energy and momentum (phase-matching) constraints contribute efficiently to measurable signals in a macroscopic sample [47].

3.5.5 Rephasing and nonrephasing phase-matching conditions

The phase-matched directions of interest for two-dimensional electronic spectroscopy are

$$\boldsymbol{k}_R = -\boldsymbol{k}_1 + \boldsymbol{k}_2 + \boldsymbol{k}_3, \tag{3.25}$$

$$\boldsymbol{k}_{NR} = +\boldsymbol{k}_1 - \boldsymbol{k}_2 + \boldsymbol{k}_3, \tag{3.26}$$

usually identified as *rephasing* (R) and *nonrephasing* (NR) directions, respectively [3, 5, 45]. In a noncollinear "box" geometry, beams are arranged such that different $\boldsymbol{k}_S$ point along distinct directions in space. A detector placed in the appropriate direction records the corresponding signal field.

Besides the rotating-wave approximation, a further simplification is obtained in the *impulsive limit* of well separated, ultrashort pulses: the interaction times are then strictly ordered and pulse overlap can be neglected [5, 43]. In this case the sign pattern in Eqs. (3.25) to (3.26) can be read directly as the side (ket/bra) on which each interaction acts (cf. the $\boldsymbol{E}^{(+)}/\boldsymbol{E}^{(-)}$ bookkeeping above and the phase-cycling geometry discussed in Sec. 3.6).

For the **rephasing** signal, $\boldsymbol{k}_R = -\boldsymbol{k}_1+\boldsymbol{k}_2+\boldsymbol{k}_3$, the first interaction is conjugated (bra-side, $-\boldsymbol{k}_1$), creating an optical coherence. The second pulse converts this coherence into a population. The third pulse generates a new coherence that radiates the signal. Because the phase evolution during the first coherence period is effectively reversed relative to the detection period, an inhomogeneously broadened ensemble can exhibit a photon-echo-like refocusing in the time domain around $t_{\text{det}} \approx t_{\text{coh}}$ [4, 5, 45].

By contrast, for the **nonrephasing** signal, $\boldsymbol{k}_{NR} = +\boldsymbol{k}_1-\boldsymbol{k}_2+\boldsymbol{k}_3$, the ordering corresponds to a ket-side first interaction ($+\boldsymbol{k}_1$), followed by a bra-side second interaction ($-\boldsymbol{k}_2$), and a ket-side third interaction ($+\boldsymbol{k}_3$). The coherence evolution does not reverse in the same way, so static disorder is not refocused and no echo is formed.

The rephasing signal is sensitive to inhomogeneous broadening and exhibits echo-like refocusing, whereas the nonrephasing signal does not rephase static disorder and emphasizes homogeneous dynamics. Their sum, properly phased, yields an absorptive 2D spectrum with well-defined peak shapes.

Other third-order directions can be accessed by alternative phase-matching or phase-cycling selections (e.g. double-quantum coherence signals, often written as $\boldsymbol{k}_S = +\boldsymbol{k}_1 + \boldsymbol{k}_2 - \boldsymbol{k}_3$ in the usual convention), but they are not required for the aims of this thesis. See Ref. [49] for a broader overview.

3.6 Phase Cycling and Selection of Pathways

Instead of spatially resolving different $\boldsymbol{k}_S$-components, one can use a collinear arrangement and separate contributions by varying the carrier phases ϕ_j and performing a discrete Fourier analysis

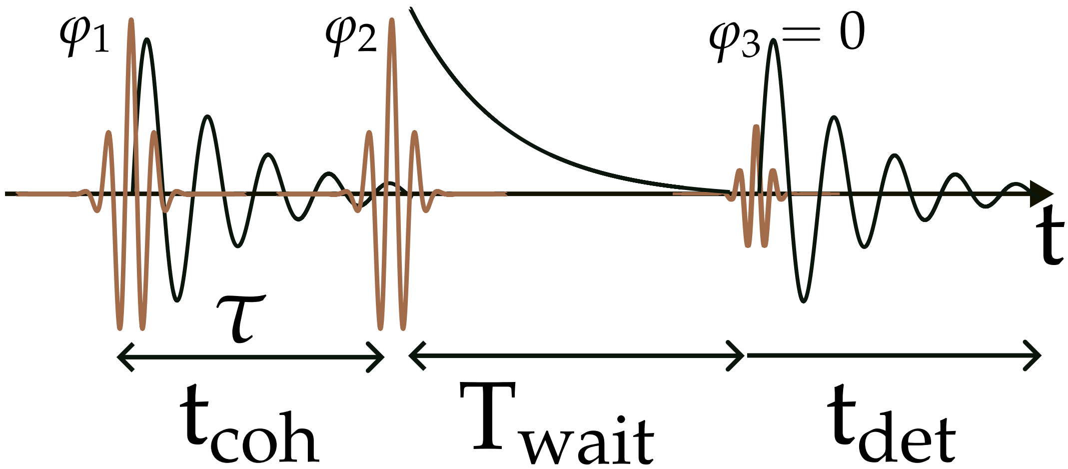

Figure 3.4: Schematic four-wave-mixing setup in a collinear, phase-cycling geometry. Spatial separation of different $\boldsymbol{k}_S$ components is replaced by varying the carrier phases of the input pulses and extracting the desired contribution via phase cycling.

with respect to (ϕ_1, ϕ_2, ϕ_3) [43, 50]. This is particularly useful for numerical simulations and for experiments where spatial selection is not practical.

Operationally, phase cycling replaces spatial separation by a set of repeated measurements (or simulation runs) in which the excitation phases are stepped through a discrete cycle. Because different Liouville-space pathways carry different phase factors, suitable linear combinations — equivalently, a discrete Fourier transform with respect to the phases — isolate the desired contribution. A common choice is to step phases in increments of $\pi/2$ (or, more generally, $2\pi/N_\phi$), which is sufficient to separate the third-order components relevant for rephasing and nonrephasing 2D spectra.

The collinear phase-cycling geometry is shown in Fig. 3.4.

3.6.1 Phase-Cycling Fourier Selection and Construction of 2D Spectra

In general, the nonlinear polarization generated by three input fields can be decomposed into wavevector-tagged components,

$$\boldsymbol{P}^{\mathrm{NL}}(\boldsymbol{r}, t) = \sum_{l,m,n} e^{i\,(l\boldsymbol{k}_1 + m\boldsymbol{k}_2 + n\boldsymbol{k}_3)\cdot\boldsymbol{r}}\, \boldsymbol{P}_{l,m,n}(t) + \text{c.c.}, \tag{3.27}$$

where the physically relevant third-order directions include the rephasing and nonrephasing combinations in Eqs. (3.25) and (3.26). In a collinear geometry these components cannot be spatially separated, but they can be isolated through their dependence on the carrier phases ϕ_1, ϕ_2, ϕ_3 (phase cycling).

At fixed observation direction (e.g. collinear), the total complex third-order polarization can be written as a Fourier series in the phases,

$$P^{(3)}(t; \phi_1, \phi_2, \phi_3) = \sum_{l,m,n} e^{i(l\phi_1 + m\phi_2 + n\phi_3)}\, P^{(3)}_{l,m,n}(t).$$

Here $P^{(3)}_{l,m,n}(t)$ collects the third-order contributions with a fixed combination of pulse phases. In particular, the rephasing and nonrephasing components correspond to

$$\text{rephasing:} \quad (l, m, n) = (-1, +1, +1), \tag{3.28}$$

$$\text{nonrephasing:} \quad (l, m, n) = (+1, -1, +1). \tag{3.29}$$

These index triples are the phase analogues of the wavevector combinations Eqs. (3.25) and (3.26) [3, 5, 50].

3.6.2 Discrete phase cycling

In practice, a finite set of phase values is used, typically $\phi_j \in \{0, 2\pi/N_\phi, \ldots, 2\pi(N_\phi - 1)/N_\phi\}$ for each j. The desired component is then obtained by discrete Fourier projection

$$P^{(3)}_{l,m,n}(t) \simeq \frac{1}{N_\phi^3} \sum_{\phi_1,\phi_2,\phi_3} e^{-i(l\phi_1+m\phi_2+n\phi_3)}\, P^{(3)}(t;\phi_1,\phi_2,\phi_3). \tag{3.30}$$

In the continuous limit, the corresponding inverse Fourier transform reads

$$P^{(3)}_{l,m,n}(t) = \frac{1}{(2\pi)^3} \int_0^{2\pi}\int_0^{2\pi}\int_0^{2\pi} d\phi_1\, d\phi_2\, d\phi_3\, e^{-i(l\phi_1+m\phi_2+n\phi_3)}\, P^{(3)}(t;\phi_1,\phi_2,\phi_3). \tag{3.31}$$

In simulations (and in this thesis), $P^{(3)}(t)$ is obtained by solving the full master equation including the time-dependent light–matter interaction at each phase setting (ϕ_1, ϕ_2, ϕ_3) and then extracting the phase-selected third-order component via Eq. (3.30) with (l, m, n) chosen according to Eqs. (3.28) and (3.29).

Remark (order selection in simulations). Besides phase cycling, another common numerical strategy is to isolate a desired nonlinear order by combining runs with different subsets of pulses turned on (and subtracting lower-order contributions). In this thesis phase cycling is used, because it selects the desired rephasing/nonrephasing pathways directly through their phase dependence.

The corresponding complex signal fields are proportional to these selected polarizations,

$$E_{S,R/NR}(t) \propto i\, P^{(3)}_{R/NR}(t), \tag{3.32}$$

where proportionality constants depend on sample geometry and propagation but do not affect the normalized 2D spectral shapes considered here. This proportionality holds within the slowly varying envelope approximation [3, 4].

3.7 Construction of Two-Dimensional Spectra

In this thesis the phase-cycled complex signal field $E_{S,R/NR}(t_{\mathrm{coh}}, T, t_{\mathrm{det}})$ is treated as the object from which 2D spectra are constructed. Experimentally, this quantity is accessed by heterodyne detection: the detected intensity contains an interference term that is linear in the signal field,

$$I_{\mathrm{det}}(\omega_{\mathrm{det}}) \propto \left|E_{\mathrm{LO}}(\omega_{\mathrm{det}}) + E_S(\omega_{\mathrm{det}})\right|^2 \approx |E_{\mathrm{LO}}(\omega_{\mathrm{det}})|^2 + 2\,\mathrm{Re}[E^*_{\mathrm{LO}}(\omega_{\mathrm{det}})E_S(\omega_{\mathrm{det}})]$$

for $|E_S| \ll |E_{\mathrm{LO}}|$ after subtraction of the LO background. The detailed discussion of homodyne/heterodyne detection, phase sensitivity, and phasing is deferred to Appendix D. The Maxwell-side propagation and phase-matching assumptions underpinning $E_S \propto P^{(3)}$ are derived in Appendix B.

3.7.1 Time variables for 2D spectroscopy

In a three-pulse experiment, it is customary to express the signal in terms of the three time delays

$$t_{\mathrm{coh}} = t^{(2)} - t^{(1)}, \qquad T = t^{(3)} - t^{(2)}, \qquad t_{\mathrm{det}} = t_{\mathrm{obs}} - t^{(3)},$$

where $t^{(1)}, t^{(2)}, t^{(3)}$ are the pulse arrival times and t_{obs} is the observation (readout) time. The quantities t_{coh}, T, and t_{det} are referred to as the coherence time, waiting time, and detection time, respectively. Scanning the waiting time T probes population dynamics and spectral diffusion during the population interval between the second and third interactions [3, 5].

Remark (positive vs. negative coherence time). In some presentations one also considers negative t_{coh} values, which correspond to reversing the order of pulses 1 and 2. In that case the rephasing photon-echo picture does not apply. Instead one obtains a nonrephasing free-induction–decay-like contribution that is complementary to the $t_{\text{coh}} > 0$ rephasing branch [51]. For each fixed waiting time T, the signal is a function of the two coherence times t_{coh} and t_{det}

$$E_{S,R/NR} = E_{S,R/NR}(t_{\text{coh}}, T, t_{\text{det}}).$$

3.7.2 Double Fourier transform and 2D spectra

After computing the time-domain polarization (**??**) and selecting the desired phase-tagged component via phase cycling (Eq. (3.31)), one obtains the corresponding complex signal field $E_{S,R/NR}(t_{\text{coh}}, T, t_{\text{det}})$ (Eq. (3.32)). For the non-perturbative numerical extraction, see Eq. (A.1) in Appendix A. The two-dimensional spectrum is then defined as the double Fourier transform of the time-domain signal with respect to $(t_{\text{coh}}, t_{\text{det}})$ at fixed T. The opposite sign in the t_{coh}-exponent distinguishes rephasing (photon-echo) and nonrephasing branches [5, 50].

In the numerical implementation, slowly varying complex envelopes are used (the optical carrier $e^{-i\omega_L t}$ is removed), so ω_{coh} and ω_{det} represent Fourier variables of these envelopes.

For the rephasing signal,

$$S_R(\omega_{\text{coh}}, T, \omega_{\text{det}}) = \int_{-\infty}^{\infty} \mathrm{d}t_{\text{det}} \int_{-\infty}^{\infty} \mathrm{d}t_{\text{coh}}\; \Theta(t_{\text{det}})\, \Theta(t_{\text{coh}})\, e^{+i\omega_{\text{coh}} t_{\text{coh}}}\, e^{-i\omega_{\text{det}} t_{\text{det}}}\, E_{S,R}(t_{\text{coh}}, T, t_{\text{det}}),$$

while for the nonrephasing signal,

$$S_{NR}(\omega_{\text{coh}}, T, \omega_{\text{det}}) = \int_{-\infty}^{\infty} \mathrm{d}t_{\text{det}} \int_{-\infty}^{\infty} \mathrm{d}t_{\text{coh}}\; \Theta(t_{\text{det}})\, \Theta(t_{\text{coh}})\, e^{-i\omega_{\text{coh}} t_{\text{coh}}}\, e^{-i\omega_{\text{det}} t_{\text{det}}}\, E_{S,NR}(t_{\text{coh}}, T, t_{\text{det}}).$$

In numerical implementations, finite integration limits (discrete time grids) are used and discrete Fourier transforms are applied with appropriate windowing. On uniform grids these are evaluated efficiently via (shifted) fast Fourier transforms [5, 50].

The *absorptive* 2D spectrum is then obtained as the real part of the sum

$$S_{\text{abs}}(\omega_{\text{coh}}, T, \omega_{\text{det}}) = \mathrm{Re}\left\{ S_R(\omega_{\text{coh}}, T, \omega_{\text{det}}) + S_{NR}(\omega_{\text{coh}}, T, \omega_{\text{det}}) \right\}.$$

This combination removes dispersive distortions that appear when considering $|S_R|$ or $|S_{NR}|$ alone and yields peak shapes directly comparable with experimental absorptive 2D spectra [4, 52].

Chapter 4

Model Systems

Having established the spectroscopy workflow in Chapter 3 and the open-system dynamics tools in Chapter 2, the concrete model Hamiltonians and dipole operators used in the numerical simulations are now specified. The discussion proceeds from minimal reference systems to more structured ensembles:

- a **qubit** as the simplest benchmark for rephasing/nonrephasing signals.
- a **dimer** (two coupled sites) with a coherent coupling constant J.
- an N**-site ensemble** with a prescribed geometry, simulated in the *ground*, *single-excitation*, and *double-excitation* manifolds $|g\rangle, \{|n\rangle\}, \{|f\rangle\}$ as introduced in Chapter 3.

Throughout, standard third-order spectroscopy concepts [3–5] and the master-equation propagation methods from Chapter 2 are used.

4.1 Qubit (Reference Model)

The simplest nontrivial benchmark is a qubit, i.e. a two-level quantum system with ground state $|g\rangle$ and excited state $|e\rangle$.

Although a strict two-level model cannot capture excited-state absorption into a separate higher manifold, it is a clean test case for the basic workflow (pulse sequence → induced polarization → rephasing/nonrephasing signals and 2D spectra) and for disentangling homogeneous dephasing, population relaxation, and inhomogeneous broadening.

4.1.1 Hamiltonian and dipole operator

The field-free system Hamiltonian is taken as

$$\hat{H}_{\mathrm{S}}^{(\mathrm{TLS})} = \omega_{eg} \, |e\rangle\langle e| \,, \tag{4.1}$$

with transition frequency ω_{eg}. The optical dipole operator (projected onto the chosen polarization direction) is written as

$$\hat{\mu}_{\mathrm{S}}^{(\mathrm{TLS})} = \mu_{ge} \, |g\rangle\langle e| + \mu_{eg} \, |e\rangle\langle g| \,, \qquad \mu_{eg} = \mu_{ge}^{*}. \tag{4.2}$$

4.1.2 System–bath coupling types

For a two-level system it is instructive to distinguish two types of system–bath coupling: *longitudinal* (or *pure-dephasing*) couplings are proportional to σ_z and commute with the bare qubit Hamiltonian in Eq. (4.1), so they do not induce transitions between energy eigenstates but instead cause the energy splitting to fluctuate in time, leading to decay of coherences without population relaxation (T_ϕ). In contrast, *transverse* couplings (e.g., σ_x, σ_y) can induce transitions and drive population relaxation (T_1).

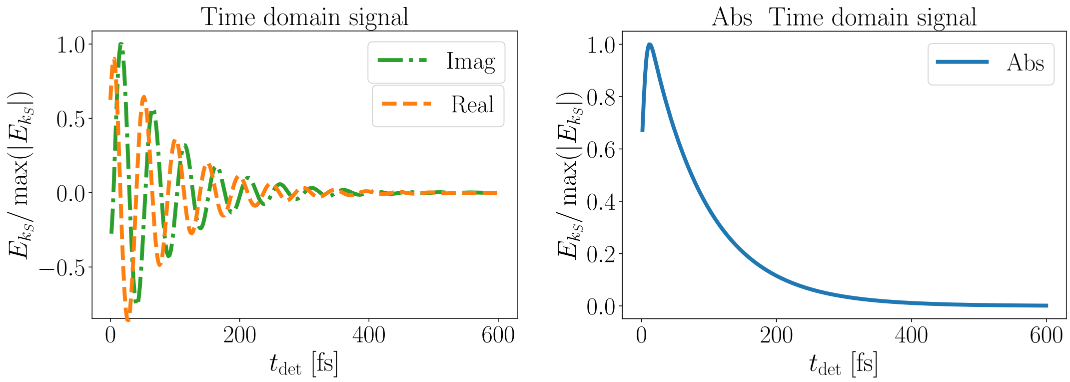

Figure 4.1: Time-domain three-pulse photon-echo (rephasing) signal of a two-level system computed from the RWA equations of motion, Eq. (4.3)–Eq. (4.5), with the paper parameters ($\tau = 300\,\mathrm{fs}$, $T = 1000\,\mathrm{fs}$, $\gamma^{-1} = 300\,\mathrm{fs}$, $T_2^* = 100\,\mathrm{fs}$, $\tau_{\mathrm{pulse}} = 15\,\mathrm{fs}$, $\omega_{eg} = 16000\,\mathrm{cm}^{-1}$, and inhomogeneous FWHM $\Delta = 200\,\mathrm{cm}^{-1}$) at resonance $\omega_L = \omega_{eg}$. This reproduces the reference result of [53].

4.1.3 Open-system dynamics and broadening scenarios

The reduced dynamics are generated by the open-system Liouvillian introduced in Chapter 2. In practice, either a Redfield-type generator or (when appropriate) a Lindblad form is used.

The qubit is examined under two broadening scenarios:

- **Homogeneous broadening**: a single transition frequency ω_{eg} with dephasing/relaxation entirely due to the bath.
- **Inhomogeneous broadening**: an ensemble of realizations with slightly different ω_{eg} values (static disorder), and spectra obtained by averaging the resulting signals as discussed in Chapter 3.

4.1.4 Results: rephasing signal and qubit 2D spectra

To illustrate the meaning of rephasing and the difference between homogeneous and inhomogeneous broadening, (i) a representative time-domain rephasing trace and (ii) sequences of TLS 2D spectra at different waiting times T are included. The figures are inserted below as placeholders. If the image files are not present yet, compilation will fall back to a framed box.

To validate the numerical workflow, a simpler phenomenological Lindblad-type evolution is first reproduced using the standard RWA equations of motion for the density-matrix elements. With the ansatz $\rho_{eg} = \sigma_{eg} e^{-i\Omega t}$, the resulting equations are

$$\dot{\sigma}_{eg} = -\big(\Gamma + i\Delta\big)\,\sigma_{eg} + i\,\mu_{eg}E(t)\,(\rho_{gg} - \rho_{ee}), \tag{4.3}$$
$$\dot{\rho}_{ee} = -\gamma\,\rho_{ee} + i\,\mu_{eg}E(t)\,\sigma_{ge} - i\,\mu_{eg}E^*(t)\,\sigma_{eg}, \tag{4.4}$$
$$\dot{\rho}_{gg} = -i\,\mu_{eg}E(t)\,\sigma_{ge} + i\,\mu_{eg}E^*(t)\,\sigma_{eg}. \tag{4.5}$$

with detuning $\Delta = \omega_{eg} - \Omega$. This is for a resonant coupling to the field ($\Delta = 0$) exactly the optical Bloch equations as presented in [53]. It uses phenomenological decay rates Γ and γ for the coherence and population, respectively. By choosing $\Gamma = \gamma/2 + 1/T_2^*$, the pure-dephasing contribution T_2^* can be separated from the population-relaxation contribution $\gamma/2$ to the total dephasing rate. In the reproduction shown in Fig. 4.1, the parameters match the paper and the polarization is sampled every 2 fs for τ and t in $[0, 600]$ fs. The field intensity is chosen so that the excited-state population remains below 1% ($E_0 = 0.05$), ensuring higher-order nonlinearities remain negligible.

The homogeneous case is shown in Fig. 4.2.

The inhomogeneous case is shown in Fig. 4.3.

Placeholder: qubit 2D spectra vs waiting time T (homogeneous broadening).
Put the file at `latex/figures/tls_2d_spectra_homogeneous.pdf`.

Figure 4.2: Sequence of rephasing/nonrephasing (or absorptive) 2D spectra of a homogeneously broadened qubit for increasing waiting time T. Solid/dashed contours indicate positive/negative contributions as in standard 2D-FTS conventions. (This placeholder should be replaced by the corresponding figure set.)

Placeholder: qubit 2D spectra vs waiting time T (inhomogeneous broadening).
Put the file at `latex/figures/tls_2d_spectra_inhomogeneous.pdf`.

Figure 4.3: Sequence of 2D spectra of an inhomogeneously broadened qubit ensemble for increasing waiting time T. Compared to Fig. 4.2, static disorder produces an elongated diagonal feature, while homogeneous dynamics govern the antidiagonal width. (This placeholder should be replaced by the corresponding figure set.)

4.2 Dimer (Two Coupled Sites)

As the next step, a dimer is considered: two optically active sites with a possible coherent coupling constant J. This is the minimal model that can produce cross peaks and coupling-induced spectral structure in 2D spectra, and it is a natural bridge to larger ensembles.

4.2.1 Manifolds and basis states

The standard excitation-manifold basis is used. The ground state is $|g\rangle$. The singly excited manifold is spanned by two localized excitations $|1\rangle$ and $|2\rangle$ ("site basis"), and the doubly excited state is $|f\rangle \equiv |12\rangle$. This matches the $|g\rangle, |n\rangle, |f\rangle$ scheme discussed in Chapter 3 with $n \in \{1, 2\}$.

4.2.2 Hamiltonian with coupling J

In the single-excitation manifold, the Hamiltonian is taken as

$$\hat{H}_{\mathrm{S}}^{(\mathrm{dimer})} = \varepsilon_1\, |1\rangle\langle 1| + \varepsilon_2\, |2\rangle\langle 2| + J\left(|1\rangle\langle 2| + |2\rangle\langle 1|\right) \;+\; (\varepsilon_1 + \varepsilon_2)\, |f\rangle\langle f|\,. \tag{4.6}$$

Setting $J = 0$ recovers two uncoupled transitions. Finite J produces excitonic eigenstates and characteristic peak splittings/cross peaks in 2D spectra.

4.2.3 Exciton basis (useful limiting picture)

For interpretation of spectra it is often useful to diagonalize the single-excitation block. Introducing the site-energy difference $\Delta\varepsilon = \varepsilon_1 - \varepsilon_2$, the two single-exciton eigenenergies are

$$\varepsilon_\pm = \tfrac{1}{2}(\varepsilon_1 + \varepsilon_2) \pm \sqrt{\left(\tfrac{\Delta\varepsilon}{2}\right)^2 + J^2}. \tag{4.7}$$

The corresponding eigenstates $|\pm\rangle$ interpolate between localized excitations (for $|J| \ll |\Delta\varepsilon|$) and fully delocalized superpositions (for $|J| \gg |\Delta\varepsilon|$). In the 2D spectrum this coherent mixing is visible already at waiting time $T = 0$ through peak splittings and, generically, non-vanishing off-diagonal structure.

4.2.4 Dipole operator

The dipole operator is written as the sum over allowed one-quantum transitions,

$$\hat{\mu}_{\mathrm{S}}^{(\mathrm{dimer})} = \sum_{n=1}^{2} \Big(\mu_{gn} \; |g\rangle\langle n| + \mu_{ng} \; |n\rangle\langle g| \Big) + \sum_{n=1}^{2} \Big(\mu_{nf} \; |n\rangle\langle f| + \mu_{fn} \; |f\rangle\langle n| \Big), \tag{4.8}$$

with $\mu_{ng} = \mu_{gn}^*$ and $\mu_{fn} = \mu_{nf}^*$. The $|n\rangle \leftrightarrow |f\rangle$ transitions are required to include excited-state absorption pathways in third-order response.

The specific dimer parameter choices and the figures to be shown for the dimer will be specified later.

4.2.5 Physical interpretation in 2D spectra: coupling vs. transfer

This dimer is the minimal setting in which the 2D spectrum can distinguish qualitatively different "excitation-transfer" mechanisms:

- **Coherent coupling (excitonic mixing).** For $J \neq 0$ the optical transitions are between $|g\rangle$ and the exciton states $|\pm\rangle$ (and, if included, between $|\pm\rangle$ and $|f\rangle$). In this case cross peaks can be present already at $T = 0$ because the eigenstates are delocalized and the Liouville pathways do not cancel exactly.

- **Incoherent population transfer.** If the sites are only weakly coupled (or even uncoupled) but the environment induces population transfer $|1\rangle \to |2\rangle$ (or both directions), then cross peaks are typically *absent at* $T = 0$ and *grow with* T as population migrates during the waiting time. In this scenario diagonal peak amplitudes decay while the corresponding cross peak rises on the same timescale.

This "$T = 0$ cross peak" versus "growth with T" distinction is a convenient rule of thumb when interpreting simulated (or measured) spectra.

4.2.6 transfer model during the waiting time

To model unidirectional or bidirectional excitation transfer within the single-excitation manifold during the waiting interval T, the coherent Hamiltonian evolution may be complemented by Lindblad jump operators of the form

$$\hat{L}_{2\leftarrow 1} = \sqrt{k_{2\leftarrow 1}} \; |2\rangle\langle 1| \,, \qquad \hat{L}_{1\leftarrow 2} = \sqrt{k_{1\leftarrow 2}} \; |1\rangle\langle 2| \,, \tag{4.9}$$

in addition to site-local relaxation/dephasing operators (Chapter 2). In the numerical workflow this affects the evolution superoperator acting between the second and third interactions and therefore controls the waiting-time dependence of diagonal and cross-peak amplitudes.

4.2.7 Coherences during T (oscillatory peak dynamics)

If the system supports excited-state coherences during the waiting time (e.g. between $|+\rangle$ and $|-\rangle$), then selected peak amplitudes can acquire oscillatory T-dependence at the exciton splitting frequency (with a decay set by the corresponding dephasing rate). In practice, this provides a direct numerical handle to separate "population transfer" signatures (monotonic rise/decay) from coherence signatures (beats) in the same model.

4.3 Generalization to an N-Site Ensemble with Geometry

Finally, an ensemble of N optically active sites ("atoms" or chromophores) placed in a prescribed geometry is considered. Motivated by protofilament arrangements in microtubules, the geometry of interest in this thesis is a discretized cylindrical surface [10]. The model is formulated in the excitation-manifold basis $|g\rangle$, $\{|n\rangle\}, \{|f\rangle\}$ already used in Chapter 3.

4.3.1 States and truncation

The ground state is $|g\rangle$. The single-excitation manifold is spanned by states $|n\rangle$, $n \in \{1, \ldots, N\}$, which represent one excitation localized at site n (or, after diagonalization, an excitonic eigenstate). If double excitations are included, states $|f\rangle \equiv |mn\rangle$ with $1 \leq m < n \leq N$ are used, representing two simultaneous excitations. In this thesis the Hilbert space is restricted to the ground state plus the single-excitation manifold, or (when needed for ESA contributions) to ground + single + double excitations.

4.3.2 What 2D spectra diagnose in ensembles

In an N-site setting, 2D spectra are particularly useful because they separate several effects that are difficult to disentangle in linear absorption:

- **Static disorder (inhomogeneous broadening)** produces elongation along the diagonal in the $(\omega_{\text{coh}}, \omega_{\text{det}})$ plane, reflecting a distribution of transition frequencies across realizations.
- **Homogeneous dephasing** controls the antidiagonal width (for fixed disorder), i.e. he width perpendicular to the diagonal, and therefore provides access to dephasing timescales even in disordered ensembles.
- **Transport and transfer** redistribute amplitude between diagonal and off-diagonal features as a function of T. The pattern of emerging cross peaks encodes which states (or spatial regions) become correlated during the waiting time.

These qualitative diagnostics motivate the use of 2D-FTS observables as a readout for geometry-dependent excitation migration in the cylindrical architectures studied below.

4.3.3 Hamiltonian

In this basis, a generic Frenkel-exciton-type Hamiltonian reads

$$\hat{H}_{\text{S}}^{(N)} = \sum_{n=1}^{N} \varepsilon_n \; |n\rangle\langle n| + \sum_{m<n}^{N} J_{mn} \left(|m\rangle\langle n| + |n\rangle\langle m| \right) + \sum_{m<n}^{N} (\varepsilon_m + \varepsilon_n) \; |mn\rangle\langle mn| + \cdots \,, \tag{4.10}$$

where ε_n are site transition frequencies and J_{mn} are coherent couplings. The ellipsis indicates additional terms in the double-excitation manifold if one wishes to include interactions beyond an additive two-exciton energy (e.g. exciton–exciton shifts).

4.3.4 Geometry and couplings

The couplings J_{mn} may be parameterized phenomenologically (e.g. nearest-neighbour) or computed from transition-dipole interactions. In the latter case one uses

$$J_{mn} \propto \frac{1}{\|\boldsymbol{r}_{mn}\|^3} \left[\boldsymbol{\mu}_m \cdot \boldsymbol{\mu}_n - 3 \, (\boldsymbol{\mu}_m \cdot \hat{\boldsymbol{r}}_{mn})(\boldsymbol{\mu}_n \cdot \hat{\boldsymbol{r}}_{mn}) \right] , \tag{4.11}$$

with site positions $\boldsymbol{r}_n$ and displacement $\boldsymbol{r}_{mn} = \boldsymbol{r}_m - \boldsymbol{r}_n$ [54–56].

For a cylindrical arrangement, let N_θ be the number of sites per ring and N_z the number of rings (so $N = N_\theta N_z$), with sites indexed by an azimuthal index and an axial index. Periodic boundary conditions can be imposed in the azimuthal direction.

4.3.5 Dipole operator

The dipole operator is the sum over allowed one-quantum transitions between adjacent excitation manifolds,

$$\hat{\mu}_{\mathrm{S}}^{(N)} = \sum_{n=1}^{N} \Big(\mu_{gn} \; |g\rangle\langle n| + \mu_{ng} \; |n\rangle\langle g| \Big) + \sum_{m<n}^{N} \sum_{\ell \in \{m,n\}} \Big(\mu_{\ell,mn} \; |\ell\rangle\langle mn| + \mu_{mn,\ell} \; |mn\rangle\langle \ell| \Big), \tag{4.12}$$

where the second sum is only needed if the double-excitation manifold is included.

This model class captures collective transport pathways and their photon-echo signatures, offering a route to connect microscopic couplings and geometry to observable 2D spectra [10].

Appendix A

Numerical Implementation

This chapter presents the computational framework developed for simulating one- and two-dimensional electronic spectroscopy of open quantum systems.

The implementation is built around the `qspectro2d` Python package, which provides a modular, configuration-driven workflow that bridges the theoretical framework established in previous chapters with practical computational methods for investigating quantum coherence phenomena in complex molecular systems. The open quantum system dynamics are solved using `QuTiP` [25].

A.1 Software Architecture and Design Philosophy

The `qspectro2d` package follows a modular design that separates concerns across distinct functional domains. The architecture enables efficiency through parallel processing and flexible configuration management. The core design is built around a **configuration-driven workflow** where YAML-based parameter specification decouples physical models from computational implementation. This approach ensures that **modular components** including atomic systems, laser pulses, environmental baths, and spectroscopic calculations remain independently configurable. The framework provides **scalable execution** with support for both local testing and high-performance computing (HPC) batch processing, complemented by **comprehensive post-processing** through automated workflows for averaging, Fourier transforms, and visualization.

The package structure comprises five main submodules: `core` (fundamental system components), `spectroscopy` (calculation engines), `config` (parameter management), `utils` (data handling utilities), and `visualization` (plotting tools).

A.2 Configuration-Driven Simulation Workflow

The simulation framework employs YAML configuration files to specify all physical and computational parameters, enabling reproducible simulations. A typical configuration defines the **atomic system** through the number of atoms, transition frequencies, geometric arrangement, and inhomogeneous broadening parameters. **Laser parameters** encompass pulse amplitudes, durations, carrier frequencies, and rotating wave approximation settings, while **environmental coupling** is characterized by bath type (Ohmic, Drude-Lorentz), temperature, coupling strength, and cutoff frequencies. Finally, **computational settings** specify the ODE solver choice (Lindblad vs. Bloch-Redfield), signal types, and time windows.

The workflow consists of three main execution phases: simulation (`calc_datas.py`), post-processing (`process_datas.py`), and visualization (`plot_datas.py`). This separation enables efficient parameter studies where computationally expensive simulations are performed once, followed by rapid exploration of different analysis and visualization options.

A.3 Third-Order Polarization and Phase Cycling

In the numerical implementation, the electric field for multiple pulses is modeled following the formalism established in Chapter 3. The total electric field is expressed as

$$\boldsymbol{E}(\boldsymbol{r}, t) = \sum_{m=1}^{3} \boldsymbol{E}_m(\boldsymbol{r}, t) + \text{c.c.},$$

where c.c. denotes the complex conjugate, and

$$\boldsymbol{E}_m(\boldsymbol{r}, t) = \boldsymbol{\chi}_m E'_m(t - \tau_m) \exp(-i(\omega_m t + \varphi_m)),$$

with frequency $\omega_m = 2\pi\nu_m$, wavevector $\boldsymbol{k}_m$, and electric field amplitude $\boldsymbol{\chi}_m$. The envelope $E'_m(t - \tau_m)$, centered at τ_m, is assumed to be Gaussian with FWHM τ_p

$$E'_m(t - \tau_m) = \exp\left(-\frac{4\ln 2(t - \tau_m)^2}{\tau_p^2}\right).$$

For numerical efficiency, the Gaussian pulse is truncated to zero once the instantaneous intensity $|E'_m(t)|^2$ falls below 10^{-4} of its peak value (approximately two FWHM from the pulse center). In these field-free intervals, the evolution is propagated with the time-independent Liouvillian, which is computationally more efficient. The core spectroscopic calculation implements phase-cycled third-order polarization measurement following the established four-wave mixing protocols detailed in Chapter 3, Sec. 3.6 [43, 50].

This thesis instead uses a more microscopic, non-perturbative (NP) approach: the full (time-dependent) equation of motion is solved with the pulses included and the induced polarization is computed via **??**. To isolate the nonlinear signal in the time domain, auxiliary runs are performed in which only a subset of pulses is active, and the lower-order (linear) contributions are subtracted from the full signal. In the weak-field regime and for centrosymmetric media (where $\chi^{(2)}$ vanishes), this yields a background-free polarization that is dominated by the desired third-order response,

$$\boldsymbol{P}^{(3)}(t) \approx \boldsymbol{P}_{\text{total}}(t) - \sum_{m=1}^{3} \boldsymbol{P}_{(m)}(t), \tag{A.1}$$

where $\boldsymbol{P}_{\text{total}}(t)$ is obtained with all three pulses present and $\boldsymbol{P}_{(m)}(t)$ denotes the polarization obtained when only pulse m is applied. In practice, the complex analytic (positive-frequency) component of the polarization is used to retain amplitude and phase information for subsequent phase cycling and signal construction.

A.4 Open Quantum System Evolution

The quantum system evolution is handled by the `SimulationModuleOQS` class, which provides a unified interface for different solver backends implementing the open quantum system dynamics developed in Chapter 2. The implementation supports both Lindblad master equation evolution (via QuTiP's `mesolve`) and Bloch-Redfield dynamics (`brmesolve`) with automatic handling of pulse overlap and time-dependent Hamiltonians.

For time-dependent pulse sequences, the total Hamiltonian takes the form established in the spectroscopy theory (Chapter 3):

$$H(t) = H_0 + \sum_i H_{\text{int},i}(t), \tag{A.2}$$

where H_0 is the bare system Hamiltonian and $H_{\text{int},i}(t)$ represents the interaction with pulse i, following the light-matter interaction formalism detailed in Chapter 3. The solver automatically detects pulse overlap regions and constructs appropriate piecewise evolution operators, ensuring accurate treatment of multi-pulse sequences regardless of timing complexity.

A.5 Inhomogeneous Broadening and Statistical Averaging

Real molecular systems exhibit inhomogeneous broadening due to environmental variations, as discussed in the spectroscopy fundamentals (Chapter 3). The implementation models this through Gaussian-distributed transition frequencies following the theoretical treatment:

$$p(\omega) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(\omega-\omega_0)^2}{2\sigma^2}\right), \tag{A.3}$$

where $\sigma = \Delta_{\text{FWHM}}/(2\sqrt{2\ln 2})$ relates the standard deviation to the experimentally relevant full width at half maximum, consistent with the broadening models established in Chapter 3. The sampling employs a rejection algorithm optimized for computational efficiency while maintaining statistical accuracy.

A.6 Parallel Processing and Scalable Execution

The framework supports both local execution for testing and HPC cluster deployment for large parameter studies. The local workflow (`calc_datas.py`) generates all combinations of coherence times and inhomogeneous samples, while the HPC workflow (`hpc_batch_dispatch.py`) distributes individual parameter combinations across compute nodes.

For each combination of parameters, the averaged response is calculated following the inhomogeneous averaging procedure established in the spectroscopy theory (Chapter 3):

$$\langle E_{k_s}(t_{\text{coh}}, t_{\text{det}})\rangle = \frac{1}{N_{\text{inhom}}} \sum_{i=1}^{N_{\text{inhom}}} E_{k_s}^{(i)}(t_{\text{coh}}, t_{\text{det}}), \tag{A.4}$$

where N_{inhom} represents the number of inhomogeneous samples as defined in the broadening treatment (Chapter 3). The implementation utilizes Python's `ProcessPoolExecutor` for local parallel execution and integrates with SLURM-based job scheduling systems for HPC deployment, enabling efficient scaling from desktop testing to production-scale parameter sweeps.

A.7 Post-Processing and Spectral Analysis

The post-processing pipeline (`process_datas.py`) handles the conversion from time-domain signals to 2D spectra. **Data aggregation** groups individual parameter combination files by inhomogeneous sample and stacks them along coherence time dimensions. **Zero-padding** extends the time-domain data to improve frequency resolution according to:

$$E_{\text{extended}}(t) = \begin{cases} E(t) & \text{for } t \in [0, T_{\text{orig}}] \\ 0 & \text{for } t \in (T_{\text{orig}}, T_{\text{extended}}] \end{cases}. \tag{A.5}$$

Subsequently, **Fourier transformation** converts the data to frequency domain. For the rephasing signal this is implemented as the (two-dimensional) Fourier transform over the coherence and detection times, In practice, windowing trades spectral resolution for reduced ringing, while zero-padding improves visual interpolation without adding new information.

$$S_R(\omega_1, T, \omega_3) \;=\; \int_0^\infty \mathrm{d}t_1 \int_0^\infty \mathrm{d}t_3\; e^{+i\omega_1 t_1}\, e^{+i\omega_3 t_3}\, S_R(t_1, T, t_3), \tag{A.6}$$

with the specific sign conventions matching those used in the spectroscopy chapter.

The frequency axes are expressed in wavenumber units (10^4 cm^{-1}) for direct comparison with experimental literature.

A.8 Model System Implementation

The `AtomicSystem` class provides flexible construction of multi-atom quantum systems with configurable geometries. For the microtubule-inspired models central to this thesis, the implementation supports **cylindrical arrangements** where atoms are positioned on cylindrical surfaces with specified chain and ring numbers. **Excitation manifold truncation** restricts the system to single and double excitation subspaces with automatic basis construction, while **dynamic coupling** enables nearest-neighbor and long-range interactions based on geometric proximity. The framework also incorporates **inhomogeneous disorder** through site-specific frequency variations sampled from realistic distributions.

The system construction automatically handles basis transformations between site and excitation manifold representations, enabling efficient evolution in the appropriate eigenbasis while maintaining physical interpretability of results.

Appendix B

Maxwell-to-Signal Derivation (Field Side)

This appendix presents the *field-side* derivation for coherent 2D-FTS/2DES, showing how Maxwell's equations lead to a heterodyne-detected signal field linear in the third-order polarization $\boldsymbol{P}^{(3)}$. It justifies the proportionality used in Sec. 3.3.

The logic is: (i) Maxwell $\Rightarrow$ a driven wave equation with $\boldsymbol{P}$ as source; (ii) phase matching and SVEA $\Rightarrow$ the emitted field in a chosen direction is proportional to the corresponding component of $\boldsymbol{P}^{(3)}$; (iii) the microscopic (matter-side) expression for $\boldsymbol{P}^{(3)}$ is derived in Appendix C.

B.1 From Maxwell's equations to a wave equation for the radiation field

In this section it is made explicit (in the same spirit as the lecture notes)

IMPORTANT: i have to actually cite the lecture

why the macroscopic polarization is the source of the emitted optical field. The key conceptual point is that only the *transverse* (radiation) part of the electromagnetic field propagates as a wave, and it is sourced by the transverse current. In condensed-matter optics, that current is conveniently written in terms of the polarization.

B.1.1 Maxwell's equations and electromagnetic potentials

The fundamental laws are introduced first. In SI units, Maxwell's equations in the presence of charge and current densities $\varrho(\boldsymbol{r}, t)$ and $\boldsymbol{j}(\boldsymbol{r}, t)$ read

$$\boldsymbol{\nabla} \times \boldsymbol{B}(\boldsymbol{r}, t) = \mu_0 \boldsymbol{j}(\boldsymbol{r}, t) + \mu_0 \varepsilon_0 \frac{\partial \boldsymbol{E}(\boldsymbol{r}, t)}{\partial t}, \tag{B.1}$$

$$\boldsymbol{\nabla} \times \boldsymbol{E}(\boldsymbol{r}, t) = -\frac{\partial \boldsymbol{B}(\boldsymbol{r}, t)}{\partial t}, \tag{B.2}$$

$$\boldsymbol{\nabla} \cdot \boldsymbol{E}(\boldsymbol{r}, t) = \frac{\varrho(\boldsymbol{r}, t)}{\varepsilon_0}, \tag{B.3}$$

$$\boldsymbol{\nabla} \cdot \boldsymbol{B}(\boldsymbol{r}, t) = 0. \tag{B.4}$$

Here $\boldsymbol{E}$ is the electric field, $\boldsymbol{B}$ the magnetic field and the constants satisfy $c^2 = 1/(\mu_0 \varepsilon_0)$.

It is often convenient to express $\mathbf{E}$ and $\mathbf{B}$ in terms of a vector potential $\mathbf{A}(\mathbf{r}, t)$ and a scalar potential $\phi(\mathbf{r}, t)$ by

$$\boldsymbol{B} = \boldsymbol{\nabla} \times \boldsymbol{A}, \qquad \boldsymbol{E} = -\frac{\partial \boldsymbol{A}}{\partial t} - \boldsymbol{\nabla}\phi. \tag{B.5}$$

This reformulation automatically satisfies Faraday's law Eq. (B.2) and the magnetic Gauss law Eq. (B.4), eliminating two equations and simplifying the system.

B.1.2 Gauge freedom and Coulomb gauge

To proceed from the potential formulation, a gauge must be chosen. While the potentials reformulation of Eq. (B.5) is useful, the potentials are not unique: for any scalar function $\chi(\boldsymbol{r}, t)$,

$$\boldsymbol{A}' = \boldsymbol{A} + \boldsymbol{\nabla}\chi, \qquad \phi' = \phi - \frac{\partial \chi}{\partial t}, \tag{B.6}$$

leaves $\boldsymbol{E}$ and $\boldsymbol{B}$ invariant. To obtain a unique solution this gauge freedom must be fixed. For optics, the Coulomb gauge is chosen

$$\boldsymbol{\nabla} \cdot \boldsymbol{A} = 0.$$

With this gauge choice, inserting the potential definitions Eq. (B.5) into electric Gauss' law Eq. (B.3) gives the Poisson equation

$$\boldsymbol{\nabla}^2 \phi(\boldsymbol{r}, t) = -\frac{\varrho(\boldsymbol{r}, t)}{\varepsilon_0}, \tag{B.7}$$

which determines ϕ instantaneously in space. This solution does *not* violate causality—it is a gauge artifact that describes the instantaneous Coulomb interaction. The causal, propagating part of the optical field is contained in the transverse (vector potential) components discussed next.

B.1.3 Transverse/longitudinal decomposition

To separate the propagating (radiative) part of the field from the instantaneous Coulomb part, the fields are decomposed into transverse and longitudinal components. Only the transverse component carries radiative degrees of freedom.

Formally, any sufficiently well-behaved vector field $\boldsymbol{F}(\boldsymbol{r})$ admits the Helmholtz decomposition into irrotational and solenoidal parts:

$$\boldsymbol{F} = \boldsymbol{F}^{\perp} + \boldsymbol{F}^{\parallel}, \tag{B.8}$$

where the transverse part is divergence-free and the longitudinal part is curl-free

$$\boldsymbol{\nabla} \cdot \boldsymbol{F}^{\perp} = 0, \qquad \boldsymbol{\nabla} \times \boldsymbol{F}^{\parallel} = 0. \tag{B.9}$$

Applying this decomposition to the electric field from Eq. (B.5),

$$\boldsymbol{E} = \boldsymbol{E}^{\perp} + \boldsymbol{E}^{\parallel}, \tag{B.10}$$

where $\boldsymbol{E}^{\perp} = -\partial_t \boldsymbol{A}$ (using Coulomb gauge) and $\boldsymbol{E}^{\parallel} = -\boldsymbol{\nabla}\phi$. Under this decomposition, Maxwell's equations Eqs. (B.1) to (B.4) split into independent transverse and longitudinal parts:

$$\boldsymbol{\nabla} \times \boldsymbol{E}^{\perp} = -\frac{\partial \boldsymbol{B}}{\partial t}, \qquad \boldsymbol{\nabla} \times \boldsymbol{E}^{\parallel} = 0, \tag{B.11}$$

$$\boldsymbol{\nabla} \cdot \boldsymbol{E}^{\parallel} = \frac{\varrho}{\varepsilon_0}, \qquad \boldsymbol{\nabla} \cdot \boldsymbol{E}^{\perp} = 0. \tag{B.12}$$

Faraday's law only constrains the transverse field because $\boldsymbol{\nabla} \times \boldsymbol{E}^{\parallel} = 0$ identically, while Gauss' law fixes the longitudinal field through the charge density. Thus, the propagating and Coulomb degrees of freedom are fully separated.

To understand which current sources what, recall that charge and current are not independent: they are linked by the continuity equation

$$\frac{\partial \varrho(\boldsymbol{r}, t)}{\partial t} + \boldsymbol{\nabla} \cdot \boldsymbol{j}(\boldsymbol{r}, t) = 0. \tag{B.13}$$

Decomposing $\boldsymbol{j} = \boldsymbol{j}^{\perp} + \boldsymbol{j}^{\parallel}$ and using $\boldsymbol{\nabla} \cdot \boldsymbol{j}^{\perp} = 0$ shows that Eq. (B.13) constrains only the longitudinal part:

$$\frac{\partial \varrho}{\partial t} + \boldsymbol{\nabla} \cdot \boldsymbol{j}^{\parallel} = 0. \tag{B.14}$$

This is the precise sense in which $\boldsymbol{j}^{\parallel}$ is tied to instantaneous Coulomb (near-field) physics, while radiative fields are sourced by $\boldsymbol{j}^{\perp}$.

Two immediate consequences are noted:

- Because $\boldsymbol{\nabla} \cdot \boldsymbol{B} = 0$, the magnetic field is purely transverse: $\boldsymbol{B} = \boldsymbol{B}^{\perp}$.
- In Coulomb gauge $\boldsymbol{\nabla} \cdot \boldsymbol{A} = 0$, the vector potential is purely transverse: $\boldsymbol{A} = \boldsymbol{A}^{\perp}$.

Combining the decomposition with the Coulomb gauge choice and Eq. (B.5), one identifies

$$\boldsymbol{E}^{\perp} = -\frac{\partial \boldsymbol{A}}{\partial t}, \qquad \boldsymbol{E}^{\parallel} = -\boldsymbol{\nabla}\phi. \tag{B.15}$$

Crucially, the radiation field relevant for propagating optics is $\boldsymbol{E}^{\perp}$. The longitudinal field $\boldsymbol{E}^{\parallel}$ is determined instantaneously by ϱ via the Poisson equation and does not represent a propagating degree of freedom.

B.1.4 Wave equation for the transverse electric field

With the transverse (radiative) field component isolated, the wave equation it satisfies is derived. Starting from Faraday's law Eq. (B.2), the curl is taken on both sides:

$$\boldsymbol{\nabla} \times (\boldsymbol{\nabla} \times \boldsymbol{E}) = -\frac{\partial}{\partial t}(\boldsymbol{\nabla} \times \boldsymbol{B}). \tag{B.16}$$

Next, eliminate $\boldsymbol{\nabla} \times \boldsymbol{B}$ using Ampère–Maxwell's law Eq. (B.1) and use the relation $\mu_0\varepsilon_0 = 1/c^2$ to obtain

$$-\boldsymbol{\nabla} \times (\boldsymbol{\nabla} \times \boldsymbol{E}) = \frac{1}{c^2}\frac{\partial^2 \boldsymbol{E}}{\partial t^2} + \mu_0 \frac{\partial \boldsymbol{j}}{\partial t}. \tag{B.17}$$

Only transverse parts contribute to the curl–curl operator because $\boldsymbol{\nabla} \times \boldsymbol{E}^{\parallel} = 0$. Likewise, the radiated field is sourced by the transverse current $\boldsymbol{j}^{\perp}$. Projection onto the transverse sector therefore gives

$$-\boldsymbol{\nabla} \times (\boldsymbol{\nabla} \times \boldsymbol{E}^{\perp}) = \frac{1}{c^2}\frac{\partial^2 \boldsymbol{E}^{\perp}}{\partial t^2} + \mu_0 \frac{\partial \boldsymbol{j}^{\perp}}{\partial t}. \tag{B.18}$$

Using the identity $\boldsymbol{\nabla} \times (\boldsymbol{\nabla} \times \boldsymbol{E}^{\perp}) = \boldsymbol{\nabla}(\boldsymbol{\nabla} \cdot \boldsymbol{E}^{\perp}) - \boldsymbol{\nabla}^2 \boldsymbol{E}^{\perp}$ and $\boldsymbol{\nabla} \cdot \boldsymbol{E}^{\perp} = 0$, one obtains the driven wave equation

$$\boldsymbol{\nabla}^2 \boldsymbol{E}^{\perp}(\boldsymbol{r}, t) - \frac{1}{c^2}\frac{\partial^2 \boldsymbol{E}^{\perp}(\boldsymbol{r}, t)}{\partial t^2} = \mu_0 \frac{\partial \boldsymbol{j}^{\perp}(\boldsymbol{r}, t)}{\partial t}. \tag{B.19}$$

B.1.5 From transverse current to polarization

The wave equation Eq. (B.19) is expressed in terms of the current $\boldsymbol{j}^{\perp}$. In condensed-matter optics, it is more convenient to work with the macroscopic polarization $\boldsymbol{P}$. On large scales, the movement of bound charges in a neutral dielectric material is described by the polarization density $\boldsymbol{P}(\boldsymbol{r}, t)$. When averaged over many particles, the current due to these bound charges equals the time derivative of the polarization,

$$\boldsymbol{j}(\boldsymbol{r}, t) = \frac{\partial \boldsymbol{P}(\boldsymbol{r}, t)}{\partial t}. \tag{B.20}$$

Together with the continuity equation Eq. (B.13), this implies the standard relation between bound charge density and polarization,

$$\varrho_b(\boldsymbol{r}, t) = -\boldsymbol{\nabla} \cdot \boldsymbol{P}(\boldsymbol{r}, t). \tag{B.21}$$

Substituting the polarization current Eq. (B.20) into Eq. (B.19) gives the wave equation in the practically useful polarization form:

$$\boldsymbol{\nabla}^2 \boldsymbol{E}^{\perp}(\boldsymbol{r}, t) - \frac{1}{c^2} \frac{\partial^2 \boldsymbol{E}^{\perp}(\boldsymbol{r}, t)}{\partial t^2} = \frac{1}{\varepsilon_0 c^2} \frac{\partial^2 \boldsymbol{P}(\boldsymbol{r}, t)}{\partial t^2}. \tag{B.22}$$

This is a crucial result: Equation (B.22) explicitly shows that the macroscopic polarization $\boldsymbol{P}$ is the source of the radiation field $\boldsymbol{E}^{\perp}$. This is the central Maxwell-side justification for spectroscopy: by measuring the optical field emitted from a material, one can extract information about the polarization induced by the incident light.

To apply this result to nonlinear spectroscopy, the polarization is decomposed as $\boldsymbol{P} = \boldsymbol{P}^{(1)} + \boldsymbol{P}_{\mathrm{NL}}$, where $\boldsymbol{P}_{\mathrm{NL}}$ contains the higher-order contributions (e.g., $\boldsymbol{P}^{(3)}$ for third-order 2D-FTS). The linear response $\boldsymbol{P}^{(1)}$ can be absorbed into an effective complex refractive index $n(\omega)$ defined by $n^2 = 1 + \chi^{(1)}$. With this redefinition, Eq. (B.22) becomes the nonlinear wave equation:

$$\boldsymbol{\nabla}^2 \boldsymbol{E}(\boldsymbol{r}, t) - \frac{n^2}{c^2} \frac{\partial^2}{\partial t^2} \boldsymbol{E}(\boldsymbol{r}, t) = \frac{1}{\varepsilon_0 c^2} \frac{\partial^2}{\partial t^2} \boldsymbol{P}_{\mathrm{NL}}(\boldsymbol{r}, t). \tag{B.23}$$

In a three-pulse coherent 2D experiment operating in the weak-field perturbative regime, only the lowest nonlinear order contributes significantly to the signal. Therefore, the relevant nonlinear source is (to leading order)

$$\boldsymbol{P}_{\mathrm{NL}}(\boldsymbol{r}, t) \approx \boldsymbol{P}^{(3)}(\boldsymbol{r}, t), \tag{B.24}$$

where $\boldsymbol{P}^{(3)}$ is cubic in the incident field.

The Maxwell-side source term has now been obtained. Two remaining tasks are: (i) connecting this source to a measurable signal field via the slowly-varying envelope approximation and phase matching, and (ii) providing the microscopic matter-side expression for $\boldsymbol{P}^{(3)}$ in terms of the incident fields.

IMPORTANT: Consider moving or expanding this phase-matching section (e.g., fold into Appendix B.3); the current placement is ad hoc.

B.2 Input fields, phase factors, and phase matching

Before proceeding to solve the wave equation Eq. (B.23), the form of the incident fields that drive the nonlinear response is specified. The incident field is represented as a sum of three pulses $j \in \{1, 2, 3\}$ with slowly varying envelopes. The carrier phases (phase kicks) ϕ_j are written explicitly for completeness. They are essential for phase cycling in collinear geometries (see Chapter 3, Sec. 3.6).

$$\boldsymbol{E}_{\mathrm{in}}(\boldsymbol{r}, t) = \sum_{j=1}^{3} \Big[\boldsymbol{E}_j(t - t_j)\, e^{i(\boldsymbol{k}_j \cdot \boldsymbol{r} - \omega_L t + \phi_j)} + \text{c.c.} \Big]. \tag{B.25}$$

In the perturbative regime, the third-order polarization $\boldsymbol{P}^{(3)}$ is cubic in the incident fields. When Eq. (B.25) is substituted into the expression for $\boldsymbol{P}^{(3)}$ (derived on the matter side), many different terms emerge with different plane-wave phase factors. Under the rotating-wave approximation (which assumes equal carrier frequencies), only the contributions oscillating near ω_L are retained. These correspond to having exactly one complex-conjugated field and two non-conjugated fields in the polarization, producing the third-order signal wavevector combinations

$$\boldsymbol{k}_S = \pm \boldsymbol{k}_1 \pm \boldsymbol{k}_2 \pm \boldsymbol{k}_3,$$

and a corresponding signed carrier-phase combination $\Phi = \pm\phi_1 \pm \phi_2 \pm \phi_3$.

Macroscopic emission from a finite sample thickness builds up coherently only when the polarization wave and the radiated wave remain approximately phase matched. This is the origin of the standard separation into signal directions. The two key third-order directions for 2D electronic spectroscopy are

$$\begin{aligned} \boldsymbol{k}_R &= -\boldsymbol{k}_1 + \boldsymbol{k}_2 + \boldsymbol{k}_3 \qquad &&\text{(rephasing / photon echo)}, \\ \boldsymbol{k}_{NR} &= +\boldsymbol{k}_1 - \boldsymbol{k}_2 + \boldsymbol{k}_3 \qquad &&\text{(nonrephasing / FID)}. \end{aligned}$$

B.3 Slowly varying envelope approximation and the emitted field

The driven wave equation Eq. (B.23) is solved to connect the nonlinear polarization to the observed signal field. The key is the slowly-varying envelope approximation (SVEA), which exploits the fact that the field envelopes vary slowly compared to the optical oscillation. Under SVEA, the emitted field in the phase-matched direction is proportional to the corresponding component of the nonlinear polarization.

A slab sample of thickness L along the z direction is considered, and a single phase-matched signal component with carrier frequency ω_L and wavevector propagating along z is selected. Starting from Eq. (B.23), the relevant Fourier component of the third-order polarization is extracted,

$$\boldsymbol{P}_{\mathrm{NL}}(\boldsymbol{r},t) \supset \boldsymbol{P}_S^{(3)}(t)\, e^{i(\boldsymbol{k}_S\cdot\boldsymbol{r}-\omega_L t)} + \text{c.c.} \tag{B.26}$$

where $\boldsymbol{P}_S^{(3)}(t)$ is the slowly-varying envelope. A corresponding forward-propagating field solution is sought in terms of an envelope,

$$\boldsymbol{E}_S(\boldsymbol{r},t) = \tfrac{1}{2}\,\boldsymbol{E}_S(z,t)\, e^{i(k'_S z-\omega_L t)} + \text{c.c.}, \tag{B.27}$$

that solves Eq. (B.23) with dispersion relation $k'_S = n(\omega_L)\omega_L/c$ (the phase velocity in the medium).

B.3.1 Slowly varying envelope approximation (SVEA)

Before solving Eq. (B.23), a key timescale separation is used. The SVEA assumes envelopes $\boldsymbol{E}_S(z,t)$, $\boldsymbol{P}_S^{(3)}(t)$ vary slowly compared to optical oscillations

$$|\partial_t \boldsymbol{E}_S| \ll \omega_L\ |\boldsymbol{E}_S|\,, \qquad \left|\partial_z^2 \boldsymbol{E}_S\right| \ll k'_S\ |\partial_z \boldsymbol{E}_S|\,. \tag{B.28}$$

This separation of timescales allows higher-order envelope derivatives to be dropped when they appear with fast oscillatory factors.

Spatial and temporal derivatives

To extract the slowly-varying envelope behavior, only the positive-frequency part is kept and the explicit time dependence is suppressed for the moment. Consider a positive-frequency component

$$\boldsymbol{E}_S^{(+)}(z) = \tfrac{1}{2}\,\boldsymbol{E}_S(z)\, e^{ik'_S z}. \tag{B.29}$$

Then

$$\partial_z \boldsymbol{E}_S^{(+)} = \tfrac{1}{2}\left(\partial_z \boldsymbol{E}_S + ik'_S \boldsymbol{E}_S\right) e^{ik'_S z}, \tag{B.30}$$

$$\partial_z^2 \boldsymbol{E}_S^{(+)} = \tfrac{1}{2}\left(\partial_z^2 \boldsymbol{E}_S + 2ik'_S\partial_z \boldsymbol{E}_S - k'^2_S \boldsymbol{E}_S\right) e^{ik'_S z} \approx \tfrac{1}{2}\left(2ik'_S\partial_z \boldsymbol{E}_S - k'^2_S \boldsymbol{E}_S\right) e^{ik'_S z}, \tag{B.31}$$

where the last step uses the SVEA condition Eq. (B.28) to drop $\partial_z^2 \boldsymbol{E}_S$.

For the time dependence, the corresponding approximation is

$$\partial_t^2\left[\boldsymbol{E}_S(z,t)e^{-i\omega_L t}\right] \approx -\omega_L^2\,\boldsymbol{E}_S(z,t)e^{-i\omega_L t}, \qquad \partial_t^2\left[\boldsymbol{P}_S^{(3)}(t)e^{-i\omega_L t}\right] \approx -\omega_L^2\,\boldsymbol{P}_S^{(3)}(t)e^{-i\omega_L t}. \tag{B.32}$$

Under SVEA, the envelopes vary slowly, so rapid oscillations at frequency ω_L dominate over envelope variations. Thus, higher time derivatives of the slowly-varying parts are negligible.

B.3.2 Envelope propagation equation

The ansätze Eq. (B.26) and Eq. (B.27) are substituted into the nonlinear wave equation Eq. (B.23). Using the SVEA approximations of the spatial and temporal derivatives (Eq. (B.31) and Eq. (B.32)) and imposing the dispersion relation $k_S'^2 = n^2(\omega_L)\omega_L^2/c^2$, the large "homogeneous" oscillatory terms cancel exactly. This leaves a first-order propagation equation for the slowly-varying envelope:

$$ik_S'\,\frac{\partial \boldsymbol{E}_S(z,t)}{\partial z} = -\frac{\omega_L^2}{\varepsilon_0 c^2}\,\boldsymbol{P}_S^{(3)}(t)\,e^{i\Delta k\,z}, \tag{B.33}$$

Here $\Delta k = k_S - k_S'$ is the phase mismatch: the difference between the wavevector of the polarization wave and the free-space dispersion relation $k_S' = n(\omega_L)\omega_L/c$. This crucial first-order propagation equation shows how the field amplitude is built up layer by layer through the material by the induced polarization.

B.3.3 Integration and sinc factor

Equation (B.33) is integrated across the sample. Assuming no signal at the entrance, $\boldsymbol{E}_S(0,t) = 0$ (the signal is generated only by the nonlinear polarization inside), integration from $z = 0$ to $z = L$ yields the signal field exiting the sample:

$$\boldsymbol{E}_S(L,t) = i\,\frac{\omega_L}{n(\omega_L)\varepsilon_0 c}\,\boldsymbol{P}_S^{(3)}(t)\,L\,e^{i\Delta k L/2}\,\mathrm{sinc}\left(\frac{\Delta k L}{2}\right). \tag{B.34}$$

This result shows that, up to sample-geometry factors and phase-matching effects (encoded in the sinc and exponential factors), the emitted signal field is directly proportional to the (phase-matched component of the) third-order polarization.

In a well-designed experiment, the phase-matching condition is satisfied: $\Delta k \approx 0$. In this limit, $\mathrm{sinc}(\Delta k L/2) \to 1$ and $e^{i\Delta k L/2} \to 1$. Therefore, from Eq. (B.34), the emitted (complex) signal field at the sample exit becomes

$$\boldsymbol{E}_S(t) \equiv \boldsymbol{E}_S(L,t) \propto i\,\frac{\omega_L}{n(\omega_L)\varepsilon_0 c}\,\boldsymbol{P}_S^{(3)}(t)\,L. \tag{B.35}$$

This is the key Maxwell-side result for 2DFT spectroscopy: in the phase-matched limit, the measured complex signal field is linear in (and shifted in phase by $\pi/2$ relative to) the corresponding component of $\boldsymbol{P}^{(3)}$. Thus, measuring the emitted field directly provides information about the nonlinear polarization response of the material.

In the frequency domain, one often summarizes this proportionality in the compact form

$$\boldsymbol{E}_S(\omega) = \kappa\,i\omega\,\boldsymbol{P}^{(3)}(\omega), \tag{B.36}$$

with a real prefactor κ that depends on propagation conventions and experimental geometry. Equation (B.36) is the key Maxwell-side link: it demonstrates that the measured field is a linear functional of $\boldsymbol{P}^{(3)}$, converting the nonlinear spectroscopy measurement into a linear inversion problem.

The Maxwell-side (field) derivation is now complete. The connection is as follows: the incident fields drive the matter system (described on the matter side by the Dyson series and response functions from Appendix C) to produce a third-order nonlinear polarization $\boldsymbol{P}^{(3)}$. This polarization, in turn, sources the radiated optical field that is measured experimentally. What remains is the microscopic expression for $\boldsymbol{P}^{(3)}$ in terms of the incident fields, which is provided in Appendix C.

Detection models (homodyne/heterodyne) and the construction of 2D spectra from the phase-selected time-domain signals are discussed in Chapter 3, Sec. 3.7.

Appendix C

Dyson Series and Response Functions (Matter Side)

This appendix derives explicit causal expressions for $\boldsymbol{P}^{(n)}(t)$ in terms of incident fields. Starting from the driven master equation for reduced density operator $\hat{\rho}_S(t)$, the strategy is: (i) set up the driven master equation, (ii) transform to the interaction picture to isolate the light–matter coupling, (iii) expand via the Dyson series, (iv) define response functions as field-independent material properties, and (v) derive convolution formulas for $\boldsymbol{P}^{(n)}$.

IMPORTANT: Revisit section order and naming: clarify why the delay-variable change sits here, how it maps to $t_{\mathrm{coh}}, T, t_{\mathrm{det}}$ in the main text, and whether a brief roadmap would help readers.

C.1 Driven master equation

To derive an explicit expression for $\boldsymbol{P}^{(n)}$, the microscopic equation of motion is introduced first. Here $\hat{\rho}_S(t)$ denotes the reduced density operator of the material system after tracing out the environment, and the equation of motion is a driven open-system master equation. It is assumed that $\hat{\rho}_S(t)$ obeys

$$\frac{\mathrm{d}\hat{\rho}_S(t)}{\mathrm{d}t} = -\mathrm{i}\,\mathcal{K}[\hat{\rho}_S(t)] \quad + \mathrm{i}\,\mathcal{V}[\hat{\rho}_S(t)]\,E(t), \tag{C.1}$$

where $\mathcal{K}$ is the field-free reduced generator obtained after eliminating the environment (e.g. via Redfield or Lindblad theory, cf. Section 2.1). This generator encodes both the system Hamiltonian evolution and dissipative/dephasing effects. The quantity $E(t)$ is the scalar field envelope along a chosen polarization direction, and

$$\mathcal{V}[\hat{A}] \equiv [\hat{\mu}, \hat{A}], \qquad \hat{\mu} \equiv \hat{\boldsymbol{\mu}} \cdot \mathbf{e}, \tag{C.2}$$

is the dipole superoperator (commutator form). The field-free propagator is

$$\mathcal{U}_0(t) \equiv \exp(-\mathrm{i}\,\mathcal{K}t),$$

meaning that in the absence of the field one has $\hat{\rho}_S(t) = \mathcal{U}_0(t - t_0)[\hat{\rho}_S(t_0)]$.

C.2 Dipole superoperator structure

Before deriving the perturbation expansion, it is useful to understand the structure of the dipole superoperator. From Eq. (C.2), the commutator can be expanded explicitly:

$$\mathcal{V}[\hat{A}] = [\hat{\mu}, \hat{A}] = \hat{\mu}\,\hat{A} - \hat{A}\,\hat{\mu}, \tag{C.3}$$

so the field interaction consists of two pieces: a *left action* of the dipole operator on the density operator and a *right action*. This "left minus right" commutator structure is the algebraic origin of the different Liouville-space (double-sided) pathways encountered in nonlinear optics: whether the dipole acts on the “ket” (left) or on the “bra” (right) determines which coherences or populations are created at each interaction (cf. also the main-text definition of the dipole superoperator in **??**).

C.3 Interaction picture formulation

To isolate the field-dependent dynamics from the field-free evolution, the interaction picture is defined with respect to the field-free generator $\mathcal{K}$. The interaction-picture density operator is defined by

$$\hat{\rho}_{S,I}(t) \equiv \mathcal{U}_0(-t)[\hat{\rho}_S(t)], \tag{C.4}$$

and the interaction-picture dipole superoperator by

$$\mathcal{V}_\mathrm{I}(t) \equiv \mathcal{U}_0(-t)\,\mathcal{V}\,\mathcal{U}_0(t).$$

By differentiating Eq. (C.4) with respect to time (chain rule), inserting the master equation Eq. (C.1), and using the commutativity of $\mathcal{U}_0$ and $\mathcal{K}$, the equation of motion simplifies to

$$\frac{\mathrm{d}\hat{\rho}_{S,I}(t)}{\mathrm{d}t} = \mathrm{i}\,\mathcal{V}_\mathrm{I}(t)[\hat{\rho}_{S,I}(t)]\,E(t). \tag{C.5}$$

This interaction-picture equation is exact for the reduced system. The field-free dynamics (including both Hamiltonian evolution and dissipation/dephasing) have been absorbed into the time-dependence of the operators $\mathcal{V}_\mathrm{I}(t)$ and the picture transformation $\mathcal{U}_0(t)$. Only the explicit light–matter coupling $\mathcal{V}_\mathrm{I}(t)E(t)$ remains on the right-hand side, making this form ideal for perturbation theory in the field strength.

C.4 Dyson series expansion

To find the solution to Eq. (C.5), the equation is converted to an integral form. Integrating Eq. (C.5) from an initial time t_0 to the current time t gives

$$\hat{\rho}_{S,I}(t) = \hat{\rho}_{S,I}(t_0) \quad + \mathrm{i}\int_{t_0}^{t} \mathrm{d}\tau\,\mathcal{V}_\mathrm{I}(\tau)[\hat{\rho}_{S,I}(\tau)]\,E(\tau). \tag{C.6}$$

Note that the initial condition is the same in both pictures: $\hat{\rho}_{S,I}(t_0) = \hat{\rho}_S(t_0)$ (the transformation is just a change of basis).

To extract the field-dependent contributions, a perturbation expansion is performed. The right-hand side of Eq. (C.6) is substituted back into itself iteratively to generate an expansion in powers of the incident field.

IDEA: maybe cite the part in chapter 2 to shorten this explanation

In direct analogy to standard time-dependent perturbation theory for wavefunctions, each term corresponds to a contribution from a fixed number of light–matter interactions. The key difference is that the density operator carries both a ket and a bra. Consequently, the light–matter interaction can act from the left or from the right. This left/right (ket/bra) structure is precisely what later appears as the commutator (“left minus right”) form of the dipole superoperator and underlies the Liouville-space pathway picture used in nonlinear spectroscopy [57].

After n iterations of substituting the integral equation into itself, collecting all terms of order n in the field, one arrives at the Dyson series expressing the full solution:

$$\hat{\rho}_{S,I}(t) = \left[\mathbb{1} + \sum_{n=1}^{\infty}(\mathrm{i})^n \int_{t_0}^{t}\mathrm{d}\tau_1 \int_{t_0}^{\tau_1}\mathrm{d}\tau_2 \cdots \int_{t_0}^{\tau_{n-1}}\mathrm{d}\tau_n\,\mathcal{V}_\mathrm{I}(\tau_1)\cdots\mathcal{V}_\mathrm{I}(\tau_n)E(\tau_1)\cdots E(\tau_n)\right][\hat{\rho}_S(t_0)]. \tag{C.7}$$

This expansion is exact to all orders in the field. The nested integration limits $t_0 < \tau^{(n)} < \cdots < \tau^{(1)} < t$ enforce the time ordering of interaction events that is required by causality. The series can be compactly written as a time-ordered exponential

$$\hat{\rho}_{S,I}(t) = \mathcal{T} \exp\left[\mathrm{i} \int_{t_0}^{t} \mathrm{d}\tau \, \mathcal{V}_{\mathrm{I}}(\tau) \, E(\tau) \right] \hat{\rho}_S(t_0),$$

where $\mathcal{T}$ is the time-ordering operator. The explicit Dyson-series form Eq. (C.7) is more useful for spectroscopy because each term of fixed order n can be isolated experimentally.

C.5 Experimentally observable time delays and causality

To connect the Dyson series to experimental observables, the time arguments are rewritten in terms of relative delays between interactions. Define:

$$\tau_1 = \tau^{(n)} - \tau^{(n-1)}, \qquad \tau_2 = \tau^{(n-1)} - \tau^{(n-2)}, \qquad \ldots, \qquad \tau_n = t - \tau^{(1)},$$

where $t_0 < \tau^{(n)} < \cdots < \tau^{(1)} < t$ are interaction times. In three-pulse spectroscopy these become the coherence time t_{coh}, waiting time T, and detection time t_{det}. Causality requires all delays positive. This change of variables transforms the nested time-ordered integrals into independent delay integrals over $[0, \infty)$, yielding the convolution structure used throughout the main text.

C.6 Polarization as a perturbation series

With the density operator expressed as a perturbation expansion, the polarization can be extracted. The macroscopic polarization is defined as the expectation value of the dipole operator (cf. **??**). Using $\hat{\rho}_S(t) = \mathcal{U}_0(t)\,[\hat{\rho}_{S,I}(t)]$ and cyclicity of the trace, this can be written in interaction-picture form as

$$\boldsymbol{P}(t) = \mathrm{Tr}\{\hat{\boldsymbol{\mu}}_{\mathrm{I}}(t)\, \hat{\rho}_{S,I}(t)\} = \sum_{n=0}^{\infty} \boldsymbol{P}^{(n)}(t),$$

where $\hat{\boldsymbol{\mu}}_{\mathrm{I}}(t)$ is the interaction-picture dipole operator defined by the field-free dynamics (equivalently, the Heisenberg-evolved dipole under $\mathcal{K}$). In Liouville-space notation this is the adjoint action of the field-free propagator on the observable, $\hat{\boldsymbol{\mu}}_{\mathrm{I}}(t) = \mathcal{U}_0^{\dagger}(t)[\hat{\boldsymbol{\mu}}]$.

$$\boldsymbol{P}^{(n)}(t) = \mathrm{Tr}\{\hat{\boldsymbol{\mu}}_{\mathrm{I}}(t)\, \hat{\rho}_{S,I}^{(n)}(t)\}$$

the contribution containing n factors of the field. Substituting the Dyson expansion Eq. (C.7) into this definition and extracting the n-th order term yields a convolution-like expression. It is often useful to rewrite $\boldsymbol{P}^{(n)}$ directly in terms of the field-free propagators $\mathcal{U}_0$ and the Schrödinger-picture superoperator $\mathcal{V}$, keeping the time ordering explicit

$$\begin{aligned} \boldsymbol{P}^{(n)}(t) = (\mathrm{i})^n \int_{t_0}^{t} \mathrm{d}\tau^{(1)} \cdots \int_{t_0}^{\tau^{(n-1)}} \mathrm{d}\tau^{(n)} \\ \mathrm{Tr}\Big\{ \hat{\boldsymbol{\mu}}\, \mathcal{U}_0(t-\tau^{(1)})\, \mathcal{V}\, \mathcal{U}_0(\tau^{(1)}-\tau^{(2)})\, \mathcal{V} \cdots \mathcal{V}\, \mathcal{U}_0(\tau^{(n)}-t_0)[\hat{\rho}_S(t_0)] \Big\} \prod_{j=1}^{n} E(\tau^{(j)}), \end{aligned} \tag{C.8}$$

This form makes the causality and structure transparent: reading from right to left inside the trace, the system starts in the initial state $\hat{\rho}_S(t_0)$, undergoes n alternating interactions (via $\mathcal{V}$) and free evolutions (via $\mathcal{U}_0$), and is finally measured via the dipole operator $\hat{\boldsymbol{\mu}}$. This physical picture underlies the interpretation of Feynman diagrams in nonlinear spectroscopy.

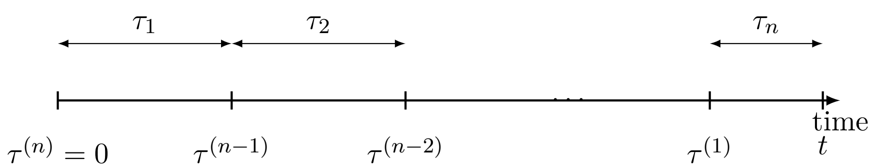

Figure C.1: Time variables in this appendix: $\tau^{(j)}$ denote absolute interaction times, while τ_j denote the nonnegative delay intervals between successive events.

C.7 Definition of the response function

The relative delay intervals τ_j defined above are the time separations between successive interactions

$$\tau_n = t - \tau^{(1)}, \qquad \tau_{n-j} = \tau^{(j)} - \tau^{(j+1)} \; (j = 1, \ldots, n-1),$$

so that $\tau_j \geq 0$ represents the time interval between the $(j+1)$-th and j-th interactions (or between measurement and the last interaction for $j = n$). Explicitly, inverting this definition gives

$$\tau^{(n)} = t - \sum_{k=1}^{n} \tau_k, \qquad \tau^{(n-1)} = t - \sum_{k=2}^{n} \tau_k, \qquad \ldots, \qquad \tau^{(1)} = t - \tau_n. \tag{C.9}$$

The Jacobian of this transformation is unity. Moreover, the nested integration region $t_0 < \tau^{(n)} < \cdots < \tau^{(1)} < t$ over absolute times maps to independent semi-infinite integrals $\tau_j \in [0, \infty)$ when the limit $t_0 \to -\infty$ is taken. This change of variables is conceptually depicted in Fig. C.1: Substituting the change of variables Eq. (C.9) into Eq. (C.8) and assuming the system starts in equilibrium $\hat{\rho}_S(t_0) = \hat{\rho}_{\text{eq}}$ at $t_0 \to -\infty$, the initial evolution becomes trivial since $\mathcal{U}_0(t)[\hat{\rho}_{\text{eq}}] = \hat{\rho}_{\text{eq}}$ for all t. The n-th order polarization then takes the standard operationally useful form:

$$\boldsymbol{P}^{(n)}(t) = \int_0^{\infty} \mathrm{d}\tau_n \cdots \int_0^{\infty} \mathrm{d}\tau_1 \, \boldsymbol{S}^{(n)}(\tau_n, \ldots, \tau_1) E(t - \tau_n)\, E(t - \tau_n - \tau_{n-1}) \cdots E\Big(t - \sum_{k=1}^{n} \tau_k\Big), \tag{C.10}$$

where the *response function* $\boldsymbol{S}^{(n)}$ is defined as follows. Expanding the product of commutators in Eq. (C.8) under the delay transformation and taking the limit $t_0 \to -\infty$, the contribution that is independent of the field shape is isolated. This intrinsic material response is:

$$\boldsymbol{S}^{(n)}(\tau_n, \ldots, \tau_1) \equiv (\mathrm{i})^n \, \mathrm{Tr}\Big\{ \hat{\boldsymbol{\mu}} \, \mathcal{U}_0(\tau_n)\, \mathcal{V} \, \mathcal{U}_0(\tau_{n-1})\, \mathcal{V} \cdots \mathcal{V} \, \mathcal{U}_0(\tau_1)\, \mathcal{V}[\hat{\rho}_{\text{eq}}] \Big\}. \tag{C.11}$$

Equation Eq. (C.11) defines the (causal) *response function*: $\boldsymbol{S}^{(n)}(\tau_n, \ldots, \tau_1)$ describes the intrinsic material response to n successive light–matter interactions separated by the delay intervals $(\tau_n, \ldots, \tau_1)$. Crucially, $\boldsymbol{S}^{(n)}$ depends only on the field-free dynamics $\mathcal{U}_0$ and the dipole operator (through $\mathcal{V}$), and is completely independent of the particular pulse shapes or envelope functions $E(t)$. This is what makes response functions a fundamental and universal characterization of nonlinear material properties. The nested-commutator structure that emerges from expanding the product in Eq. (C.11) is essential for understanding nonlinear spectroscopy. Since each interaction produces a left and a right action on the density operator as shown in Eq. (C.3). Expanding the product of n commutators therefore generates 2^n distinct Liouville-space contributions or "pathways" before additional physical selections (e.g., rotating-wave approximation, phase matching/phase cycling) restrict the subset that contributes to a given measured signal. These pathways are naturally represented by double-sided Feynman diagrams, where the two branches represent the ket and bra evolution and each field interaction is drawn as an arrow acting from the left or from the right [3].

A final important point: while the Dyson series (Eq. (C.7)) itself is strictly time ordered with $\tau_1 > \tau_2 > \cdots > \tau_n$, in a real experiment the three pulses can arrive in any order. Different pulse orderings correspond to different time-ordered contributions to $\boldsymbol{P}^{(n)}$. The experimentally measured total response is obtained by summing the relevant orderings for the chosen pulse sequence.

Two points are emphasized. First, causality is explicit: $\boldsymbol{S}^{(n)}$ is only needed for nonnegative delays, and the fields are evaluated at times earlier than t. Second, $\boldsymbol{S}^{(n)}$ depends only on field-free dynamics and the dipole operator. It is independent of the particular pulse shapes.

C.8 Tensor structure and polarization components

If one does not fix a single polarization direction, the response becomes a rank-(n+1) tensor. Introduce component dipole superoperators

$$\mathcal{V}_\beta[\hat{A}] \equiv [\hat{\mu}_\beta, \hat{A}], \qquad \beta \in \{x, y, z\},$$

and define

$$S^{(n)}_{\alpha\beta_1\cdots\beta_n}(\tau_n, \ldots, \tau_1) = (\mathrm{i})^n \operatorname{Tr}\Big\{\hat{\mu}_\alpha\, \mathcal{U}_0(\tau_n)\, \mathcal{V}_{\beta_n}\, \mathcal{U}_0(\tau_{n-1})\, \mathcal{V}_{\beta_{n-1}} \cdots \mathcal{U}_0(\tau_1)\, \mathcal{V}_{\beta_1}[\hat{\rho}_{\mathrm{eq}}]\Big\}.$$

Then $P^{(n)}_\alpha(t)$ is obtained by contracting this tensor with the field components $E_\beta(t)$. Choosing a fixed unit polarization $\mathbf{e}$ and writing $E_\beta(t) = e_\beta E(t)$ reduces the tensorial form back to Eq. (C.10).

C.9 Explicit linear and third-order response

The linear (first-order) response is briefly discussed as it represents the simplest case of light-matter interaction, illustrating fundamental absorption and dispersion processes. The third-order response is examined in detail, as it forms the foundation for two-dimensional Fourier transform (2DFT) spectroscopy.

C.9.1 Linear response

With $n = 1$, the first-order (linear) polarization is

$$\boldsymbol{P}^{(1)}(t) = \int_0^\infty \mathrm{d}\tau_1\, \boldsymbol{S}^{(1)}(\tau_1)\, E(t - \tau_1),$$

where the linear response function $\boldsymbol{S}^{(1)}(\tau_1)$ from Eq. (C.11) with $n = 1$ is

$$\boldsymbol{S}^{(1)}(\tau_1) = \mathrm{i} \operatorname{Tr}\{\hat{\boldsymbol{\mu}}\, \mathcal{U}_0(\tau_1)\, \mathcal{V}[\hat{\rho}_{\mathrm{eq}}]\}.$$

This linear response describes absorption and dispersion of the incident field.

C.9.2 Third-order response

For the third-order term central to 2D spectroscopy, $n = 3$ is used in Eq. (C.10):

$$\boldsymbol{P}^{(3)}(t) = \int_0^\infty \mathrm{d}\tau_3 \int_0^\infty \mathrm{d}\tau_2 \int_0^\infty \mathrm{d}\tau_1\, \boldsymbol{S}^{(3)}(\tau_3, \tau_2, \tau_1)\, E(t - \tau_3)\, E(t - \tau_3 - \tau_2)\, E(t - \tau_3 - \tau_2 - \tau_1), \quad \text{(C.12)}$$

where the third-order response function $\boldsymbol{S}^{(3)}(\tau_3, \tau_2, \tau_1)$ from Eq. (C.11) with $n = 3$ is

$$\boldsymbol{S}^{(3)}(\tau_3, \tau_2, \tau_1) = (\mathrm{i})^3 \operatorname{Tr}\Big\{\hat{\boldsymbol{\mu}}\, \mathcal{U}_0(\tau_3)\, \mathcal{V}\, \mathcal{U}_0(\tau_2)\, \mathcal{V}\, \mathcal{U}_0(\tau_1)\, \mathcal{V}[\hat{\rho}_{\mathrm{eq}}]\Big\}. \quad \text{(C.13)}$$

This third-order response captures all two-quantum coherence and collective effects in the material. In a multi-pulse 2D experiment, the incident field is decomposed into three pulse envelopes: $E(t) = E_1(t)+E_2(t)+E_3(t)$. When substituted into Eq. (C.12), this generates many terms. Experimentally, the desired third-order signal (e.g., the rephasing pathway) is isolated by phase matching or phase cycling. The response function Eq. (C.13) provides the causal kernel underlying that selection [57].

Remark on frequency-domain. For completeness, the frequency domain is sometimes convenient. After decomposing each incident field into positive and negative frequency components (complex fields and their conjugates), the convolution in Eq. (C.12) transforms into an algebraic product in frequency space. A compact bookkeeping form for third-order mixing is

$$P_i^{(3)}(t) = \varepsilon_0 \sum_{jkl} \sum_{\sigma_1,\sigma_2,\sigma_3=\pm 1} \chi^{(3)}_{ijkl}\big(-\omega_S; \sigma_1\omega_1, \sigma_2\omega_2, \sigma_3\omega_3\big)\, E_{1,j}^{(\sigma_1)} E_{2,k}^{(\sigma_2)} E_{3,l}^{(\sigma_3)}\, e^{-i\omega_S t},$$

with $\omega_S = \sigma_1\omega_1 + \sigma_2\omega_2 + \sigma_3\omega_3$. This form shows how the third-order susceptibility tensor $\chi^{(3)}$ governs the frequency mixing. Fixing all field polarization directions reduces this general tensorial expression to a scalar that matches expressions used throughout the main text.

It is worth pausing to note what has been accomplished. Starting from the microscopic equation of motion (Eq. (C.1)), invoking the Dyson series to generate a perturbative expansion, transforming to experimentally natural time intervals, and assuming equilibrium initial conditions, an explicit and causal expression for $\boldsymbol{P}^{(n)}(t)$ has been derived as a convolution of a system-dependent response function with the incident fields. In nonlinear spectroscopy, this is not merely a formal mathematical result: each term of fixed order corresponds to a real, experimentally isolable physical process (selectable by phase matching or phase cycling), and the response function $\boldsymbol{S}^{(n)}$ provides the system-specific, causal "memory kernel" through which the incident fields probe the material.

This appendix stops here. Detection models (homodyne/heterodyne) and the construction of 2D spectra from the phase-selected time-domain signals are discussed in Chapter 3, Sec. 3.7.

Appendix D

Detection Schemes and Complex-Signal Reconstruction in 2D Spectroscopy

This appendix collects the experimental detection-side remarks that motivate why coherent two-dimensional Fourier-transform spectroscopy (2D-FTS/2DES) is commonly described in terms of a *complex* signal field. The Maxwell-side link from a nonlinear polarization source to an emitted phase-matched field (including the heterodyne cross term that renders the measurement linear in the signal field) is derived in Appendix B.

D.1 Homodyne and heterodyne detection

Experimentally there are two common detection schemes that differ in sensitivity and in whether phase information is retained: homodyne and heterodyne detection [58].

Detectors measure intensities. In 2DES, the outgoing field is typically sent through a spectrometer and recorded as a function of detection frequency ω_{det} (spectrally dispersed detection). Formally, the recorded signal is proportional to the detected intensity integrated over the detector response time,

$$S_{\mathrm{det}}(\omega_{\mathrm{det}}) \propto \int I_{\mathrm{det}}(\omega_{\mathrm{det}}, t)\, \mathrm{d}t. \tag{D.1}$$

Homodyne (self-heterodyne) detection. In *homodyne* detection, one records the intensity of the signal field itself,

$$I_{\mathrm{HOD}}(\omega_{\mathrm{det}}) \propto \left|E_S(\omega_{\mathrm{det}})\right|^2. \tag{D.2}$$

This scales quadratically with the signal field (and thus with the underlying polarization), mixes different nonlinear orders, and does not directly retain the signal phase. Many linear experiments can nonetheless be viewed as a special "self-heterodyne" limit where the weak generated field interferes with a strong transmitted reference field that is automatically present.

Heterodyne detection (explicit LO). In background-free four-wave-mixing geometries the signal is emitted into a distinct phase-matched direction, so no strong reference beam automatically co-propagates with it. One therefore introduces an explicit *local oscillator* (LO) field E_{LO} that overlaps with the signal on the detector. Writing the spectrally resolved total field as

$$E_{\mathrm{tot}}(\omega_{\mathrm{det}}) = E_{\mathrm{LO}}(\omega_{\mathrm{det}}) + E_S(\omega_{\mathrm{det}}), \tag{D.3}$$

the measured spectral intensity is

$$I_{\mathrm{HET}}(\omega_{\mathrm{det}}) \propto \left|E_{\mathrm{tot}}(\omega_{\mathrm{det}})\right|^2 = |E_{\mathrm{LO}}(\omega_{\mathrm{det}})|^2 + |E_S(\omega_{\mathrm{det}})|^2 + 2\ \mathrm{Re}\left[E_{\mathrm{LO}}^*(\omega_{\mathrm{det}})E_S(\omega_{\mathrm{det}})\right]. \tag{D.4}$$

For $|E_S| \ll |E_{\mathrm{LO}}|$ the $|E_S|^2$ term can be neglected, and the LO background $|E_{\mathrm{LO}}|^2$ can be measured and subtracted. It is common to define a normalized heterodyne signal,

$$S_{\mathrm{HET}}(\omega_{\mathrm{det}}) \equiv \frac{I_{\mathrm{HET}}(\omega_{\mathrm{det}}) - |E_{\mathrm{LO}}(\omega_{\mathrm{det}})|^2}{|E_{\mathrm{LO}}(\omega_{\mathrm{det}})|^2} \approx 2\ \mathrm{Re}\left[\frac{E^*_{\mathrm{LO}}(\omega_{\mathrm{det}})E_S(\omega_{\mathrm{det}})}{|E_{\mathrm{LO}}(\omega_{\mathrm{det}})|^2}\right], \tag{D.5}$$

which makes explicit that heterodyne detection renders the measurement linear in the signal field.

Phase sensitivity and phasing. Writing the LO as $E_{\mathrm{LO}}(\omega_{\mathrm{det}}) = A(\omega_{\mathrm{det}})e^{i\phi_{\mathrm{LO}}(\omega_{\mathrm{det}})}$ shows that the recorded cross term depends on the (in general unknown) LO spectral phase,

$$S_{\mathrm{HET}}(\omega_{\mathrm{det}}) \propto \mathrm{Re}\left\{e^{-i\phi_{\mathrm{LO}}(\omega_{\mathrm{det}})}\, E_S(\omega_{\mathrm{det}})\right\}. \tag{D.6}$$

If $\phi_{\mathrm{LO}}(\omega_{\mathrm{det}})$ is not properly controlled or calibrated, absorptive and dispersive contributions are mixed in the measured spectrum. In practice one addresses this by spectral interferometry with a known LO, or by an equivalent analytic-signal reconstruction plus an overall *phasing* procedure [59].

Remark (pulse shaping and direct access to $S^{(3)}$). In the impulsive limit (short, well-separated pulses), the heterodyne-detected signal can become directly proportional to the response function evaluated at the experimental delays, $S^{(3)}(t_{\mathrm{det}}, T, t_{\mathrm{coh}})$, rather than a broad convolution with long pulse envelopes [3].

Remark (what is actually measured, and how to recover a complex signal). Even with heterodyne detection, the detector output at fixed waiting time T and detection frequency is a *real* function of the coherence time. In an idealized description one may write the measured (rephasing) trace as

$$S_R^{(\mathrm{meas})}(\omega_{\mathrm{det}}; T, t_{\mathrm{coh}}) = \mathrm{Re}\left[e^{i\phi}\, E_{S,R}(\omega_{\mathrm{det}}; T, t_{\mathrm{coh}})\right], \tag{D.7}$$

where ϕ is an (unknown) global phase set by the relative signal/LO optical paths. A corresponding expression holds for the nonrephasing branch.

In experiments, one reconstructs a complex signal by spectral interferometry with a known LO (or equivalent analytic-signal methods) and then fixes the remaining global phase by a standard *phasing* procedure (often using a pump–probe projection). In this thesis, complex $E_{S,R/NR}(t_{\mathrm{coh}}, T, t_{\mathrm{det}})$ is obtained directly from phase cycling in the simulations and the experimental reconstruction is therefore not discussed in detail.

Bibliography

[1] H.-P. Breuer and F. Petruccione, *The theory of open quantum systems*, 1st ed. (Oxford University PressOxford, Jan. 25, 2007).

[2] U. Weiss, *Quantum dissipative systems*, 4th ed. (WORLD SCIENTIFIC, Mar. 2012).

[3] S. Mukamel, *Principles of Nonlinear Optical Spectroscopy* (Oxford University Press, New York, 1995), 543 pp.

[4] D. M. Jonas, "Two-dimensional femtosecond spectroscopy", Annu. Rev. Phys. Chem. **54**, 425–463 (2003).

[5] M. Cho, *Two-dimensional optical spectroscopy* (CRC Press, Boca Raton, 2009), 378 pp.

[6] A. Gelzinis et al., "Two-dimensional spectroscopy for non-specialists", Biochim. Biophys. Acta (BBA) - Bioenerg. **1860**, 271–285 (2019).

[7] Á. Rivas et al., "Markovian master equations: a critical study", New Journal of Physics **12**, 113032 (2010).

[8] J. Segarra-Martí et al., "Towards Accurate Simulation of Two-Dimensional Electronic Spectroscopy", Topics in Current Chemistry **376**, 24 (2018).

[9] H. Sun et al., "Two-dimensional spectroscopy of open quantum systems: nonequilibrium green's function formulation", J. Phys. Chem. Lett. **16**, 2008–2015 (2025).

[10] A. P. Kalra et al., "Electronic Energy Migration in Microtubules", ACS Cent. Sci. **9**, 352–361 (2023).

[11] C. W. Kim et al., "Analog quantum simulation of the dynamics of open quantum systems with quantum dots and microelectronic circuits", PRX Quantum **3**, 40308 (2022).

[12] T. D. Ladd et al., "Quantum computers", Nature **464**, 45–53 (2010).

[13] M. Schlosshauer, *Decoherence and the quantum-to-classical transition*, Frontiers Collection (Springer, Berlin, Heidelberg, 2007).

[14] R. Feynman and F. Vernon, "The theory of a general quantum system interacting with a linear dissipative system", Ann. Phys. **24**, 118–173 (1963).

[15] Y. Tanimura, "Numerically "exact" approach to open quantum dynamics: the hierarchical equations of motion (HEOM)", J. Chem. Phys. **153**, 20901 (2020).

[16] A. Redfield, "The theory of relaxation processes", in *Advances in Magnetic and Optical Resonance*, Vol. 1 (Elsevier, 1965), pp. 1–32.

[17] Á. Rivas et al., "Quantum non-markovianity: characterization, quantification and detection", Rep. Prog. Phys. **77**, 94001 (2014).

[18] L. Li et al., "Concepts of quantum non-markovianity: a hierarchy", Phys. Rep. **759**, 1–51 (2018).

[19] G. Lindblad, "On the generators of quantum dynamical semigroups", Commun. Math. Phys. **48**, 119–130 (1976).

[20] D. Manzano, "A short introduction to the Lindblad master equation", AIP Adv. **10**, 025106 (2020).

[21] F. Campaioli et al., "Quantum Master Equations: Tips and Tricks for Quantum Optics, Quantum Computing, and Beyond", PRX Quantum **5**, 020202 (2024).

[22] M. Reed and B. Simon, *Methods of modern mathematical physics* (Academic Press, New York, 1972).

[23] A. Vezvaee et al., "Fourier transform noise spectroscopy", npj Quantum Inf. **10**, 52 (2024).

[24] A. D'Abbruzzo et al., "A time-dependent regularization of the Redfield equation", SciPost Phys. **15**, 117 (2023).

[25] N. Lambert et al., “QuTiP 5: the quantum toolbox in python”, Phys. Rep. **1153**, 1–62 (2026).

[26] A. A. Clerk et al., “Introduction to quantum noise, measurement, and amplification”, Rev. Mod. Phys. **82**, 1155–1208 (2010).

[27] R. Alicki and K. Lendi, “Recent developments”, in *Quantum dynamical semigroups and applications*, edited by R. Alicki and K. Lendi (Springer, Berlin, Heidelberg, 2007), pp. 109–121.

[28] E. B. Davies, “Markovian master equations”, Commun. Math. Phys. **39**, 91–110 (1974).

[29] A. S. Trushechkin et al., “Open quantum system dynamics and the mean force gibbs state”, AVS Quantum Sci. **4**, 12301 (2022).

[30] J. Thingna, “Steady-state transport properties of anharmonic systems”, PhD thesis (National University of Singapore, Department of Physics, Singapore, 2013), 169 pp.

[31] G. I. Hagstrom and P. J. Morrison, “Caldeira–leggett model, landau damping, and the vlasov–poisson system”, Physica D, Special Issue: Fluid Dynamics: From Theory to Experiment **240**, 1652–1660 (2011).

[32] A. J. Leggett et al., “Dynamics of the dissipative two-state system”, Rev. Mod. Phys. **59**, 1–85 (1987).

[33] R. Korol et al., “High-frequency tails in spectral densities”, J. Phys. Chem. A **129**, 3587–3596 (2025).

[34] A. Das et al., “Quantum brownian motion: drude and ohmic baths as continuum limits of the rubin model”, Phys. Rev. E **102**, 62130 (2020).

[35] Y. Tanimura, “Reduced hierarchy equations of motion approach with drude plus brownian spectral distribution: probing electron transfer processes by means of two-dimensional correlation spectroscopy”, J. Chem. Phys. **137**, 22A550 (2012).

[36] N. Lambert et al., “QuTiP-BoFiN: A bosonic and fermionic numerical hierarchical-equations-of-motion library with applications in light-harvesting, quantum control, and single-molecule electronics”, Phys. Rev. Research **5**, 013181 (2023).

[37] D. A. Skoog et al., *Principles of instrumental analysis* (Cengage Learning, Jan. 27, 2017), 990 pp.

[38] J. J. Balmer, “Notiz über die spectrallinien des wasserstoffs”, Ann. Phys. **261**, 80–87 (1885).

[39] J. R. Rydberg, “XXXIV. On the structure of the line-spectra of the chemical elements”, Lond. Edinb. Dublin Philos. Mag. J. Sci. **29**, 331–337 (1890).

[40] N. Bohr, “I. On the constitution of atoms and molecules”, Lond. Edinb. Dublin Philos. Mag. J. Sci. **26**, 1–25 (1913).

[41] P. R. Berman, *Principles of laser spectroscopy and quantum optics* (Princeton University Press, Princeton, 2011), 1 p.

[42] P. Dirac, “The quantum theory of the emission and absorption of radiation”, Proc. R. Soc. Lond., Contain. Pap. Math. Phys. Character **114**, 243–265 (1927).

[43] P. Hamm, “Principles of nonlinear optical spectroscopy: a practical approach”, (2005).

[44] U. Bangert et al., “High-resolution two-dimensional electronic spectroscopy reveals the homogeneous line profile of chromophores solvated in nanoclusters”, Nat. Commun. **13**, 3350 (2022).

[45] E. Fresch et al., “Two-dimensional electronic spectroscopy”, Nat. Rev. Methods Primers **3**, 84 (2023).

[46] C. Cohen-Tannoudji et al., eds., *Atom-photon interactions: basic processes and applications*, A Wiley-Interscience Publication (Wiley, New York, NY, 2008), 656 pp.

[47] R. W. Boyd, “Chapter 6 - nonlinear optics in the two-level approximation”, in *Nonlinear Optics (third Edition)*, edited by R. W. Boyd, Third Edition (Academic Press, Burlington, 2008), pp. 277–328.

[48] D. Groll et al., “Fundamentals of heterodyne wave mixing spectroscopy: a tutorial”, Nano Futures **9**, 42601 (2025).

[49] M.-R. Cai et al., “Extracting double-quantum coherence in two-dimensional electronic spectroscopy under pump–probe geometry”, Rev. Sci. Instrum. **95**, 33006 (2024).

[50] D. Green et al., “Vibrational coherences in half-broadband 2D electronic spectroscopy: spectral filtering to identify excited state displacements”, J. Chem. Phys. **160**, 234104 (2024).

[51] N. S. Ginsberg et al., "Two-dimensional electronic spectroscopy of molecular aggregates", Acc. Chem. Res. **42**, 1352–1363 (2009).

[52] F. D. Fuller and J. P. Ogilvie, "Experimental implementations of two-dimensional fourier transform electronic spectroscopy", Annu. Rev. Phys. Chem. **66**, 667–690 (2015).

[53] T. Mančal et al., "Two-dimensional optical three-pulse photon echo spectroscopy. I. Nonperturbative approach to the calculation of spectra", The Journal of Chemical Physics **124**, 234504 (2006).

[54] D. J. Griffiths, *Introduction to electrodynamics*, 4. ed., international ed, Always Learning (Pearson, Boston, 2013), 599 pp.

[55] R. H. Lehmberg, "Radiation from an N -Atom System. I. General Formalism", Phys. Rev. A **2**, 883–888 (1970).

[56] B. Masters, "Paths to Förster's resonance energy transfer (FRET) theory", Eur. Phys. J. H **39**, 87–139 (2014).

[57] P. Hamm and M. Zanni, *Concepts and methods of 2D infrared spectroscopy* (Cambridge University Press, Cambridge, 2011).

[58] D. Abramavicius et al., "Coherent multidimensional optical spectroscopy of excitons in molecular aggregates; quasiparticle versus supermolecule perspectives", Chem. Rev. **109**, 2350–2408 (2009).

[59] J. D. Hybl et al., "Two-dimensional electronic spectroscopy", Chem. Phys. Lett. **297**, 307–313 (1998).